

Embryology of the Kidney

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KEY POINTS

- The development of the kidney depends on the triadic interactions among three types of progenitors: the nephron progenitor, the ureteric bud, and the stromal progenitor.
- The epithelial cells that comprise the tubular structures of the kidney are derived from two distinct progenitors: the ureteric bud, which branches and gives rise to collecting ducts, and the nephron progenitor, which undergoes a mesenchyme-to-epithelial transition to form glomerular podocytes, parietal epithelial cells, proximal tubules, loops of Henle, distal tubules, and connecting tubules.
- Nephrogenesis and nephron endowment require a balance between self-renewal of nephron progenitors and epithelial differentiation.
- The timing of incorporation of nephron progenitors into nascent nephrons predicts their positional identity within the highly patterned mature nephron.
- Collecting duct epithelia differentiate into principal cells and intercalated cells through lateral inhibition mediated by Notch signaling.
- Stromal cells regulate ureteric branching morphogenesis, nephrogenesis, and vascular development.
- Endothelial cells track the development of the ureteric epithelium and establish the renal vasculature through a combination of vasculogenic and angiogenic processes.
- Single-cell RNA sequencing has identified representative marker genes and developmental trajectories of multiple cell types, not only in mouse kidneys but also in human kidneys.
- Identification of the origins of the renal progenitors enabled generation of kidney organoids, and single-cell RNA-sequencing revealed similarities and differences between the organoids and the embryonic kidney *in vivo*.

MAMMALIAN KIDNEY DEVELOPMENT

ANATOMIC OVERVIEW OF THE MAMMALIAN KIDNEY

The kidney is a sophisticated, highly vascularized organ that plays a central role in overall body homeostasis. In humans, the kidneys filter as much as 180 liters of blood per day, receiving as much as ~20% of the total cardiac output. Renal filtration of blood removes metabolic waste products (e.g., urea, ammonia, and byproducts of bile from the liver) as urine while concomitantly adjusting the levels of water, electrolytes, and pH of tissue fluids. Additionally, the

kidneys regulate blood pressure via the renin-angiotensin-aldosterone system; secrete erythropoietin, which stimulates erythrocyte production; and contribute to the activation of vitamin D to control calcium and phosphate balance.

The filtration function of the kidneys is accomplished by basic units called nephrons, which consist of glomeruli and renal tubules (Fig. 1.1). On average, humans have 1 million nephrons per adult kidney, but the range of total nephrons is highly variable across human populations (200,000–2.5 million).¹ Each mouse kidney may contain up to 12,000–16,000 nephrons depending on the strain.² This wide range in nephron number is influenced by genetic background, fetal nutrition and environment, and maturity at birth.^{3,4}

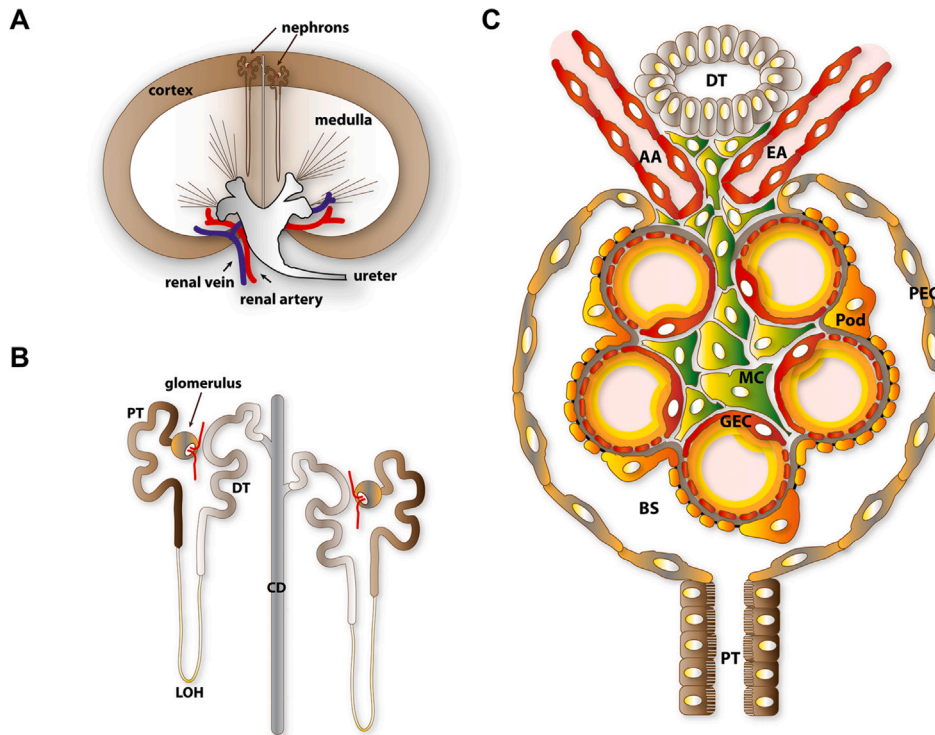


Fig. 1.1 Anatomic organization of the kidney. (A) Spatial distribution of nephron within the metanephric kidney. Glomeruli, the filtration compartments of the nephrons, are found in the cortex. (B) Segmental structure of nephrons. The vascularized glomerulus is found at the proximal end and is connected through a series of renal tubules where urinary filtrate composition is refined through resorption and secretion. (C) Cellular organization of the glomeruli. AA, Afferent arteriole; BS, Bowman space; CD, collecting duct; DT, distal tubule; EA, efferent arteriole; GEC, glomerular endothelial cell; LOH, loop of Henle; MC, mesangial cell; PEC, parietal epithelial cell; Pod, podocyte; PT, proximal tubule. (Reproduced with permission from Scott RP, Quaggin SE. The cell biology of renal filtration. *J Cell Biol.* 2015;209:100–210.)

Nephron endowment can be clinically important as markedly reduced nephron numbers raise the susceptibility risk to hypertension and chronic kidney disease.⁵⁻⁹ At the core of the nephron is the renal corpuscle or glomerulus (see Fig. 1.1). The glomerulus consists of a porous and highly convoluted capillary bed composed of fenestrated glomerular endothelial cells. These glomerular capillaries are circumscribed by morphologically elaborate and interdigitating cells called podocytes. These capillaries are further structurally supported by pericytes called mesangial cells. Blood filtration occurs through this capillary tuft, generating primary urine that collects within the Bowman capsule, an enclosure formed by parietal epithelial cells. From the Bowman capsule, urine drains through a series of tubules starting with the proximal tubules, the loop of Henle, the distal tubules, and finally the collecting ducts, the latter of which is not included in the definition of a nephron. These renal tubules, as well as collecting ducts, are responsible for dynamic resorption and secretion processes that help recycle filtered small molecules; they also adjust water, electrolyte, and acid-base balance by fine-tuning the composition of the final urine output before it exits the ureter and is excreted via the bladder. Supporting the main functions of the nephrons are interstitial fibroblasts and a heterogeneous network of extraglomerular vasculature.

DEVELOPMENTAL ORIGIN OF THE KIDNEY

The vertebrate kidney derives from the intermediate mesoderm (IM), which represents an anatomically defined region along the dorsoventral axis—positioned between the paraxial mesoderm, forming the somites in the dorsal part of the embryo, and the lateral plate mesoderm, which differentiates into ventrally located skeletomuscular and vascular tissues.¹⁰⁻¹² Mammalian kidneys develop in three successive stages, generating three distinct excretory

structures known as the pronephros, mesonephros, and metanephros (Fig. 1.2). The pronephros and mesonephros are vestigial structures in mammals and degenerate before birth; the metanephros is the definitive mammalian kidney. Many of the signaling pathways and genes that play important roles in the metanephric kidney appear to play parallel roles during the development of the pronephros and mesonephros. The pronephros consists of pronephric tubules and the pronephric duct (also known as the precursor to the Wolffian duct) and develops from the rostral-most region of the urogenital ridge at 22 days' gestation in humans and 8.5 days post coitum (embryonic stage E8.5) in mice (Table 1.1). Throughout the rest of this chapter, most timelines of kidney development are with reference to the mouse. The pronephros serves as the principal excretory organ of the larval stages of fishes and amphibians. The mesonephros develops caudal to the pronephric tubules in the midsection of the urogenital ridge. The mesonephros becomes the functional excretory apparatus in lower vertebrates (adult fish and amphibians) and may perform a filtering function during embryonic life in mammals. Before its degeneration, endothelial, peritubular myoid, and steroidogenic cells from the mesonephros migrate into the adjacent adrenogonadal primordia, which ultimately form the adrenal gland and gonads.¹³ Abnormal mesonephric migration leads to gonadal dysgenesis, a fact that underscores the intricate association between these organ systems during development and explains the common association of gonadal and renal defects in congenital syndromes.^{14,15}

DEVELOPMENT OF THE METANEPHROS FROM THE THREE TYPES OF PROGENITORS

The metanephros, which is formed at the caudal end of the urogenital ridge, is the third and final kidney stage, representing the definitive adult kidney of higher

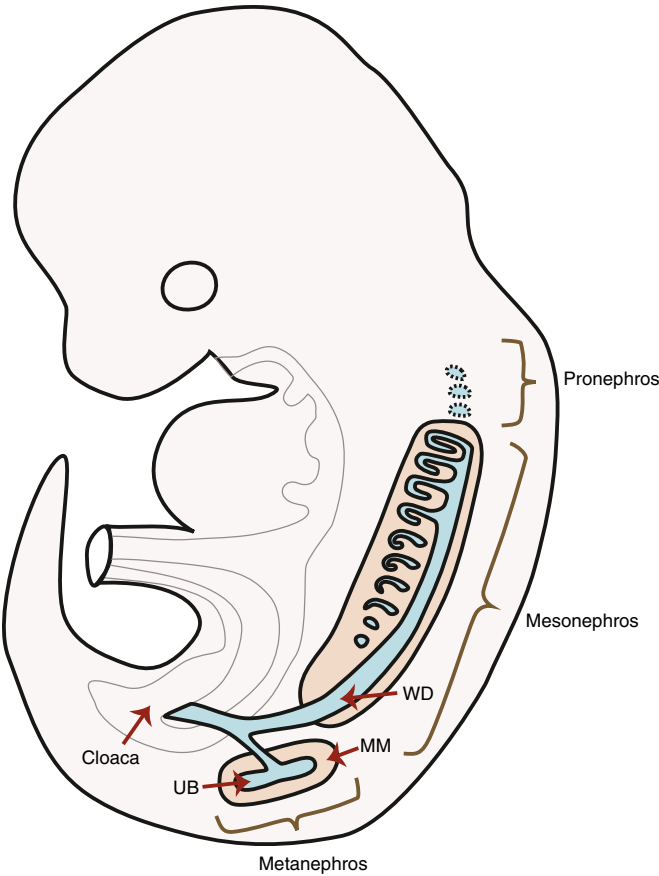


Fig. 1.2 Three types of kidney primordia during mammalian development. Schematic of mouse E11.5 embryo. The pronephros and mesonephros develop in a rostral-to-caudal direction, and the tubules are aligned adjacent to the Wolffian or nephric duct (WD). The metanephros develops by interaction of the ureteric bud epithelium (UB), outgrowth from the caudal end of the WD, and metanephric mesenchyme (MM). The pronephros and mesonephros are vestigial structures in mice and humans and are regressed during development.

vertebrates. The metanephros development relies on triad interactions among the three types of progenitors: the nephron progenitor (NP) and stromal progenitor that reside in the metanephric mesenchyme (MM) and the ureteric bud (UB) ^{16,17} (see Fig. 1.2). The NP in the MM forms the nephron (i.e., glomerular podocytes, parietal epithelial cells, proximal tubules, loops of Henle, distal tubules, and connecting tubules), while the UB gives rise to collecting ducts and the ureter. The stromal progenitor in the MM differentiates into cortical and medullary interstitial cells, as well as mesangial cells and renin-producing cells. The UB is first visible as an outgrowth at the caudal end of the Wolffian duct approximately between the fourth and fifth weeks of gestation in humans or E10.5 in mice. The MM becomes histologically distinct from the surrounding mesenchyme and is found adjacent to the UB. Upon invasion of the UB into the MM, signals from the MM cause the UB to branch into a T-shaped form (at around E11.5 in mice) and then undergo iterative dichotomous branching, giving rise to the urinary collecting duct system (Fig. 1.3). Simultaneously, the UB sends reciprocal signals to NPs in the MM, which are induced to condense along the surface of the UB tips.

Table 1.1 Timelines of Human and Mouse Kidney Development

Stage/Event	Human ^a	Mouse ^b
Pronephros		
Emergence	22nd day	E8.5
Disappearance by	25th day	E10
Mesonephros		
Emergence	24th day	E10
Disappearance by	16th week	E14
Metanephros		
Ureteric bud induction	28th–32nd day	E10.5
Nephrogenesis	44th day	E13
Glomerulogenesis	8th–9th week	E14
Cessation of nephrogenesis	36th week	P3
Gestation (Total Length)	40 weeks	19–21 days

^aHuman timelines refer to gestational periods.
^bMouse timelines are indicated as either embryonic days post coitum (E) or postnatal (P).

NEPHRON DIFFERENTIATION FROM THE NEPHRON PROGENITOR

After condensation, a subset of NP aggregates adjacent to and below the tips of the branching UB. These cell clusters, known as pretubular aggregates, undergo mesenchymal-to-epithelial conversion to become the renal vesicle (Fig. 1.4).

The renal vesicle undergoes patterned segmentation and proceeds through a series of morphologic changes that include gradual recruitment of NPs to form the glomerulus and components of the nephrogenic tubules from the proximal convoluted tubule, loop of Henle, and distal tubule. The renal vesicles undergo differentiation, passing through morphologically distinct stages starting from the comma-shaped stage and proceeding to the S-shaped body, capillary loop, and mature stages. Each step involves precise proximal-to-distal patterning and structural transformations, ultimately forming the glomerulus and components of the nephrogenic tubules from the proximal tubule, loop of Henle, and distal tubule (see Fig. 1.4). Remarkably, this process is repeated 600,000 to 1 million times in each developing human kidney as new nephrons are sequentially born at the tips of the UB, namely the nephrogenic zone, throughout fetal life.

THE NEPHROGENIC ZONE

The nephrogenic zone is an active site of nephrogenesis. It is morphologically identified as a narrow band beneath the renal capsule where the branching UB tips are found together with nascent nephrons (pretubular aggregates, renal vesicles, comma-shaped bodies, and S-shaped bodies), self-renewing NPs, and stromal progenitors (see Fig. 1.4). Kidney growth and nephrogenesis occur in a radial fashion, with new branches of the ureteric tree and newer nephrons added at the outermost periphery of the

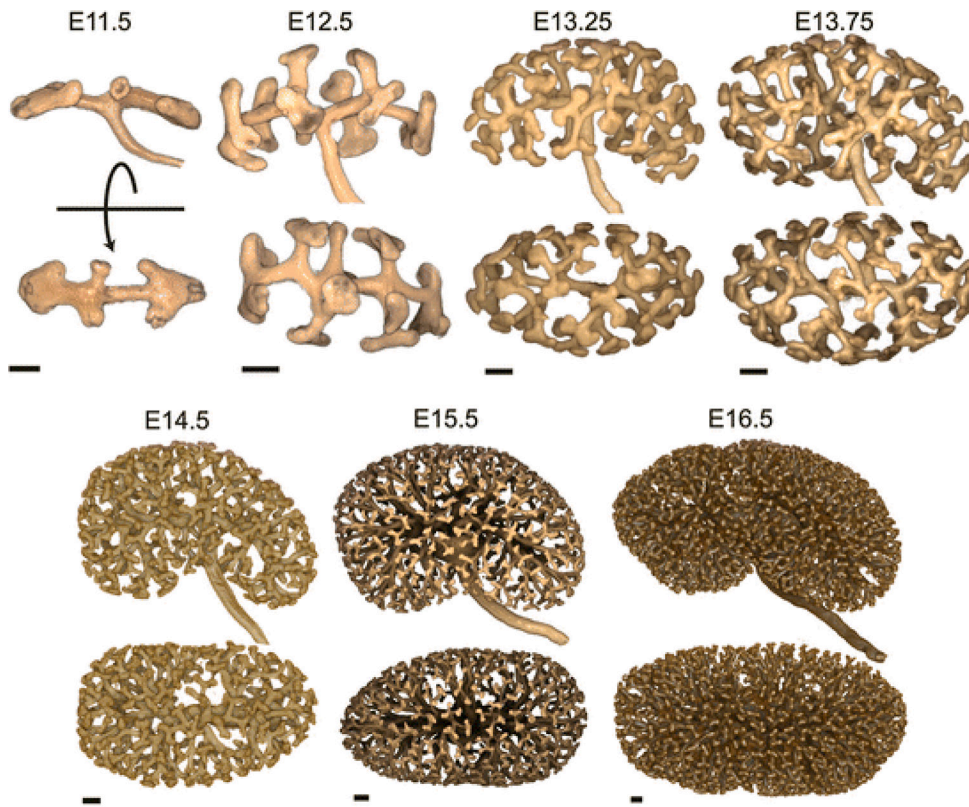


Fig. 1.3 Ureteric branching morphogenesis. Rapid reiterative branching of the ureteric bud epithelium within a 5-day period in mice as imaged with a pan-cytokeratin antibody using optical projection tomography. By E16.5, the renal pelvis, formed by the widening and coalescence of the earliest branches of the ureteric tree, is already apparent. (Reproduced with permission from Short KM, Smuth I. Imaging, analyzing and interpreting branching morphogenesis in the developing kidney. *Results Probl Cell Differ.* 2017;60:233–256.)

developing cortex. Accordingly, within the developing kidney, the most mature nephrons are found in the innermost layers of the cortex and the most immature nephrons are found in the most peripheral regions (see Fig. 1.1). The nephrogenic zone becomes progressively thinner with the gradual depletion of nephrogenic progenitors and disappears when the remaining NPs have completely epithelialized by 36 weeks' gestation in humans and postnatal day 3 (P3) in mice. The kidney eventually divides into two major compartments: an outer region called the cortex and an inner region called the medulla. The glomeruli and proximal and distal tubules are in the cortex, along with the distal part of the nephron, which connects directly to the collecting ducts. The loop of Henle and rest of the collecting duct network comprise the epithelial structures found in the medulla.

NEPHRON SEGMENT PATTERNING ALONG THE PROXIMAL-DISTAL AXIS

The glomerulus develops from the most proximal end of the renal vesicle, farthest from the tip of the UB.^{18,19} Distinct cell types of the glomerulus are first identified in the S-shaped stage, where presumptive podocytes appear as a columnar epithelial cell layer (see Fig. 1.4). Parietal epithelial cells differentiate and flatten to form the Bowman capsule, a structure that surrounds the urinary space and is continuous with the proximal tubular epithelium. A vascular cleft develops, separating the presumptive podocyte layer from more distal cells that will form the proximal tubule. Simultaneously, endothelial cells migrate into the cleft to form capillary loops. The tubular portion of the nephron is segmented in a proximal-distal order into the proximal convoluted tubule, descending and ascending loops of

Henle, and distal convoluted tubule. The distal tubule is adjacent to the collecting duct, a derivative of the UB.

Although all segments of the nephron are present at birth and filtration occurs before birth, maturation of the tubule continues in the postnatal period. Increased expression levels of transporters, switches in transporter isoforms, alterations in paracellular transport mechanisms, and permeability and biophysical properties of tubular membranes have all been observed to occur postnatally.²⁰

GLOMERULOGENESIS BY THE INTERACTION OF PODOCYTES, ENDOTHELIAL CELLS, AND MESANGIAL CELLS

During glomerulogenesis, endothelial cells produce the glomerular basement membrane together with podocytes, a major component of the mature filtration barrier. Initially the podocytes are connected by intercellular tight junctions at their apical surface.²¹ As glomerulogenesis proceeds, the podocytes flatten and spread out to cover the increased surface area of the growing glomerular capillary bed. They develop microtubular-based primary processes and actin-based secondary foot processes.^{22–24} Foot processes of neighboring podocytes interdigitate and elongate. As podocytes mature, intercellular epithelial tight junctions linking become restricted to the basal aspect of the podocyte, relocate from the cell body to the foot processes and are eventually replaced by a modified adherens junction-like structure known as the slit diaphragm.^{21,25} Slit diaphragms are signaling hubs serving as the final layer of the glomerular filtration barrier.²⁶ Mesangial cell ingrowth follows the migration of endothelial cells and is required for development and patterning of the capillary loops that are found in normal glomeruli. The endothelial cells also

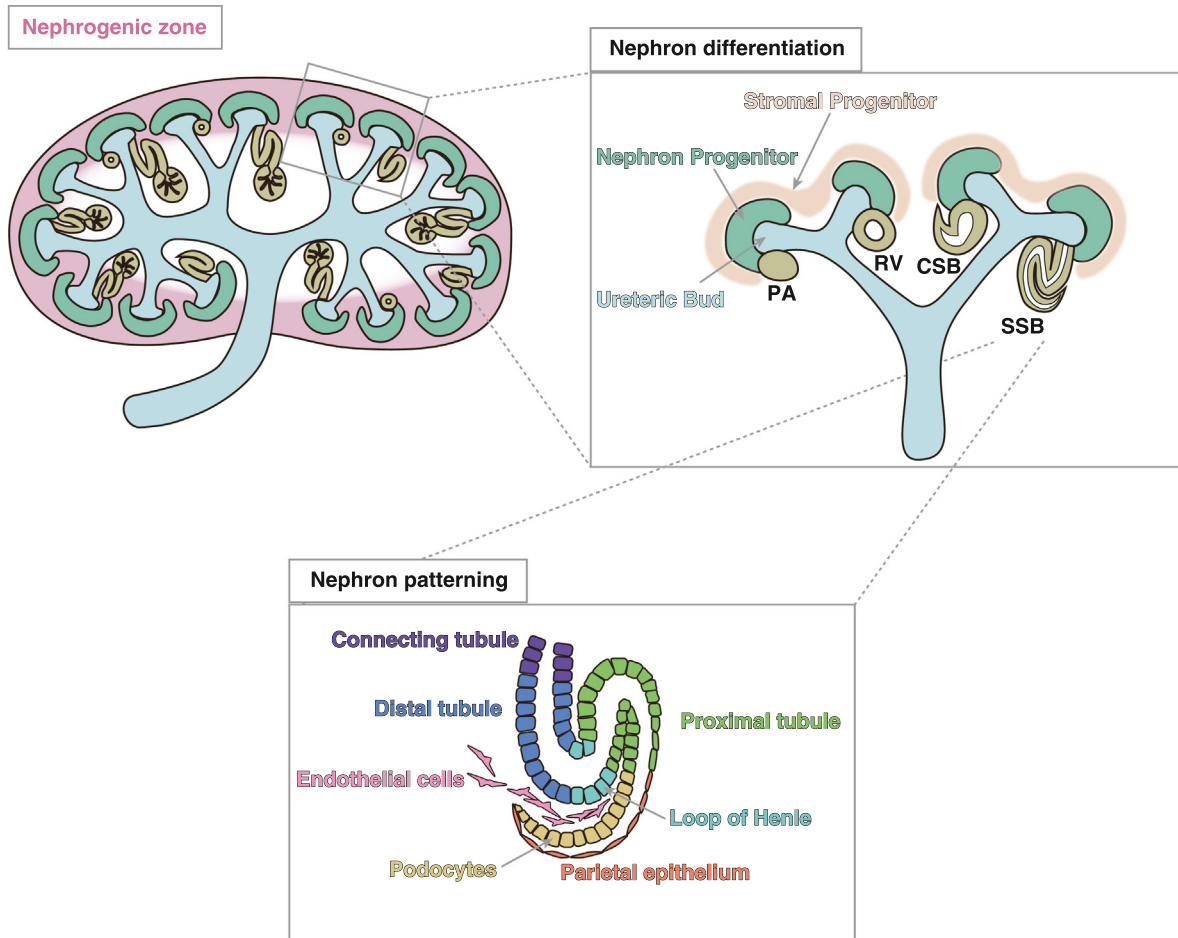


Fig. 1.4 Overview of nephrogenesis. Reciprocal interaction among the nephron progenitor, stromal progenitor, and ureteric bud within the nephrogenic zone enables continuous branching of the ureteric epithelium, self-renewal of nephron progenitor, and differentiation into epithelialized nephron structures. Successive proximo-distal patterning within the nascent nephron structure results in the formation of an S-shaped body with distinct gene expression patterning coding for the future nephron domains. CSB, comma-shaped body; PA, pretubular aggregate; RV, renal vesicle; SSB, S-shaped body.

flatten considerably, and capillary lumens are formed due to apoptosis of a subset of endothelial cells.²⁷ At the capillary loop stage, glomerular endothelial cells develop fenestrae, which are semipermeable transcellular pores common in capillary beds exposed to high hemodynamic flux.

In the mature-stage glomerulus, the podocytes, fenestrated endothelial cells, and intervening glomerular basement membrane comprise the filtration barrier that separates the urinary from the blood space. Together, these components provide a size- and charge-selective barrier that permits free passage of small solutes and water but prevents the loss of larger molecules such as proteins. The mesangial cells are found between the capillary loops, where they are required to provide ongoing structural support to the capillaries and possess smooth muscle cell-like characteristics that have the capacity to contract, which may account for some of the dynamic properties of the glomerulus.

COLLECTING DUCT DIFFERENTIATION FROM THE URETERIC BUD

The collecting duct system is composed of hundreds of tubules through which the filtrate produced by the nephrons is conducted out of the kidney, to the ureter, and then to the bladder. Water and salt resorption and excretion, ammonia

transport, and H^+ ion secretion required for acid-base homeostasis also occur in the collecting ducts, under different regulatory mechanisms and using different transporters and channels that are active along tubular portions of the nephron. The collecting ducts are the product of branching morphogenesis from the single UB, which enables the formation of arborized structure with a single exit (see Fig. 1.3). Considerable remodeling is involved in forming collecting ducts from branches of the UB.²⁸ The branching is highly patterned, with the first several rounds of branching being somewhat symmetric, followed by additional rounds of asymmetric branching, in which a main trunk of the collecting duct continues to extend toward the nephrogenic zone, while smaller buds branch as they induce new nephrons within the nephrogenic zone. The mouse kidney has a single papilla and calyx, whereas a human kidney has 8 to 10 papillae, each of which drains into a minor calyx, with several minor calyces draining into a smaller number of major calyces.

INTERSTITIAL DIFFERENTIATION FROM THE STROMAL PROGENITOR

For decades in classic embryologic studies of kidney development, emphasis has been placed on the reciprocal inductive signals between MM and UB. However, in recent

years, interest in the stromal cell as a key regulator of nephrogenesis has arisen.^{19,29-31} The stromal progenitor in the MM differentiates into cortical and medullary interstitial cells, as well as mesangial cells and renin-producing cells³² (Fig. 1.5). The stromal progenitor cells also provide signals required for UB branching and patterning of the developing kidney. Disruption or loss of these stromal cells leads to impairment of UB branching, reduction in nephron number, disrupted nephron patterning with failure of cortical-medullary boundary formation, and maldevelopment of the renal vasculature. A reciprocal signaling loop from the UB exists to properly pattern stromal cell populations. Loss of these UB-derived signals leads to a buildup of stromal cells beneath the capsule that are several layers thick. As nephrogenesis proceeds, stromal cells differentiate into peritubular interstitial cells and pericytes that are required for vascular remodeling and the production of extracellular matrix responsible for proper nephron formation.³¹

DEVELOPMENT OF THE RENAL VASCULATURE

The microcirculations of the kidney include the specialized glomerular capillary system, responsible for production of the ultrafiltrate and the vasa recta bundles and peritubular capillaries involved in the countercurrent mechanism for urine osmoregulation. At E11.5 in mice, the UB is tracked by a primitive vessel that elaborates in synchrony with both UB branching and nephrogenesis. Capillaries form networks around the developing nephric tubules by E13.5, and the hilar artery and first-order interlobar renal artery branches can be identified by E14.5. These branches will form the corticomedullary arcades and interlobular arteries that branch from these arcades. Further branching produces the glomerular afferent arterioles. From E13.5 onward, endothelial cells migrate into the vascular cleft of developing glomeruli, where they undergo differentiation to form the glomerular capillary loops. The presence of endothelial cell-containing glomeruli becomes apparent at E14.5. The efferent arterioles carry blood away from the glomerulus to a system of fenestrated peritubular capillaries that are in close contact with the adjacent tubules and receive filtered water and solutes reabsorbed from the filtrate.³³ These capillaries have few pericytes. In comparison, the vasa rectae, which surround the medullary tubules and are involved in urinary concentration, are also fenestrated but have more pericytes. They arise from the efferent arterioles of deep glomeruli.³⁴ The peritubular capillary system surrounding the proximal tubules is well developed in the late fetal period, whereas the vasa rectae mature 1 to 3 weeks postnatally.

MODEL SYSTEMS TO STUDY KIDNEY DEVELOPMENT

THE KIDNEY ORGAN CULTURE SYSTEM

From the late 1950s, the method of growing mouse embryonic kidneys as floating cultures on top of filters, or more recently on transwell inserts (Fig. 1.6), a technique pioneered by Clifford Grobstein and improved by Lauri Saxen, accelerated the advancement of the kidney developmental biology field. This classic method, which remains widely in use, has the advantages that the kidney explants are cultivated within an easily manipulated, controlled environment, and there is a

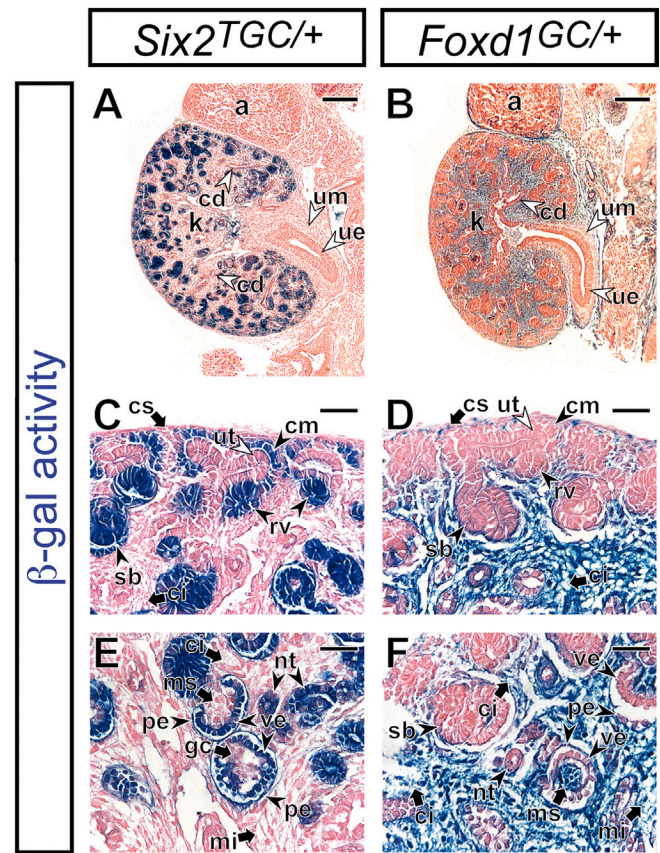


Fig. 1.5 The nephron and stromal progenitors. Lineage tracing analysis in E14.5 mouse kidneys showing derivatives of *Six2*-expressing nephron progenitors (A, C, and E) and the *Foxd1*-expressing stromal progenitors (B, D, and F), stained for β -galactosidase activity (blue). a, Adrenal gland; cd, collecting duct; ci, renal cortical interstitium; cm, cap mesenchyme; cs, renal cortical stroma; gal, galactosidase; gc, glomerular capillary; k, kidney; mi, renal medullary interstitium; ms, mesangium; nt, nephrogenic tubule; pe, parietal epithelium; rv, renal vesicle; sb, S-shaped body; ue, ureter epithelium; um, ureter mesenchyme; ut, ureteric tip; ve, visceral epithelium (podocyte). (Reproduced with permission from Kobayashi A, Mugford JW, Krautzbeger Am, et al. Identification of a multipotent self-renewing stromal progenitor population during mammalian kidney organogenesis. *Stem Cell Reports*. 2014;3:650–662.)

possibility of visualizing the pattern of kidney growth by real-time fluorescence microscopy. Although vascularization and functional maturation are largely restricted in embryonic kidney explants, in vitro cultured kidneys display remarkable recapitulation of ureteric branching and epithelialization and segmental patterning of the MM. Historically, kidney explant cultures provided crucial proof for the principle of reciprocal tissue induction in organogenesis. They have been used to demonstrate that the UB and MM exchange inductive cues, driving branching morphogenesis of the UB and epithelialization of the MM.^{35,36}

As originally shown by Grobstein, Saxen, and colleagues, the two major components of the metanephric kidney, the MM and UB, could be separated from each other, and the isolated mesenchyme could be induced to form nephron-like tubules by other embryonic tissues, the best example of which is the embryonic neural tube, or more recently by Wnt-expressing feeder cells (Fig. 1.6).³⁵⁻³⁹ When the neural

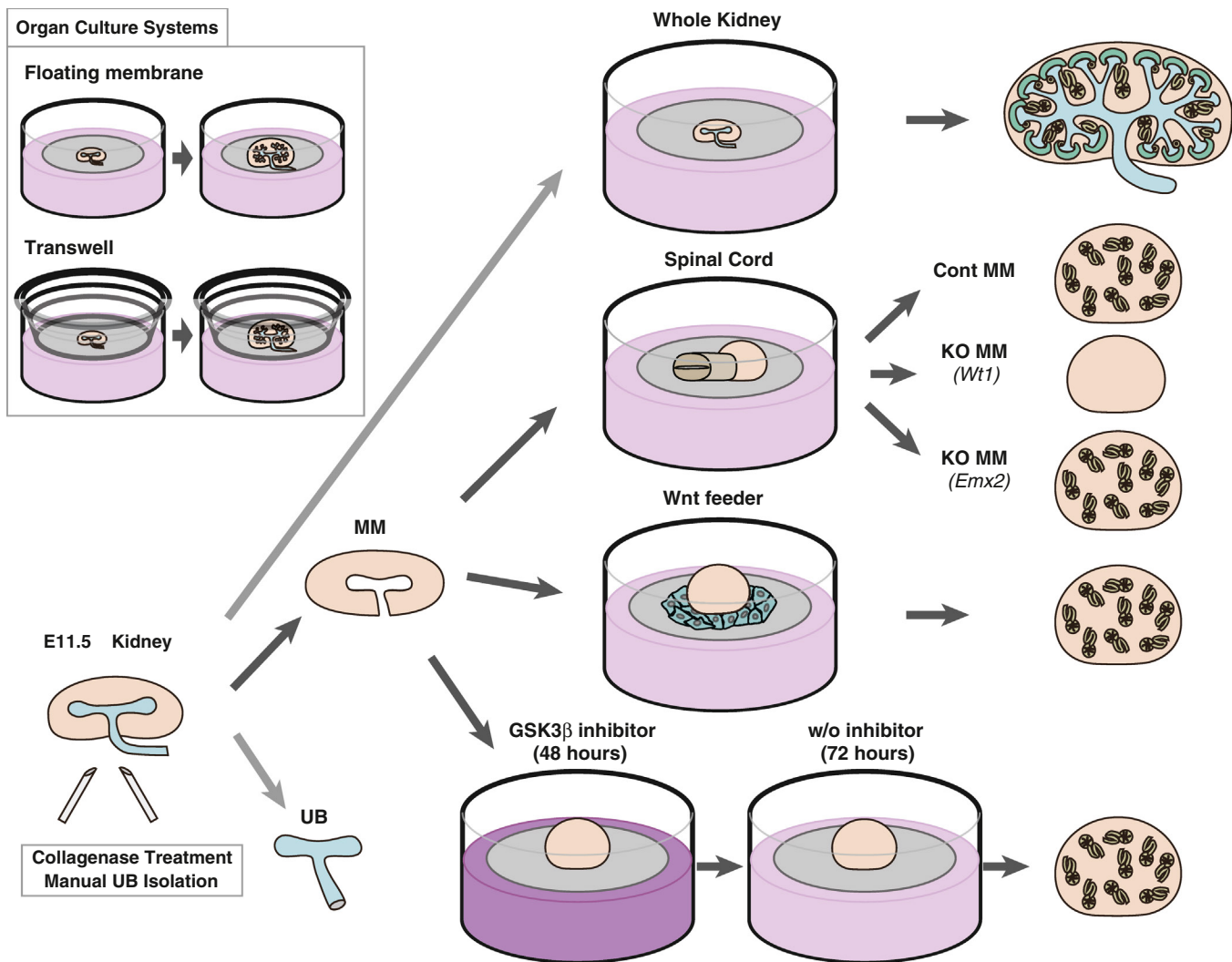


Fig. 1.6 Metanephric organ explants. The top-left panels depict the schematic of kidney organ culture system with the floating membrane (upper panels) or transwell inserts (lower panels). Embryonic kidney explants are grown at the air–growth medium interface. E11.5 isolated embryonic kidney can be cultured as a whole (top right) or a combination of external Wnt sources and manually isolated MM (right middle and lower panels). Schematics of kidneys grown after 7 days of culture are shown on the right. The intact kidney explant retains the nephron progenitors in the periphery, while the progenitors within the isolated MM are exhausted when cultured with external Wnt signal sources.

tube is used to induce the separated mesenchyme, there is differentiation of the mesenchyme into tubules but not significant tissue expansion. In contrast, intact metanephric rudiments can grow more extensively, displaying both sustained UB branching and early induction of nephrons even when cultured for a week (see Fig. 1.6).

The isolated mesenchyme experiment has proven useful in the analysis of renal agenesis phenotypes, where there is no outgrowth of the UB. In these cases, the mesenchyme can be placed in contact with the neural tube to determine whether it has the intrinsic ability to differentiate. Most often, when renal agenesis is due to the mutation of a transcription factor acting in a cell-autonomous fashion in the MM (e.g., *Wt1*), tubular induction is not rescued by the neural tube (see Fig. 1.6).⁴⁰ In the converse situation, in which renal agenesis is caused by loss of a gene function in the UB (e.g., *Emx2*), it is usually possible for the embryonic neural tube to induce tubule formation in isolated mesenchyme (see Fig. 1.6).⁴¹ Therefore the organ culture induction assay can

be used to test hypotheses concerning whether a particular gene is autonomously required in the UB or MM.

Recently, various recombinant signaling ligands and chemical inhibitors specific for signal transduction pathways have become available. This has made it possible to add them to organ cultures and study the roles of specific signaling molecules and their downstream pathways in kidney development with greater time resolution, which is difficult to achieve in *in vivo* studies. Examples are the use of drugs to modulate the ERK/MAP kinase, PI3K/Akt, Notch signaling, and Fgf and Wnt signaling pathways in renal explant cultures.^{42–47} Most notably, the pleiotropic roles of Wnt signaling in nephron differentiation,^{37–39,45,48} NP self-renewal,^{49–53} and nephron patterning^{46,54} have been extensively studied through *in vitro* experiments. These studies leverage the capabilities of *in vitro* culture experiments to dissect multiple parameters, such as concentration, duration, and downstream pathways of Wnt signaling.

GENETIC MOUSE MODELS

Many of the genetic pathways regulating kidney development are conserved between mice and humans. Due to the relative ease of manipulating the genome of the mouse, its small size, and shorter gestation period, the mouse has become the primary model organism for the study of mechanism of human development and diseases. Anatomic and functional features of the human kidney are highly similar in the mouse, albeit at a smaller scale, while a number of genes identified as essential for normal mouse kidney development are also known to be associated with congenital anomalies of the kidney and urinary tract (CAKUT) and other kidney diseases.⁵⁵⁻⁵⁷

The wealth of our understanding of metanephric kidney development over the past few decades is owed to the harnessing of homologous gene recombination to introduce targeted mutations and novel alleles into specific genes in cultured mouse embryonic stem cells (ESCs). This paved the way for the creation of numerous genetically engineered mice that have become highly valuable tools for the study of renal developmental biology and the etiology of certain genetic diseases of the kidneys and urinary tract (Table e1.1). The technology is applied in many ways. Its simplest way is creation of null mutations within the germline that generate gene knockout mouse models. The limitation of this method is that certain genes are essential for early development and their inactivation in the germline can cause premature lethality, thus precluding the analysis of the function of those genes in organogenesis. An improvement to this is the use of a conditional gene-targeting strategy, allowing for the creation of conditional alleles. This involves introducing small recognition sites for recombinase enzymes of which the Cre recombinase is the most routine now (Fig. 1.7). A conditional “floxed” allele of the target gene locus is created by incorporating two *loxP* sites within two separate introns, flanking the exons that can be excised or recombined. In principle, normal transcription from the locus is expected before recombination of the floxed allele. The Cre recombinase is engineered under the control of a tissue-specific promoter. Breeding between tissue-specific transgenic Cre animals and those harboring a conditional allele for a particular gene ultimately results in a cell- or tissue-type specific inactivation of the gene of interest. This strategy has also been refined in some cases, so the Cre expression is temporally regulated using a drug such as doxycycline or tamoxifen. A number of Cre lines are now available to target genes specifically within different subpopulations of renal cells and progenitors. The transgenic Cre lines driven by the *Hoxb7*, *Six2*, *Foxd1*, and *Nphs2* promoters are among the most highly cited gene excision drivers for targeted genetic inactivation in the ureteric, nephrogenic, stromal, and podocyte cell lineages.^{32,58-62} Lately, the introduction of the more precise CRISPR-Cas9 gene editing tool, an ingenious application of the adaptive immune response of prokaryotes against bacteriophages, is facilitating the way researchers create customized mutant cells and animal models to study various aspects of renal development (Fig. 1.8).⁶³⁻⁶⁶ The complex of guide RNAs and Cas9 proteins can be introduced directly into fertilized oocytes to rapidly generate mutant mice, so there is no need to manipulate mouse ESCs. This method is becoming increasingly

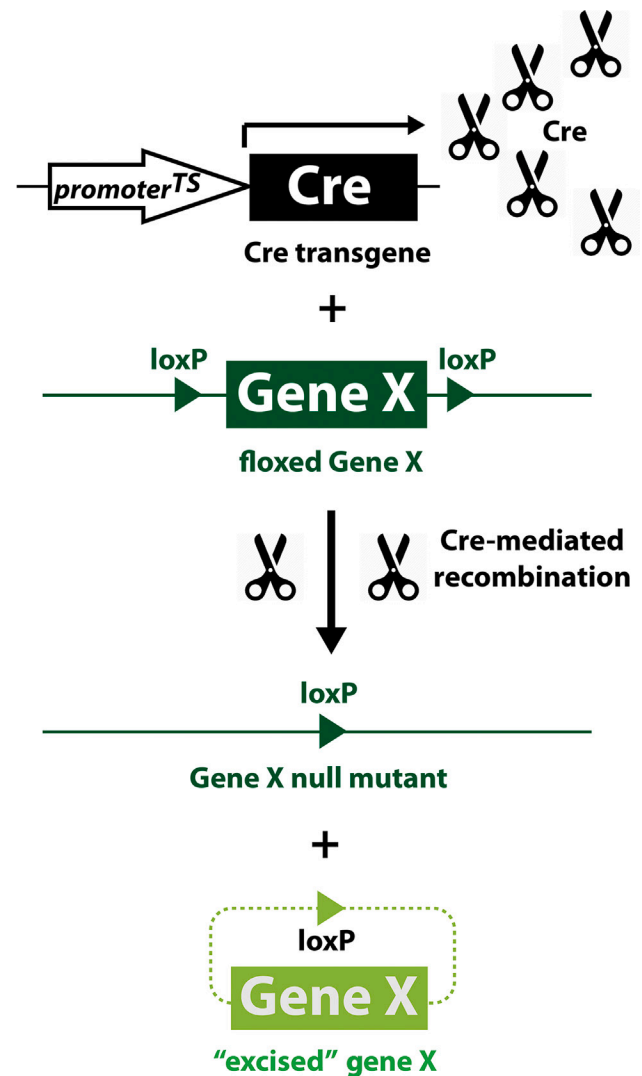


Fig. 1.7 Cre-loxP homologous gene recombination system. Simplified overview of Cre recombinase-mediated homologous recombination for generation of tissue-specific conditional null mutations of a target gene. Cre recombinase expression is engineered to be driven under the control of tissue-specific promoter (promoter^{TS}). The target gene (or typically certain exons within the target chromosomal locus) is flanked with *loxP* sites (recognition sites for Cre recombinase). Cre activity in specific cell types mediates the excision of the *loxP*-flanked (floxed) target gene, creating a null allele.

popular, although great care must be taken to avoid off-target mutations.

Significant innovations in high-throughput gene sequencing are also facilitating the identification of monogenic mutations associated with heritable kidney diseases.^{67,68} Putative disease-causing genes identified from genome-wide association studies can then be validated in corresponding gene-targeted mouse models. Similarly, a gene whose mutation has been identified to cause kidney maldevelopment and dysfunction in mice can be investigated for incidence of mutations across the human genome (e.g., *Arhgdia* in nephrotic syndrome).⁶⁹⁻⁷² Other genome-wide approaches that have led to the discovery of novel genes in kidney development and disease include gene trap consortia^{73,74} and genome-wide transcriptome and proteome projects.⁷⁵⁻⁷⁷

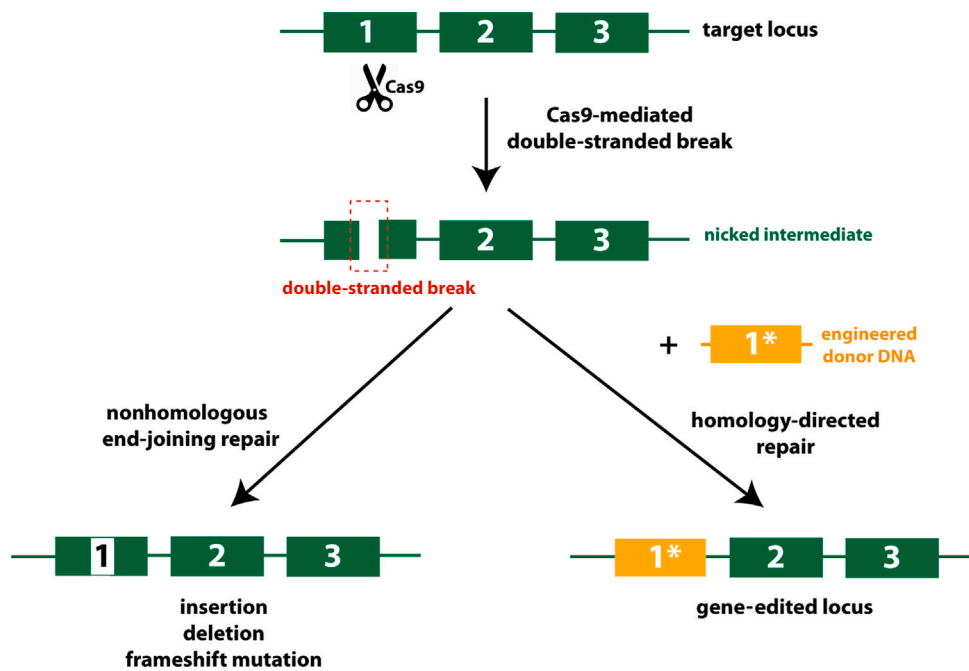


Fig. 1.8 Gene editing using the CRISPR-Cas9 system. Simplified overview for targeted introduction of mutations using the CRISPR-Cas9 system. Site-specific cleavage of a target gene locus with the Cas9 endonuclease creates a double-strand DNA break. The nicked target gene can be repaired by an error-prone, nonhomologous repair mechanism resulting in nucleotide insertion, deletion, or frameshift mutations. Introduction of a homologous engineered donor DNA results in a homologous substitution into the target locus, which then results in a precisely edited gene.

IMAGING AND LINEAGE TRACING STUDIES

Reporter mouse lines are animal models engineered to express transgenes encoding a fluorescent protein tag or an enzyme (e.g., bacterial β -galactosidase) in a cell lineage-specific manner. In the simplest model, a single transgene comprising the reporter coding region is placed downstream of a cell-specific promoter. Such a model is useful to identify and follow distinct cell lineages where the chosen promoter is active. *Hoxb7-EGFP* was the first fluorescent transgene developed to visualize renal development.⁷⁸ Enhanced green fluorescent protein (EGFP), placed under the control of the *Hoxb7* promoter, specifically labels the Wolffian duct and the ureteric epithelial lineage. *Hoxb7-EGFP* has therefore proven to be invaluable in studying the rates and pattern of ureteric branching morphogenesis and ureteral development, including disruption of these events in the context of particular mutant backgrounds.^{59,79-81} An alternative model uses two separate transgenes, namely a cell-specific Cre driver and an independent reporter transgene (e.g., R26R). With the R26R reporter model, expression of fluorescent protein gene (e.g., tdTomato) or β -galactosidase transgene is switched on by a Cre-mediated removal of an upstream stop codon.⁸² Inducible reporter systems incorporate genetic regulatory elements that allow for both cell-specific promoter- and drug-dependent activation of the reporter gene. The most commonly used strategies for controlled transgene expression in mice are the Tet-operon/repressor bi-transgenic and the estrogen receptor ligand-binding domain (Cre-ERT2) systems. These inducible reporter systems allow for timed pulse labeling of cell lineages, resulting in permanent tagging of progenitor cells and their direct derivatives. Inducible reporter systems have become valuable tools for the mapping of cell fates. Inducible reporter systems have been used to establish that the *Six2*⁺ cells are the progenitors of all epithelial structures of the mature nephron (see Fig. 1.5).⁶⁰ In some cases, the Cre transgene is engineered with an associated

fluorophore expression cassette (e.g., GFP-Cre fusion gene) under the control of a shared promoter, thus allowing easy identification of cells with targeted gene mutations. A wide variety of reporter and Cre transgenes (see Table e1.1) are now available to characterize the development and organization of multiple compartments of the kidney.

Reporter mice expressing fluorescent tags also come in handy for segregating and isolating particular cell types on the basis of fluorescence, facilitating global gene expression analysis, elucidation of transcriptional regulatory networks, and epigenetic interactions that orchestrate particular morphogenetic events (Table 1.2). Purified populations of fluorescence-tagged cells can be used for high-throughput discovery of transcriptomes, genome-wide DNA-binding protein interactions, histone modifications, and nucleosome complexes through next-generation sequencing methods such as RNA-seq (RNA sequencing) and ChIP-seq (chromatin immunoprecipitation sequencing).⁸³⁻⁸⁵

KIDNEY ORGANOIDS

Recent advances in synthetic biology using an “understanding by creating” approach have enabled the ex vivo generation of kidney tissues from pluripotent stem cells (PSCs), namely kidney organoids⁸⁶⁻⁸⁸ (Fig. 1.9). These PSC-derived kidney organoids can form many renal structures including the glomerulus, proximal tubule, loop of Henle, distal tubule, collecting ducts, interstitium, and a primitive endothelial network. This breakthrough, in turn, has served as a tool to validate the efficacy of lineage-specifying signaling molecules identified through in vivo studies. The feasibility of screening using in vitro experimental setups has significantly enhanced our understanding of kidney lineage specification with unprecedented resolution, particularly in analyzing signal concentration, combinations, timing, and duration aspects that are not readily apparent in in vivo analyses. In addition, organoids with gene-edited PSCs using CRISPR-Cas9 technology (see Fig. 1.8) provide a

Table e1.1 Summary of Knockout and Transgenic Models for Kidney Development

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
Renal Agenesis			
<i>Celsr1</i>	Renal agenesis, hydroureter, hydronephrosis	Spina bifida, unilateral renal agenesis, hydronephrosis	354
<i>Ctnnb1</i> (β -catenin)	Renal agenesis or severe renal hypoplasia, premature differentiation of UB epithelia (UB-selective)	Mental retardation, multiple cancers, eye defects	171,526
<i>Emx2</i>	Renal agenesis	Schizencephaly (cerebral cleft abnormalities)	41
<i>Emx2, Pax2</i>	Duplicated kidneys and ureter, ureteral obstruction	CAKUT, VUR	527
<i>Esrp1</i>	Renal agenesis, renal hypoplasia		528
<i>Etv4, Etv5</i>	Renal agenesis or severe renal hypodysplasia		230,232
<i>Eya1</i>	Renal agenesis	Branchiootorenal syndrome (brachial fistulae, deafness)	142,156
<i>Fgf9, Fgf20</i>	Renal agenesis		270
<i>Fgf10, Gdnf, Gfra1</i>	Renal agenesis		241
<i>Fgfr1, Fgfr2</i>	Renal agenesis (MM-selective)		167
<i>Fras1, Frem1, Frem2</i>	UB failure, defect of GDNF expression	Fraser syndrome (cryptophthalmos, syndactyly, CAKUT); Manitoba oculotrichoanal (MOTA) syndrome	184,185,201,203, 204
<i>Gata3</i>	Renal agenesis, gonad dysgenesis (null mutation)	Hypoparathyroidism, sensorineural deafness, and renal dysplasia (HDRS) syndrome; autoimmune disease	169,170,172,529
<i>Gdf11</i>	UB failure, skeletal defects		151,530
<i>Gdnf, Gfra1, Ret</i>	Renal agenesis or rudimentary kidneys, aganglionic megacolon	Hirschsprung disease, Multiple endocrine neoplasm type IIA/B (MEN2A/MEN2B), and familial medullary thyroid carcinoma (FMTC)	148-150,154,181,183, 531,532,533,534
<i>Gen1</i>	Renal agenesis, duplex kidneys, hydronephrosis, ureteral obstruction		535
<i>Gli3</i>	Renal agenesis, severe renal agenesis, absence of renal medulla and papilla	Pallister-Hall (PH) syndrome (polydactyly, imperforate anus, abnormal kidneys, defects in the gastrointestinal tract, larynx, and epiglottis)	336,337
<i>Greb1l</i>	Renal agenesis	CAKUT	536,537,538
<i>Grem1</i>	Renal agenesis; apoptosis of the MM		152
<i>Grlh2</i>	Occasional unilateral renal agenesis, CD barrier dysfunction, diabetes insipidus	Autosomal dominant deafness, ectodermal dysplasia	539,540
<i>Grip1</i>	Renal agenesis	Fraser syndrome (cryptophthalmos, syndactyly, CAKUT)	205,206,207
<i>Hnf1b</i>	Renal agenesis, renal hypoplasia, hydroureter, duplex kidneys	CAKUT, diabetes mellitus, renal cysts, renal carcinoma	541
<i>Hoxa11, Hoxd11</i>	Distal limbs, vas deferens	Radioulnar synostosis with amegakaryocytic thrombocytopenia	542
<i>Hs2st1</i>	Lack of UB branching and mesenchymal condensation		543
<i>Isl1</i>	Renal agenesis, renal hypoplasia, hydroureter (MM-selective)		490
<i>Itga8</i> ($\alpha 8$ Integrin)	Renal agenesis, renal hypodysplasia	Fraser syndrome (cryptophthalmos, syndactyly, CAKUT)	188
<i>Itgb1</i> ($\beta 1$ Integrin)	Renal agenesis, disrupted UB branching, hypoplastic collecting duct system (collecting duct-selective); podocyte dedifferentiation (podocyte-selective)	Fraser syndrome (cryptophthalmos, syndactyly, CAKUT)	199,544,545

Continued on following page

Table e1.1 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Kif26b</i>	Renal agenesis, failed UB attraction to the MM		186
<i>Lamc1</i>	UB failure, delayed nephrogenesis, water transport defects		546
<i>Lhx1 (Lim1)</i>	Renal agenesis (null mutant); renal hypoplasia, UB branching defect, hydronephrosis, distal ureter obstruction (UB-selective); arrested nephrogenesis, nephron-patterning defects (MM-selective)	Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome (müllerian duct agenesis)	147,510,547
<i>Lrp4</i>	Delayed UB induction, failed MM induction, syndactyly, oligodactyly	Cenani-Lenz syndrome	548,549,550,551
<i>Mark2 (Par1b), Mark3 (Par1a)</i>	Renal hypoplasia, proximal tubule dilation, immature glomeruli		552
<i>Npnt</i>	Delayed UB association with the MM		187
<i>Osr1</i>	Lack of MM, adrenal gland, gonads, defects in formation of pericardium and atrial septum		143,11,256,257
<i>Pax2</i>	Renal hypoplasia, VUR	CAKUT, VUR, optic nerve colobomas	553,141
<i>Pax2, Pax8</i>	Defect in intermediate mesoderm transition, failure of pronephric duct formation	CAKUT, VUR, optic nerve colobomas	554
<i>Pbx1</i>	Unilateral renal agenesis, expansion of nephrogenic precursors	CAKUT, hearing loss, abnormal ears	47,555,556,557
<i>Ptf1a</i>	Failure of UB induction, anal atresia, persistent cloaca, skeletal malformation	Pancreatic and cerebellar agenesis; diabetes mellitus	558,559,560
<i>Rara, Rarb</i>	Renal hypoplasia, dysplasia, hydronephrosis, skeletal and multiple visceral abnormalities		15,19,80
<i>Sall1</i>	Renal agenesis, severe renal hypodysplasia	Townes-Brock syndrome (anal, renal, limb, ear anomalies)	146,561
<i>Shh</i>	Bilateral or unilateral renal agenesis, unilateral ectopic dysplastic kidney, defective ureteral stromal differentiation	Vertebral defects, anal atresia, cardiac defects, tracheoesophageal fistula, renal anomalies, and limb abnormalities (VACTERL) syndrome	58
<i>Six1</i>	Lack of UB branching and mesenchymal condensation	Branchiootorenal syndrome	142,156
<i>Sox8, Sox9</i>	Renal genesis, renal hypoplasia	Campomelic dysplasia (limb and skeletal defects, abnormal gonad development)	562
<i>Tln1, Tln2</i>	Renal agenesis		563
<i>Wnt5a</i>	Renal agenesis, renal dysplasia, duplex kidneys, hydronephrosis	CAKUT	564,565,566
<i>Wt1</i>	Renal and gonadal agenesis, severe lung, heart, spleen, adrenal, and mesothelial abnormalities	Wilms tumor, aniridia, genitourinary abnormalities, and retardation (WAGR) syndrome; Denys-Drash syndrome	40,164,287,567
Hypoplasia/Dysplasia/Low Nephron Mass			
<i>Adamts1</i>	Hypoplasia of the renal medulla, hydronephrosis		282,284
<i>Adamts1, Adamts4</i>	Hypoplasia of the renal medulla, hydronephrosis		568
<i>Agtr2</i>	Various collecting duct defects	CAKUT	400,396,397
<i>Ald1a2 (Raldh2)</i>	Renal hypoplasia, hydronephrosis, ectopic ureter		172
<i>Bmp1ra (Alk3)</i>	Hypoplasia of renal medulla, fewer UB branches (UB-selective)	Juvenile polyposis syndrome	569
<i>Bmp4</i>	Severe renal hypodysplasia	Microphthalmia, orofacial cleft	246
<i>Bmp7</i>	Renal hypoplasia, reduced MM survival		263
<i>Cask</i>	Renal hypoplasia and dysplasia, premature depletion of nephrogenic precursor cells	Microcephaly, mental retardation	272,570

Table e1.1 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Cdc42</i>	Renal hypoplasia, oligonephronia, defects in mesenchyme-to-epithelial transition (CM-selective)		362,571
<i>Cfl1</i>	Renal hypodysplasia, ureter duplication		572
<i>Chd4 (Mi-2β)</i>	Renal hypodysplasia	Sifrim-Hitz-Weiss syndrome (autosomal dominant intellectual disability disorder with variable congenital defects in cardiac, skeletal, and urogenital systems)	281,573
<i>Ctnnb1 (β-catenin)</i>	Severely hypoplastic kidney, lack of nephrogenic zone and S-shaped body (CM-selective)	Mental retardation, multiple cancers, eye defects	252
<i>Dchs1, Dchs2</i>	Renal hypoplasia, impaired ureteric branching, expansion of nephrogenic precursor zone	Van Maldergem syndrome 1—periventricular nodular heterotopia, intellectual disability, deafness, renal hypoplasia, tracheal anomalies, and skeletal dysplasia	360,363,574
<i>Dicer1</i>	Renal hypoplasia, dysplasia, cysts (UB-selective); renal hypoplasia characterized by premature termination of nephrogenesis (MM-selective)		575,576
<i>Dkk1</i>	Overgrown renal papilla (renal tubule and collecting duct restricted)		330
<i>Dlg1</i>	Renal hypoplasia and dysplasia, premature depletion of nephrogenic precursor cells		272,570
<i>Dstn (Dextrin)</i>	Renal hypodysplasia, ureter duplication		572
<i>Egfr</i>	Hypoplasia of the renal papilla, moderate polyuria, and urine concentration defects		328
<i>Egln1 (Phd2), Egln3 (Phd3)</i>	Renal hypoplasia, oligonephronia, abnormal postnatal nephron formation, glomerulosclerosis (stroma-specific)	Familial erythrocytosis; abnormally high EPO levels; high altitude adaptation hemoglobin (HALAH)	577
<i>Esrrg</i>	Agensis of renal papilla		578
<i>Fat4</i>	Renal hypoplasia, impaired ureteric branching, failed nephrogenesis (mesenchyme-to-epithelial transition), expansion of nephrogenic precursor zone (stroma-selective)	Van Maldergem syndrome 2 (VMLDS2): dysmorphic faces, tracheomalacia, microtia, intellectual disability, and skeletal dysplasia	359,360,363,574
<i>Fgf7</i>	Small kidneys, reduction in nephron number		333
<i>Fgf8</i>	Renal dysplasia, arrested nephrogenesis at pretubular aggregate stage (MM-selective)	Kallmann syndrome, hypogonadism	249,250
<i>Fgf10</i>	Renal hypoplasia, multiorgan developmental defects, including the lungs, limb, thyroid, pituitary, and salivary glands		332
<i>Fgfr1, Fgfr2</i>	Renal agenesis (MM-selective)		167
<i>Fgfr2</i>	Renal hypoplasia, hydronephrosis (UB-selective)		59
<i>Foxc2</i>	Renal hypoplasia	Lymphedema-distichiasis syndrome	243,579
<i>Foxd1</i>	Accumulation of undifferentiated CM, attenuated UB branching, stromal patterning defects		30,357,358
<i>Frs2</i>	Mild renal hypoplasia (UB-selective)		580
<i>Fzd4, Fzd8</i>	Impaired UB branching, renal hypoplasia		581
<i>Hdac1, Hdac2</i>	Renal hypoplasia, renal dysplasia, arrest of nephrogenesis at the renal vesicle stage		582,583

Continued on following page

Table e1.1 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Lats1, Lats2</i>	Renal hypoplasia, impaired UB branching and UB tip specification, impaired nephrogenesis, and renal interstitial differentiation		361,584
<i>Lgr4</i>	Severe renal hypoplasia and oligonephronia; renal cysts	Aniridia-genitourinary anomalies; mental retardation	585,586
<i>Lmx1b</i>	Renal dysplasia, skeletal abnormalities	Nail-patella syndrome	294,295
<i>Map2k1 (Mek2), Map2k (Mek1)</i>	Renal hypodysplasia, megaureter	Cardiofaciocutaneous syndrome	228
<i>Mdm2</i>	Renal hypoplasia and dysplasia, severely impaired UB branching and nephrogenesis (UB-selective); depletion of nephrogenic precursors (MM-selective)		587,588
<i>Mf2</i>	Renal hypoplasia, oligonephronia		589
<i>Mitf</i>	Oligonephronia	Microphthalmia, Waardenburg syndrome type 2A	590
<i>Nf2</i>	Renal hypoplasia, renal dysplasia		584
<i>Notch1, Notch2</i>	Loss of nephron derivatives, nephron segmentation defects	Alagille syndrome (cholestatic liver disease, cardiac disease, kidney dysplasia, renal cysts, renal tubular acidosis)	262
<i>Pbx1</i>	Reduced UB branching, expansion of nephrogenic precursors, delayed mesenchyme-to-epithelial transformation, dysgenesis of adrenal gland and gonads		47,555
<i>Plxnb2</i>	Renal hypoplasia and ureter duplication		591
<i>Pou3f3 (Brn1)</i>	Impaired development of distal tubules, loop of Henle, and macula densa; distal nephron-patterning defect		283
<i>Prr</i>	Renal hypoplasia, renal dysplasia, oligonephronia		592,593
<i>Psen1, Psen2</i>	Severe renal hypoplasia, severe defects in nephrogenesis		594
<i>Ptgs2</i>	Oligonephronia		595
<i>Rbpj</i>	Severe renal hypoplasia, oligonephronia, loss of proximal nephron segments, tubular cysts (MM-selective)		596,276
<i>Sall1</i>	Severe renal hypoplasia, cystic dysplasia of nephrogenic derivatives (tubules and glomeruli)	Townes-Brocks branchiootorenal-like syndrome	364
<i>Shp2</i>	Severe impairment of UB branching, renal hypoplasia		227
<i>Six1</i>	Hydronephrosis, hydroureter, abnormal development of ureteral smooth muscle		417
<i>Six2</i>	Renal hypoplasia and premature depletion of nephrogenic precursors (homozygous loss); increased UB branching and augmentation of nephron endowment (haploinsufficiency)		253,597
<i>Tbx18</i>	Hydronephrosis, hydroureter, abnormal development of ureteral smooth muscle		411,417
<i>Tfap2b</i>	MM failure, craniofacial and skeletal defects		598
<i>Trp53 (p53)</i>	Oligonephronia, precocious depletion of nephrogenic precursors	Multiple cancers	599
<i>Trps1</i>	Impaired UB branching, renal hypoplasia	Trichorhinophalangeal syndrome (skeletal defects)	600
<i>Vangl2</i>	Impaired UB branching and renal hypoplasia	Neural tube defects	355
<i>Wnt4</i>	Failure of MM induction		252

Table e1.1 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Wnt7b</i>	Complete absence of medulla and renal papilla (UB-selective)		325
<i>Wnt9b</i>	Vestigial kidney, failure of MM induction; cystic kidney (CD-selective)		37,348
<i>Wnt11</i>	Impaired ureteric branching, renal hypoplasia		231
<i>Yap</i>	Renal hypoplasia renal dysplasia, hydronephrosis, severe disruption of UB branching (UB-selective), oligonephronia, defects in mesenchyme-to-epithelial transition (CM-selective)	Coloboma, hearing impairment, cleft palate, cognitive deficit, hematuria	362,571,584
Mislocalized or Ectopic UB/Increased UB Branching			
<i>Bmp4</i>	Duplex ureter, hydroureter, renal hypodysplasia	Microphthalmia, orofacial cleft	246
<i>Cer1</i>	Increased ureteric branching, altered spatial organization of ureteric branches		601
<i>Cfl1</i>	Renal hypodysplasia, ureter duplication		572
<i>Foxc1</i>	Duplex kidneys, ectopic ureters, hydronephrosis, hydroureter		243
<i>Gata3</i>	Ectopic ureteric budding, duplex kidneys, hydroureter (UB-selective)	Hypoparathyroidism, sensorineural deafness, and renal dysplasia (HRS) syndrome; autoimmune disease (rheumatoid arthritis)	169,170,172,529
<i>Hnf1b, Pax2</i>	Renal hypoplasia, duplex kidneys, ectopic ureters, megaureter, hydronephrosis	CAKUT	602
<i>Hspb11 (lft25)</i>	Duplex kidneys		603
<i>lft27</i>	Duplex kidneys		603
<i>Lzts2</i>	Duplex kidneys/ureters, hydronephrosis, hydroureter		604
<i>Plxnb1</i>	Increased ureteric branching		605
<i>Plxnb2</i>	Renal hypoplasia and ureter duplication		591
<i>Robo2</i>	Increased UB branching	CAKUT, VUR	244,245
<i>Ror2</i>	Duplex ureter, hydronephrosis	Skeletal dysplasia, shortened limbs, brachydactyly, facial dysmorphism (brachydactyly, type B1, Robinow syndrome)	566,606
<i>Sema3a</i>	Increased ureteric branching (UB-selective)		607
<i>Slit2</i>	Increased UB branching	CAKUT, VUR	244,245
<i>Spry1</i>	Supernumerary UBs, multiple ureters		79,237
<i>Wnt5a</i>	Duplex ureter, hydronephrosis	Robinow syndrome (skeletal dysplasia, shortened limbs, brachydactyly, facial dysmorphism)	566,606
Renal Cysts			
<i>Angpt1, Angpt2</i>	Interstitial medullary cysts, urinary concentration defects		444
<i>Aqp11</i>	Abnormal vacuolization of proximal tubules; polycystic kidneys		608
<i>Arhgap35 (GRLF1)</i>	Glomerular cysts		609
<i>Bcl2</i>	Renal hypoplasia and cysts		610
<i>Bicc1</i>	Polycystic kidneys		611
<i>Bpck</i>	Polycystic kidneys, hydrocephalus	Meckel syndrome 3	612
<i>Erbb4</i>	Renal cysts (overexpression in renal tubules) Dilated and mispolarized tubules, increased renal fibrosis (renal tubule deletion)		613

Continued on following page

Table e1.1 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Fat4</i>	Renal cysts, disrupted hair cell organization in inner ear	Van Maldergem syndrome (mental retardation, abnormal craniofacial features, deafness, limb malformations, renal hypoplasia)	353,614
<i>Glis3</i>	Polycystic kidney, neonatal diabetes	Congenital hypothyroidism, diabetes mellitus, hepatic fibrosis, congenital glaucoma	615,616
<i>Gpc3</i>	Disorganized tubules and medullary cysts	Simpson-Golabi-Behmel syndrome	617,618,619
<i>Hnf1b</i>	Polycystic kidney disease (tubule-selective)	CAKUT, diabetes mellitus, renal cysts, renal carcinoma	349,352,541
<i>Ift88 (Orpk)</i>	Polycystic kidneys; defective left-right asymmetric patterning	ARPKD	620,621
<i>Ilk</i>	Medullary cysts (stroma-specific)		622
<i>Invs</i>	Polycystic kidneys, inverted viscera	Nephronophthisis	623,624
<i>Kif3A</i>	Polycystic kidney disease (tubule-selective)		625
<i>Maib (Kreisler)</i>	Decreased glomeruli, cysts, and tubular dysgenesis	Musculoaponeurotic fibrosarcoma	626,627
<i>Mks1</i>	Renal hypoplasia and cysts	Meckel syndrome (multicystic dysplasia, neural tube defect)	628
<i>Pkd1, Pkd2</i>	Renal cysts	ADPKD, ARPKD	629
<i>Pten</i>	Abnormal ureteric bud branching, cysts (UB-selective)	Cowden disease, Bannayan-Riley-Ruvalcaba syndrome, various tumors	229
<i>Sall1</i>	Nephrogenic tubule and glomerular cysts (stroma specific)	Townes-Brocks branchiootorenal-like syndrome	364
<i>Taz</i>	Polycystic kidneys, emphysema		630,631
<i>Tek (Tie2)</i>	Interstitial medullary cysts, urinary concentration defects	Cutaneous and mucosal venous malformation, congenital glaucoma	444
<i>Vhl</i>	Renal cysts (tubule-selective)	Von Hippel-Lindau syndrome	504
<i>Xylt2</i>	Polycystic kidneys and liver		632
<i>Zeb2</i>	Glomerular cysts	Mowat-Wilson syndrome (Hirschsprung disease with associated mental retardation)	633
Later Phenotypes (Tubular, Vascular, and Glomerular Defects)			
<i>Ace</i>	Atrophy of renal papillae, vascular thickening and hypertrophy, perivascular inflammation	Chronic systemic hypotension	401,402
<i>Actn4</i>	Glomerular developmental defects, FSGS	SRNS	634,635
<i>Adam10</i>	Loss of principal cells of the CD, hydronephrosis, polyuria	Alzheimer disease; reticulate acropigmentation of Kitamura	339
<i>Agtr</i>	Atrophy of renal papillae, vascular thickening and hypertrophy, perivascular inflammation	Chronic systemic hypotension	399,407
<i>Agtr1a (AT1A)</i>	Hypertrophy of juxtaglomerular apparatus and expansion of renin cell progenitors, mesangial cell hypertrophy	Chronic systemic hypotension	636
<i>Agtr1a/Agtr1b (AT1A/AT1B)</i>	Atrophy of renal papillae, vascular thickening and hypertrophy, perivascular inflammation	Chronic systemic hypotension	406
<i>Ampd</i>	Podocyte foot process effacement, proteinuria	Minimal change nephropathy	637
<i>Angpt1</i>	Simplification and dilation of glomerular capillaries; detachment of glomerular endothelium from the GBM; loss of mesangial cells; loss of ascending vasa recta (compound deletion with Angpt2)		440,444
<i>Angpt2</i>	Cortical peritubular capillary abnormalities (null allele); apoptosis of glomerular capillaries, proteinuria (transgenic overexpression); loss of ascending vasa recta (compound deletion with Angpt1)		443,442,444

Table e1.1 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Arhgdia</i> (<i>RhoGDIα</i>) <i>Bmp7</i>	Podocyte effacement and proteinuria Hypoplastic kidney, impaired maturation of nephron, reduced proximal tubules (podocyte-selective)	SRNS, FSGS	69,72,638 639
<i>Cd151</i>	Podocyte foot process effacement, disorganized GBM, tubular cystic dilation	Nephropathy (FSGS) associated with pretibial epidermolysis bullosa and deafness	640,641
<i>Cd2ap</i> <i>Cdc42</i>	Podocyte effacement, proteinuria Congenital nephrosis; impaired formation of podocyte foot processes (podocyte-selective)	FSGS	642 643
<i>Cmas</i>	Congenital nephrosis; impaired formation of podocyte foot processes, defective sialylation		644
<i>Col4a1</i> , <i>Col4a3</i> , <i>Col4a4</i> , <i>Col4a5</i>	Disorganized GBM, proteinuria	Alport syndrome	645,646,647,648
<i>Coq6</i>	Nephrotic syndrome and deafness	SRNS, FSGS, sensorineural deafness	649,650
<i>Crb2</i>	Podocyte effacement and proteinuria	SRNS, FSGS	651
<i>Crk1</i> , <i>Crk2</i> , <i>CrkL</i>	Albuminuria, altered podocyte cytoarchitecture (podocyte-selective)		652
<i>Cxcl12</i> (<i>SDF1</i>), <i>Cxcr4</i> , <i>Cxcr7</i>	Petechial hemorrhage in the kidneys, glomerular aneurysm, fewer glomerular fenestrations, reduced mesangial cells, podocyte foot process effacement, mild renal hypoplasia	WHIM (warts, hypogammaglobulinemia, infections, and myelokathexis) syndrome	450,451,653
<i>Dicer1</i>	Podocyte damage, albuminuria, end-stage renal failure (podocyte-selective); reduced renin production, renal vascular abnormalities, striped fibrosis (renin cell-selective)	Pleuropulmonary blastoma	654,655,656,657
<i>Dnm1</i> , <i>Dnm2</i> (<i>Dynammin</i> 1/2)	Podocyte foot process effacement and proteinuria (podocyte-selective)		658
<i>Dot1l</i>	Increased intercalated at the expense of principal CD cells; polyuria		343,659
<i>Efnb1</i> (<i>Ephrin B1</i>)	Podocyte foot process effacement and proteinuria (podocyte-selective)		660
<i>Efnb2</i> (<i>Ephrin B2</i>)	Dilation of glomerular capillaries		448
<i>Egln1</i> (<i>Phd2</i>), <i>Egln3</i> (<i>Phd3</i>)	Renal hypoplasia, oligonephronia, abnormal postnatal nephron formation, abnormally elevated erythropoietin production, dilation of renal blood vessels, glomerulosclerosis (stroma-specific)	Familial erythrocytosis; abnormally high EPO levels; high altitude adaptation hemoglobin (HALAH)	577
<i>Elf5</i>	Paucity in principal CD cells		341
<i>Fat1</i>	Foot process fusion, failure of foot process formation, proteinuria	SRNS, FSGS, hematuria with neurologic defects; glioblastoma, colorectal cancer, head and neck cancer	661,662
<i>Fermt2</i> (<i>Kindlin-2</i>)	Rac1 hyperactivation, podocyte effacement and proteinuria		663
<i>Flt1</i> (<i>Vegfr1</i>)	Nephrotic syndrome		664
<i>Foxc1</i> and <i>Foxc2</i>	Impaired podocyte differentiation, dilated glomerular capillary loop, poor mesangial migration; proteinuria and glomerulosclerosis	Anterior segment dysgenesis/Axenfeld-Rieger syndrome (iris hypoplasia and defective cornea); lymphedema-distichiasis syndrome (lower limb swelling and extra eyelashes)	75,665,666
<i>Foxi1</i>	Tubular acidosis; absence of CD intercalated cells	Tubular acidosis and deafness	667,342
<i>Fyn</i>	Podocyte foot process effacement, abnormal slit diaphragms, proteinuria		668,669

Continued on following page

Table e1.1 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Gata3</i>	Impaired maintenance of mesangial cells, dilation of glomerular capillaries, glomerulosclerosis and mesangial matrix expansion, proteinuria		670
<i>Gnas</i> (<i>Gαs</i>)	FSGS, mesangial expansion, proteinuria, urinary concentration defect (renin cell-specific)	Pseudohypoparathyroidism, McCune-Albright syndrome, endocrine tumors	671,672
<i>Gne</i> (<i>Mnk</i>)	Hyposialylation defect, foot process effacement, GBM splitting, proteinuria, and hematuria		673
<i>Grlh2</i>	Occasional unilateral renal agenesis, CD barrier dysfunction, diabetes insipidus	Autosomal dominant deafness, ectodermal dysplasia	539,540
<i>Ilk</i>	Nephrotic syndrome (podocyte-selective); collecting duct obstruction (UB-selective)		674,675
<i>Insr</i>	Podocyte effacement, GBM alteration, proteinuria (podocyte-selective)	Diabetic nephropathy	676
<i>Itga3</i> (<i>Integrin α3</i>)	Reduced UB branching, glomerular defects, poor foot process formation		329,331
<i>Itga6</i> (<i>Integrin α6</i>)	Collecting duct dilation and dysplasia	Epidermolysis bullosa, collecting duct dysplasia	677
<i>Itgb1</i> (<i>Integrin β1</i>)	Podocyte loss, capillary and mesangial degeneration, glomerulosclerosis (podocyte-selective)		544,545
<i>Itgb4</i> (<i>Integrin β4</i>)	CD dilation and dysplasia	Epidermolysis bullosa	677
<i>Kdr</i> (<i>Flk1/Vegfr2</i>)	Thrombotic microangiopathy, ascites; renal papilla dysplasia, urine concentration defects, loss of peritubular papillary capillaries (stroma specific)	Capillary infantile hemangioma	678,436
<i>Kirrel</i> (<i>Neph1</i>)	Abnormal slit diaphragm function, FSGS		74
<i>Lama5</i>	Defective glomerulogenesis, abnormal GBM, poor podocyte adhesion, loss of mesangial cells		679
<i>Lamb2</i>	Proteinuria prior to the onset of foot process effacement	Pierson syndrome	306,680
<i>Lmx1b</i>	Impaired differentiation of podocytes, cytoskeletal disruption in podocytes	Nail-patella syndrome	681,682,683
<i>Mafb</i> (<i>Kreisler</i>)	Abnormal podocyte differentiation		291
<i>Magi2</i>	Podocyte effacement and proteinuria	SRNS, FSGS	684,685,686
<i>Mib1</i>	Loss of principal CD cells, polyuria, urine concentration defects, sodium wasting, hydronephrosis	Left ventricular noncompaction	338
<i>Mpp5</i> (<i>Pals1</i>)	Tubular cysts, podocyte effacement, and proteinuria		687
<i>Mpv17</i>	Nephrotic syndrome		688
<i>Mtor</i>	Proteinuria, podocyte autophagy defects (podocyte-selective)		689
<i>Myo1E</i>	Podocyte foot process effacement and proteinuria	SRNS	690,691,692
<i>Nck1</i> , <i>Nck2</i>	Failure of foot process formation (podocyte-selective)		62
<i>Nid1</i>	Abnormal GBM		693
<i>Notch1</i> , <i>Notch2</i>	Lack of glomerular endothelial and mesangial cells (standard knockout) lack of podocytes and proximal tubular cells (MM-selective); impaired nephrogenesis (cap mesenchyme-selective)	Alagille syndrome (cholestatic liver disease, cardiac disease, kidney dysplasia, renal cysts, renal tubular acidosis)	374,375,276,694
<i>Nphs1</i> (<i>Nephrin</i>)	Absent slit diaphragms, podocyte effacement, proteinuria	Congenital nephrosis of the Finnish type, childhood-onset steroid-resistant nephritic syndrome, childhood- and adult-onset FSGS	695

Table e1.1 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Nphs2</i> (<i>Podocin</i>)	Congenital nephrosis, FSGS, vascular defects	SRNS, FSGS	696,697
<i>Npnt</i>	Mislocalization of $\alpha 8\beta 1$ integrin, mesangial hyperproliferation and sclerosis (nephron and podocyte-specific deletion)		698
<i>Nrp1</i>	Glomerular aneurysm, impaired mesangial development, glomerulosclerosis		379
<i>Par1a</i> , <i>Par1b</i>	Proximal tubule dilation and immature glomeruli		552
<i>Pdgfb</i> , <i>Pdgfrb</i>	Lack of mesangial cells, ballooned glomerular capillary loop		377,378
<i>Pik3c3</i> (<i>Vps34</i>)	FSGS, defects in vesicular trafficking (podocyte-selective)		699,700
<i>Podxl</i>	Podocyte effacement, reduced glomerular capillary fenestrae, abnormal GBM, anuria	Congenital FSGS, omphalocele, and microcoria	701,702
<i>Prkci</i> (<i>aPKCλ/ι</i>)	Defect of podocyte foot processes, nephrotic syndrome (podocyte-selective)		703,704
<i>Ptpro</i> (<i>GLEPP1</i>)	Broadened podocyte foot processes with altered interdigitation patterns	SRNS	673,705
<i>Rab3A</i>	Albuminuria, disorganization of podocyte foot process structure		706
<i>Rbpj</i>	Decreased renal arterioles, absence of mesangial cells, and depletion of renin cells (stromal cell-selective)		341,376,394
	Reduction in juxtaglomerular cells, impaired renin synthesis (renin cell-selective)		
<i>Rhpn1</i>	Loss of principal CD cells (UB-specific)		707
	FSGS, podocyte foot process effacement, GBM thickening		
<i>Robo2</i>	Abnormal pattern of podocyte foot process interdigitation, focal effacement of foot processes, proteinuria	CAKUT, VUR	708
<i>Slc5a2</i> (<i>SGLT2</i>)	Elevated urinary excretion of glucose, calcium, and magnesium	Glucosuria	709
<i>Sh3gl1</i> , <i>Sh3gl2</i> , <i>Sh3gl3</i> (<i>Endophilin 1/2/3</i>)	Podocyte foot process effacement and proteinuria, neuronal defects		658
<i>Sirpa</i>	Irregular podocyte foot process interdigitation, mild proteinuria		710
<i>Sox4</i>	Oligonephronia, podocyte effacement, GBM defects (MM-selective)		711
<i>Sox17</i> , <i>Sox18</i>	Vascular insufficiency in kidneys and liver; ischemic atrophy of renal and hepatic parenchyma; defective postnatal a	HLT (hypotrichosis-lymphedema-telangiectasia) syndrome (hair, vascular, and lymphatic disorder)	452,453
<i>Sv2b</i>	Podocyte foot process effacement and proteinuria		712
<i>Synj1</i>	Podocyte foot process effacement and proteinuria; neuronal defects		658
<i>Tcf21</i> (<i>Pod1</i>)	Lung and cardiac defects, sex reversal and gonadal dysgenesis, vascular defects, disruption in UB branching, impaired podocyte differentiation, dilated glomerular capillary, poor mesangial migration		14,290,292
<i>Tek</i> (<i>Tie2</i>)	Loss of ascending vasa recta and medullary capillary plexus, urinary concentration defects	Cutaneous and mucosal venous malformations, congenital glaucoma	444

Continued on following page

Table e1.1 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Tfcp2l1</i>	Loss of CD intercalated cells		340
<i>Tjp1</i> (ZO-1)	Podocyte effacement and proteinuria		713
<i>Trp63</i> (TP63)	Loss of CD intercalated cells	ADULT (acro-dermato-ungual-lacrima-tooth) syndrome; limb-mammary syndrome	714
<i>Trpc6</i>	Protected from angiotensin-mediated or proteinuria or complement-dependent glomerular injury (null mutation); podocyte foot process effacement and proteinuria (transgenic overexpression in the podocyte lineage)	SRNS, FSGS	715,716,717,718, 713,719,720
<i>Vangl2</i>	Immature and poorly branched glomerular tuft	Neural tube defects	355
<i>Vegfa</i>	Endotheliosis, disruption of glomerular filtration barrier formation, nephrotic syndrome (podocyte-selective); peritubular capillary rarefaction and polycythemia (tubule-specific)		435,385,437,722
<i>Vhl</i>	Glomerulonephritis (podocyte-selective)	Von Hippel-Lindau syndrome	449
<i>Wasl</i> (N-wasp)	Podocyte effacement, proteinuria		723
<i>Wnt7b</i>	Impaired development medullary microvasculature		724
<i>Wnt11</i>	Glomerular cysts		725

ADPKD, Autosomal dominant polycystic kidney disease; ARPKD, autosomal recessive polycystic kidney disease; CAKUT, congenital anomalies of the kidney and urinary tract; CD, collecting duct; CM, cap mesenchyme; FSGS, focal segmental glomerulosclerosis; GBM, glomerular basement membrane; GDNF, glial cell-derived neurotrophic factor; MM, metanephric mesenchyme; SRNS, steroid-resistant nephrotic syndrome; UB, ureteric bud; VUR, vesicoureteral reflux.

Table 1.2 Mouse Strains for Conditional Gene Targeting and Lineage Marking of Cells

Gene Promoter	Cre	rtTA	Cre-ERT2	Fluorescent Reporter	Renal Expression	Extrarenal Expression	Reference
<i>11Hsd2</i>	✓			✓	Principal cells of collecting duct, connecting tubules	Amygdala, cerebellum, colon, ovary, uterus, epididymis, salivary glands	478
<i>Aqp2</i>	✓		✓		Principal cells of collecting duct	Testis, vas deferens	479
<i>Atp6v1b1</i>	✓			✓	Collecting ducts (intercalated cells), connecting tubule		480,481
<i>Bmp7</i>	✓		✓	✓	Cap mesenchyme		482
<i>Cdh16</i> (Ksp-Cadherin)	✓		✓		Renal tubules, collecting ducts, ureteric bud, Wolffian duct, mesonephros	Müllerian duct	483,484
<i>Cited1</i>			✓	✓	Cap mesenchyme		485
<i>Emx1</i>	✓		✓	✓	Renal tubules (proximal and distal tubules)	Cerebral cortex, thymus	486
<i>Foxd1</i>	✓		✓	✓	Stromal cells		487
<i>Gdnf</i>			✓	✓	Cap mesenchyme		488
<i>Ggt1</i>	✓				Cortical tubules		489
<i>Hoxb6</i>	✓		✓		Metanephric mesenchyme	Lateral mesoderm, limb buds	490,491
<i>Hoxb7</i>	✓	✓		✓	Ureteric bud, wolffian duct, collecting ducts, distal ureter	Spinal cord, dorsal root ganglia	58
<i>Kap</i>	✓				Proximal tubules	Brain	492,493
<i>Klf3</i>			✓		Collecting ducts	Gonads	GUDMAP ^a
<i>Nphs1</i>	✓	✓		✓	Podocytes	Brain	494,495
<i>Nphs2</i>	✓	✓	✓		Podocytes		496
<i>Osr1</i>	✓		✓	✓	Metanephric mesenchyme	Intermediate mesoderm	497,137
<i>Osr2</i>	✓				Condensing metanephric mesenchyme; glomeruli	Palatal mesenchyme	498
<i>Pax2</i>	✓			✓	Pronephric duct, wolffian duct, ureteric bud, cap mesenchyme	Inner ear, midbrain, cerebellum, olfactory bulb	499
<i>Pax3</i>	✓				Metanephric mesenchyme	Neural tube, neural crest	496,500,501
<i>Pax8</i>	✓	✓	✓	✓	Renal tubules (proximal and distal tubules) and collecting ducts (Tet-On inducible system)		502,503
<i>Pck1</i>	✓		✓	✓	Proximal tubules	Liver	504
<i>Pdgfb</i>			✓		Endothelium	Systemic vascular endothelium	505
<i>Pdgfrb</i>	✓	✓	✓	✓	Mesangial cells, vascular smooth muscles	Pericytes, vascular smooth muscles	379,448,506
<i>Prox1</i>	✓		✓	✓	Ascending vasa recta, lymphatic vessels	Systemic lymphatic vasculature	507,508,509
<i>Rarb</i>	✓			✓	Metanephric mesenchyme		510
<i>Ren1</i>	✓		✓	✓	Juxtaglomerular cells, afferent arterioles, mesangial cells	Adrenal gland, testis, sympathetic ganglia	511
<i>Ret</i>			✓		Ureteric bud, collecting ducts	Dorsal root ganglion, neural crest	512
<i>Sall1</i>			✓	✓	Metanephric mesenchyme (tamoxifen-inducible system)	Limb buds, central nervous system, heart	513
<i>Six2</i>	✓	✓	✓	✓	Cap mesenchyme		60
<i>Slc22a6</i>			✓		Proximal tubules		503
<i>Slc5a2</i>	✓				Proximal tubules		514
<i>Sox18</i>			✓		Cortical and medullary vasculature	Blood vessels and precursor of lymphatic endothelial cells	515,516,517

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Table 1.2 Mouse Strains for Conditional Gene Targeting and Lineage Marking of Cells (Cont'd)

Gene Promoter	Cre	rtTA	Cre-ERT2	Fluorescent Reporter	Renal Expression	Extrarenal Expression	Reference
<i>Spink3</i>	✓				Medullary tubules (distal or connecting tubules?)	Mesonephric tubules, pancreas, lung, liver, gastrointestinal tract	500,501,518
<i>T (Brachyury)</i>	✓		✓		Whole kidney (both ureteric bud and metanephric mesenchyme)	Pan-mesodermal	249
<i>Tbx18</i>	✓				Ureteral mesenchyme	Heart, limb buds	519
<i>Tcf21</i>	✓				Metanephric mesenchyme, cap mesenchyme, podocytes, stromal cells	Epicardium, lung mesenchyme, gonad, spleen, adrenal gland	326
<i>Tek</i>	✓	✓	✓	✓	Endothelium	Systemic vasculature endothelium	520,521,522
<i>Tie1</i>	✓	✓			Endothelium	Systemic vasculature endothelium	523
<i>Umod</i>	✓		✓		Thick ascending limbs of loops of Henle	Testis, brain	524
<i>Wnt4</i>	✓		✓	✓	Renal vesicles, nascent nephrons (comma- and S-shaped bodies)	Lungs, developing gonads	60,525

^aGenitourinary Development and Molecular Anatomy Project (GUDMAP); <http://www.gudmap.org>.

Cre (noninducible Cre recombinase transgene); rtTA (reverse tetracycline transactivator, tetracycline-inducible expression system); Cre-ERT2 (Cre-estrogen receptor ligand binding domain fusion transgene, tamoxifen-inducible expression system); fluorescent reporter (promoter-driven expression of a fluorescent protein, such as green fluorescent protein and its variants.)

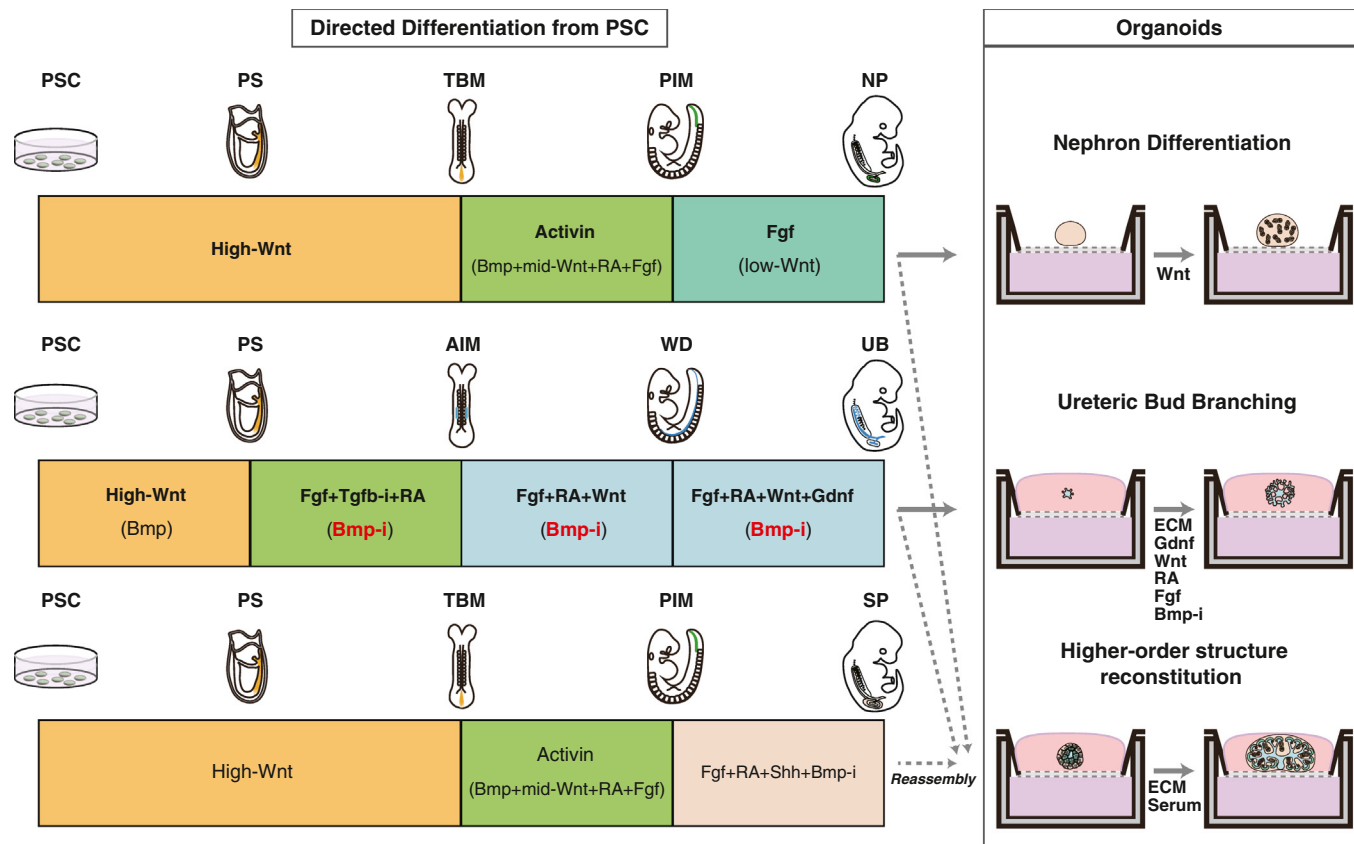


Fig. 1.9 Generation of kidney organoid based on development. Representative-directed differentiation protocols of pluripotent stem cells (PSCs) toward nephron progenitor (NP), ureteric bud (UB), and stromal progenitor (SP) lineages are summarized. Induced progenitors are further differentiated to form nephron organoids (right upper panel) or ureteric bud organoids (right middle panel), or they are reassembled to reconstitute higher-order kidney structures (right lower panel). Commonly used signaling molecules among multiple reports are shown in bold. A factor specifically used in human PSC differentiation is shown in red. AIM, Anterior intermediate mesoderm; ECM, extracellular matrix; PIM, posterior intermediate mesoderm; PS, primitive streak; PSC, pluripotent stem cell; RA, retinoic acid; TBM, tail bud mesenchyme; WD, Wolffian duct.

unique opportunity for a time- and cost-effective strategy to interrogate novel gene functions relevant to embryonic kidney development in not only mice but also humans. In addition, the selective induction protocol of either NP or UB lineage allowed the cell-autonomous role of commonly expressed genes, such as *PAX2*, in nephron and the collecting duct development to be addressed.^{89,90} Such gene-edited organoids or patient PSC-derived organoids are being used to model various genetic diseases including podocyte diseases, ciliopathies, and polycystic diseases.⁹¹⁻⁹⁵ These efforts will eventually lead to discovery of curative drugs. Another long-term goal of kidney organoid research is kidney replacement therapy. However, the currently available kidney organoids are immature and lack ureters and a proper vascular network.^{17,96} A deeper understanding of the developmental processes in vivo will be required to generate transplantable kidneys ex vivo.

SINGLE-CELL ANALYSIS OF THE EMBRYONIC KIDNEY AND SYNTHETIC KIDNEY

The kidney, including the embryonic kidney, contains numerous cell types, with each cell type consisting of a limited number of cells. Conventional transcriptomic techniques, which usually require thousands of cells, result in the averaged property of the whole kidney or sorted limited fractions of cell types. The newly emerging scRNA-seq technology has revolutionized the field of kidney development research. It can measure transcriptomic information of a few tens of thousands of cells, with a depth of a few thousand genes per single cell, facilitating the collection of transcriptomic data from entire tissues or embryos. Moreover, advances in bioinformatics tools have enabled computational retrospective recovery of cell types, eliminating the need to presort the target cell of interest and even allowing identification of previously unknown subpopulations.^{97,98} Indeed, this technique has allowed the identification of multiple cell types in the embryonic and adult kidney and many representative genes expressed in each cell type including not only nephrons and collecting ducts but also interstitial cells, mesangial cells, and the vasculature.^{99,100} The scope of the analysis has been extended from mouse to human embryonic kidney so that in addition to genes conserved across species, many human-specific gene features have been identified.¹⁰¹ Taking advantage of the relatively slow pace of human development, scRNA-seq analysis more precisely identifies the transitional cell states from NPs to nephron formation via S-shaped bodies.¹⁰² scRNA-seq analysis has also been applied to human kidney organoids, allowing precise identification of cell types in the organoids. Comparison of these data with human embryonic kidney data revealed the common or differentially expressed genes in the organoids and showed that the currently available organoids are immature and correspond to the embryonic stage.⁹⁶ Thus scRNA-seq has become one of the standard gene expression analyses for renal research.

The weakness of scRNA-seq is the loss of positional information, as dissociation of the tissue into individual cells is required for analysis. Some spatial scRNA-seq techniques, which detect gene expression in histological sections, have been developed and applied to some renal research including kidney injury models.¹⁰³ However, current spatial

scRNA-seq has not achieved analysis at the single-cell resolution. Instead, new hybridization techniques using a few hundred or thousands of known gene probes have been developed.¹⁰⁴ Further advances in single-cell analysis are expected in the next decade.

NONMAMMALIAN MODEL SYSTEMS FOR KIDNEY DEVELOPMENT

Organisms separated by millions of years of evolution from humans still provide useful models to study the genetic basis and function of mammalian kidney development. This stems from the fact that all these organisms possess excretory organs designed to remove metabolic wastes from the body and that genetic pathways involved in other aspects of invertebrate development may serve as templates to dissect pathways in mammalian kidney development. The excretory organs of invertebrates differ greatly in their structure and complexity and range in size from a few cells in *Caenorhabditis elegans* to several hundred cells in *Drosophila*, to the more recognizable kidneys in amphibians, birds, and mammals. In *C. elegans*, the excretory system consists of a single large H-shaped excretory cell, a pore cell, a duct cell, and a gland cell.^{105,106} *C. elegans* provides many benefits as a model system: the availability of powerful genetic tools including “mutants by mail,” a short life and reproductive cycle, a publicly available genome sequence and resource database (<http://www.wormbase.org>), the ease of performing genetic enhancer-suppressor screens in worms, and the fact that they share many genetic pathways with mammals. Major contributions to our understanding of the function of polycystic and cilia-related genes have been made from studying *C. elegans*. The *Pkd1* and *Pkd2* homologs, *LOV1* and *LOV2* of *C. elegans*, are involved in cilia development and mating behavior.^{107,108} Elucidation of the genetic interactions of the *Kirrel* (*Neph1*) ortholog and nephrin-like molecules *SYG1* and *SYG2* in synapse formation in *C. elegans* also provided clues to the roles of these genes in glomerular and slit diaphragm formation and function in mammals.¹⁰⁹

Similar to *C. elegans*, the relative ease of large-scale genetic screens and phenotypic characterization in *Drosophila* makes it another valuable complementary model for understanding the genetic basis of developmental processes. The excretory system of *Drosophila* consists of two parts, the nephrocytes and the malpighian tubules, which are functionally analogous to podocytes and renal tubules, respectively. A fundamental difference from vertebrate kidneys is that the nephrocytes and malpighian tubules are not physically connected. Nephrocytes either surround the heart (epicardial nephrocytes) or the esophagus (garland cells) and have elaborate membrane invaginations that closely resemble the glomerular filtration barrier. Remarkably, mutations of *Drosophila* homologs of genes known to be essential to form slit diaphragms and maintain podocyte functions also impair nephrocyte morphology and filtration functions.¹¹⁰⁻¹¹³ Similarly, conserved genes that regulate normal patterning and function of Malpighian tubules and vertebrate renal tubules have been identified.¹¹⁴⁻¹¹⁹ Functional readouts such as impaired nephrocyte filtration or uptake of tracers, ultrastructural analysis, and mortality screens can be executed efficiently in *Drosophila*, which can

facilitate the characterization of novel gene functions that are vital for renal filtration.¹²⁰⁻¹²²

The pronephros is the functional kidney of the larva of some fishes (except for jawless fishes, which only develop the pronephros) and amphibians, while the mesonephros serve as the kidney in adults of these aquatic animals. The pronephros of the zebrafish (*Danio rerio*) larva consists of two tubules connected to a fused, single, midline glomerulus. The zebrafish pronephric glomerulus expresses many of the same genes found in mammalian glomeruli (e.g., *Vegfa*, *Nphs1*, *Nphs2*, and *Wt1*) and contains podocytes and fenestrated endothelial cells.¹²³ Advantages to the zebrafish as a model system include its short reproductive cycle, transparency of the larvae with easy visualization of defects in pronephric development without sacrificing the organism, the availability of the genome sequence, the ability to rapidly knockout/knockdown gene function, and the ability to perform functional studies of filtration using fluorescently tagged labels of varying sizes.¹²⁴ These features make zebrafish amenable to both forward and reverse genetic screens. The pronephros of the clawed frog *Xenopus laevis* has also been used as a simple model to study early events in nephrogenesis.^{125,126} Similar to the fish, the pronephros of the clawed frog consists of a single glomus, paired tubules, and a duct. The fact that *X. laevis* embryos develop rapidly outside the body (all major organ systems are formed by 6 days of age); the ease of injecting DNA, mRNA, and protein; ability to perform grafting; and in vitro culture experiments establish the frog as a valuable model system to dissect early inductive and patterning cues.^{127,128} The availability of transgenic zebrafish and *Xenopus* lines that express the fluorescent protein EGFP in pronephric and mesonephric kidneys provides an opportunity to visualize real-time kidney development and function.¹²⁹⁻¹³¹

MOLECULAR GENETICS OF KIDNEY DEVELOPMENT IN MAMMALS

ORIGIN OF THE KIDNEY: INTERMEDIATE MESODERM REDEFINED IN TIME AND SPACE

The IM is the origin of anatomically distinct kidney primordia including the pronephros, mesonephros, and metanephros (see Fig. 1.2). Therefore a deeper understanding of how the IM forms through body patterning and gives rise to the corresponding kidney primordia along both the temporal and anteroposterior axes is critical. This understanding is also key to the in vitro generation of functional metanephric kidney progenitors from PSCs.

Early insights into IM formation and subsequent Wolffian duct development were gained from studies in nonmammalian vertebrates, taking advantage of the greater feasibility of tissue manipulation techniques. Lineage tracing by direct tissue labeling and transplantation experiments in gastrulation-stage avian embryos revealed that the dorsoventral axis of the trunk is derived from prepatterned along the anteroposterior positioning within the primitive streak.¹³² Consequently, the paraxial mesoderm, IM, and lateral plate mesoderm originate from the anterior, middle, and posterior parts of the primitive streak, respectively, and migrate out to form the body trunk¹⁰ (Fig. 1.10).

Injection of dye into the *c-Pax2/c-Sim1*-expressing domain of IM cells adjacent to the 10th somite resulted in labeling of the entire Wolffian duct. This suggests that the Wolffian duct originates from the anteriorly restricted domain of the IM, then proliferates and migrates caudally along the elongation of the body axis.^{133,134} Another study proposed the involvement of retinoic acid signaling and the downstream *Hoxb4* gene in defining the anterior boundary of this Wolffian duct precursor domain, the pronephric anlage, within the IM.¹³⁵

In mice, the IM is first recognized by the *Osr1*-positive bilateral stripe domain around E8.5, the 6 to 8 somite stage. Subsequently, genes such as *Pax8*, *Pax2*, *Lhx1*, and *Gata3* begin to be expressed in 10 to 12 somite stage embryos in a more restricted domain located adjacent to the 6th to 8th somite level, referred to as the pronephric anlage, which serves as the origin of the Wolffian duct (see Fig. 1.10).¹³⁶ This *Pax2*-expressing cell cluster extends caudally, undergoes a mesenchymal-to-epithelial transition, and finally forms the Wolffian duct by E9.5. Thus along the relative positioning within the embryonic body trunk, the early *Osr1*-expressing IM domain at E8.5 can be redefined as the anterior IM (see Fig. 1.10).⁸⁶ The formed Wolffian duct is characterized by *Pax2/Lhx1/Cdh1* expression and now shuts down *Osr1*, reaching the urogenital sinus at the caudal end of the trunk.

The initial directed differentiation studies attempting to generate kidney organoids sometimes misidentified this early-stage *Osr1+/Pax2+/Lhx1+* IM as the common origin of the entire kidney including the MM. However, at E9.5, adjacent and ventral to the Wolffian duct, there is a cordlike mesenchymal tissue that expresses *Osr1+/Wt1+*. This late-stage IM, also called the nephrogenic cord, extends from the presumptive forelimb level (7th–12th somites) to the hindlimb level (23rd–28th somites), giving rise to the future mesonephros and metanephros. In particular, the *Hox11* gene family is expressed in the region of the presumptive hindlimb level, presumably determining the future metanephric mesenchyme domain,¹³⁷ which is redefined as the posterior IM (see Fig. 1.10).⁸⁶

TIMING OF DIFFERENTIATION FROM BRACHYURY-EXPRESSING (T+) MESODERM DETERMINES THE ANTEROPOSTERIOR AXIS IN THE IM

In vivo lineage tracing experiments using inducible Cre-loxP mouse lines have provided deeper insights into the mechanisms underlying anteroposterior axis formation in the IM. When *Osr1*-positive cells at approximately E8.5 to E9.5 were labeled by tamoxifen injection into a dam at E8.5, these cells contributed to all kidney parenchymal cells, namely derivatives of all three metanephric progenitors: NPs, UB, and stromal progenitors. In contrast, *Osr1*-positive cells labeled around E9.5 to E10.5 contributed to the nephron and stromal progenitor lineages but not to the UB-derived collecting ducts.¹² In another study, *brachyury* (*T*)-expressing nascent mesoderm cells were tracked in light of the mechanisms of embryonic body axis elongation. During gastrulation (E6.5 to E8.5), the caudal *T*-expressing primitive streak contributes to the formation of anterior (or rostral) trunk structures up to the level of the forelimb, including the heart and foregut-derived endodermal organs. During the following days, *T*-positive cells continue to self-renew

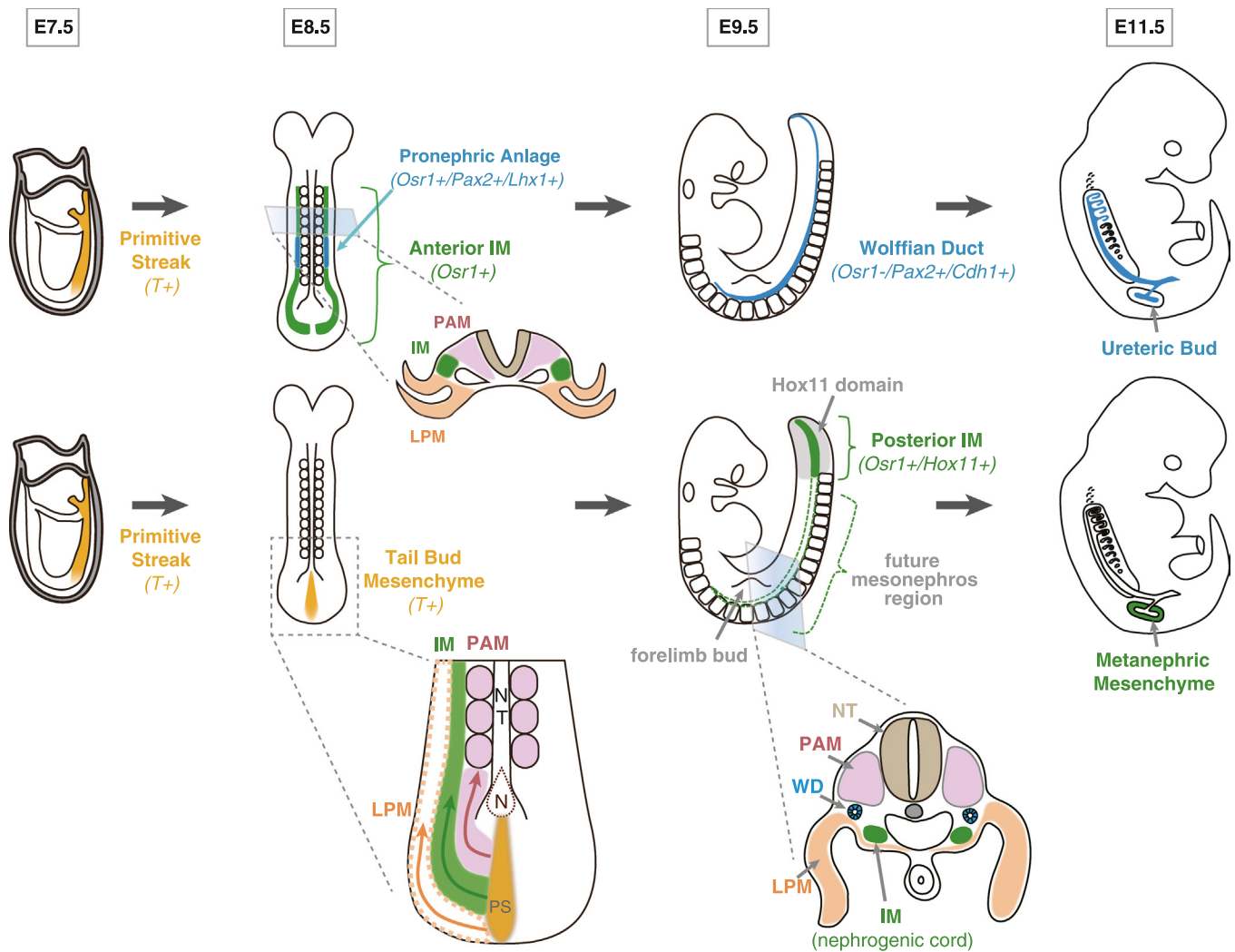


Fig. 1.10 Developmental origin of metanephric kidney progenitors. Model for lineage segregation of the ureteric bud (UB) and metanephric mesenchyme (MM). The UB differentiates from E7.5 T-positive primitive streak through the pronephric anlage, a part of anterior intermediate mesoderm (IM) at E8.5. The MM originates from the E7.5 T-positive primitive streak, then stays in the T-positive tail bud mesenchyme until E8.5, and finally differentiates into posterior IM at E9.5. The section image of the E8.5 and E9.5 embryos is shown on the side panels. The magnified image of the tailbud mesenchyme region shows the cell migration from the primitive streak. LPM, Lateral plate mesoderm; PAM, paraxial mesoderm; N, notochord; NT, neural tube; WD, Wolffian duct.

at the caudal end of the embryo, known as the tailbud mesenchyme (or axial progenitor). They continuously differentiate in an additive manner to form the caudal trunk tissues beyond the forelimb level of the embryo, eventually achieving elongation of the body axis.¹³⁸ Temporal labeling of T-expressing nascent mesoderm around E7.5 to E8.5 contributed to all three metanephric progenitor lineages. However, T-expressing tail bud mesenchyme labeled around E8.5 to E9.5 contributed only to nephron and stromal progenitors but not to the UB lineage.⁸⁶ These results demonstrate that the ureteric bud originates from T+ nascent mesodermal cells around E7.5 and differentiates into T/*Osr1*+ early IM, namely the anterior IM, around E8.5. Conversely, the precursors of nephron and stromal progenitors maintain a T+ status until at least E8.5 and transition to T/*Osr1*+ late IM around E9.5, corresponding to the posterior IM. Thus the origins of the nephron/stromal progenitors and the ureteric bud are spatially and temporally distinct, and the timing of differentiation from

T+ nascent mesoderm determines the anteroposterior axis in the IM.

In summary, the early-onset IM or anterior IM forms the pronephric anlage at the level of the forelimb and migrates caudally, giving rise to the Wolffian duct, whereas the late-onset IM forms the metanephric mesenchyme via the posterior IM and eventually meets the Wolffian duct, which has migrated from its anterior origin (see Fig. 1.10).

INTERACTION OF THE URETERIC BUD AND METANEPHRIC MESENCHYME

The classic studies with the organ culture system that started in the 1950s^{36,139,140} demonstrated that epithelialization of the MM requires a UB-derived factor.³⁷ However, the modern era of studies on the early development of the kidney began with the observation of renal agenesis phenotypes in gene-targeted or knockout mice, the earliest among these being the knockout of several transcription factors including *Wt1*, *Pax2*, *Eya1*,

Osr1 (*Odd1*), *Six1*, *Sall1*, *Lhx1* (*Lim1*), and *Emx2*.^{40,41,141-147} The knockout of genes for several secreted signaling molecules such as GDNF (glial cell line–derived neurotrophic factor), GDF11 (growth differentiation factor 11), gremlin, and the receptors Ret and GFR α 1 also resulted in renal agenesis, at least in the majority of embryos.¹⁴⁸⁻¹⁵⁴ These studies revealed that GDNF is the major factor secreted from the MM to attract the Ret-expressing UB toward the MM and that Wnt9b is the UB-derived factor to initiate epithelialization of the NPs in the MM¹⁵⁵ (Fig. 1.11).

Clinical Relevance

Perturbation of cell–cell communication during embryonic kidney development can have wide-ranging detrimental consequences including renal agenesis, CAKUT, proteinuria, kidney cysts, defective urine osmoregulation, acidosis, and predisposition to hypertension and chronic kidney diseases.

LINEAGE SPECIFICATION TOWARD THE METANEPHRIC MESENCHYME

Because many of the transcription factors in the MM are required to maintain GDNF expression, their deletion often leads to insufficient UB attraction toward the MM.¹⁵⁵ Two exceptions are the *Osr1* (*Odd1*)- and *Eya1*-mutant embryos, where the MM itself is absent, suggesting that *Osr1* and *Eya1* represent early determinants of the MM (see Fig. 1.11).^{11,142} *Osr1* marks the IM from which the mesenchymal cells within the mesonephros and metanephros are derived and is subsequently downregulated upon epithelial differentiation. Mice lacking *Osr1* do not form the MM and do not express several other factors required for metanephric kidney formation including *Eya1*, *Six2*, *Pax2*, *Sall1*, or *Gdnf*.¹¹

Eya1-mediated specification of the MM cell fate is thought to occur via interaction with another transcription factor, *Six1*. *EYA1* and *SIX1* mutations are found in humans with branchiootorenal (BOR) syndrome.¹⁵⁶ *Eya1* and *Six1* form a regulatory complex that is involved in transcriptional regulation.^{157,158} Interestingly, *Eya1* was shown to have an intrinsic phosphatase activity that regulates the activation of the *Eya1/Six1* complex.^{158,159} Moreover, *Eya* and *Six* family genes are coexpressed in several tissues in mammals, *Xenopus* and *Drosophila*, further supporting a functional interaction between these genes.^{142,144,145,160,161} Direct transcriptional targets of this complex appear to include the pro-proliferative factor *Myc*.¹⁵⁸ In the *Eya1*-deficient urogenital ridge the putative MM is completely absent.¹⁶² Consistent with this finding, *Six1* is either absent or poorly expressed in the presumptive location of the MM of *Eya1*-null embryos.^{158,160-162} *Eya1* is expressed in the *Six1*-null mesenchyme, suggesting that *Eya1* is upstream of *Six1*.^{144,145}

The transcription factor *Wt1* is another essential regulator of early MM development. *Wt1* expression is weak in the uninduced MM but increases in the condensed cap mesenchyme surrounding the branching UB tips. *Wt1* expression remains throughout nephrogenesis but eventually becomes restricted to the presumptive podocytes at the proximal end of the S-shaped nephron. Mature podocytes continue to express *Wt1* at high levels. Genetic loss of *Wt1* in mice prevents UB outgrowth and causes apoptosis of the MM, whereas human mutations of *WT1* have been linked to renal tumors.^{40,163} Among the numerous identified transcriptional targets of *Wt1* known to be required for kidney development are *Bmp7*, *Pax2*, and *Sall1*.¹⁶⁴ It has also been shown that *Wt1* regulates antagonistic fibroblast growth factor (FGF) and bone morphogenetic protein (BMP)/SMAD (contraction of homologous Sma and MAD genes of worms and fruitflies) signaling pathways, effectively

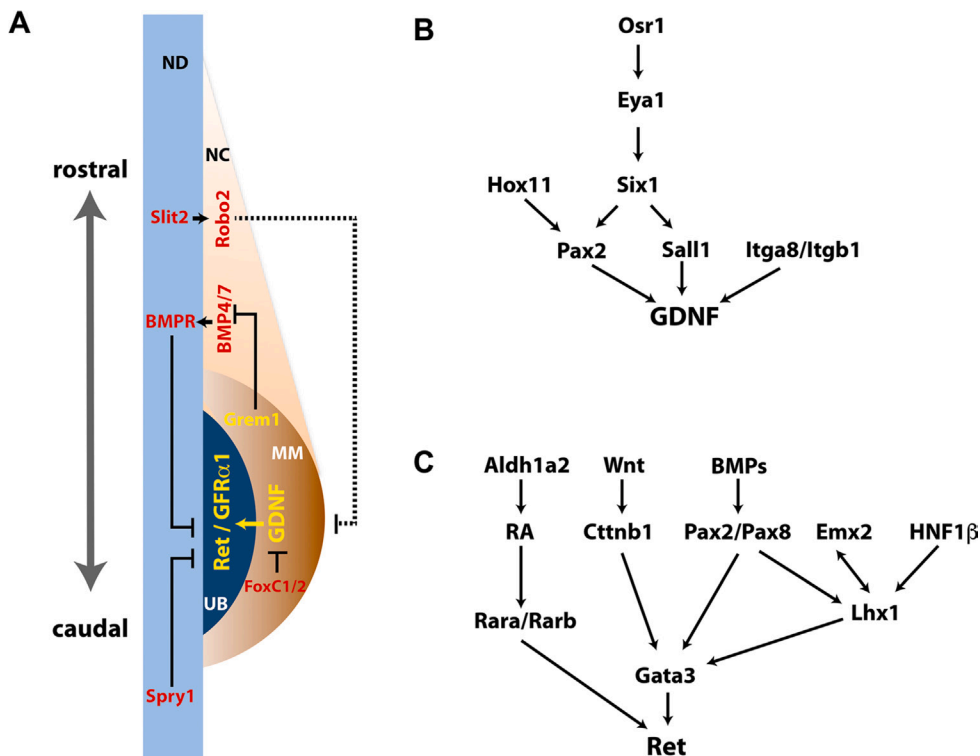


Fig. 1.11 Genetic interactions during early metanephric kidney development. (A) Regulatory interactions control the strategically localized expression of glial cell–derived neurotrophic factor (GDNF) and Ret and the subsequent induction of the ureteric bud. The anterior part of GDNF expression is restricted by *Foxc1/2* and *Slit2/Robo2* signaling. *Spry1* suppresses the postreceptor activity of Ret. *BMP4/7*–*BMPR* signaling inhibits the response to GDNF, an effect counteracted by *Grem1*. The genetic regulatory network controls the expression of (B) GDNF and (C) Ret. *BMPR*, Bone morphogenetic protein receptor; *MM*, metanephric mesenchyme; *NC*, nephrogenic cord; *ND*, nephric duct; *UB*, ureteric bud.

promoting the proliferation and survival of the MM.^{165,166} The absence of *Wt1* significantly downregulates the expression of the genes for several FGF ligands, including *Fgf8*, *Fgf10*, *Fgf16*, and *Fgf20*, which support mesenchymal proliferation.^{164,166} The impairment of FGF signaling upon loss of *Wt1* is exacerbated by the upregulation of BMP/SMAD signaling, which promotes apoptosis.¹⁶⁶ This is thought to occur through the loss of *Bmper* expression, a direct target of *Wt1*, which inhibits BMP4 signaling.

At the signaling factor level, compound loss of FGFR1 and FGFR2 in the MM leads to renal agenesis, presumably originating from the misactivation of critical transcription factors such as *Six2*, *Sall1*, and *Pax2* as early as E10.5.¹⁶⁷

IN VITRO INDUCTION OF NEPHRONS VIA NEPHRON PROGENITORS

The discovery of the distinct origins of UB and nephron/stromal progenitors has significantly advanced research on the directed differentiation of PSCs⁸⁶ (Fig. 1.9). The *T+* tailbud mesenchyme at E8.5, redefined as the source of NPs, was known to differentiate and be maintained under active Wnt signaling in vivo (Fig. 1.10). Accordingly, *T+* tailbud mesenchyme-like cells were induced from mouse ESCs by prolonged exposure to high concentrations of a Wnt signaling agonist (60 hours). The subsequent combined application of activin, retinoic acid, Bmp, and a moderate level of Wnt agonist was found to synergistically induce the expression of *Osr1*, *Wt1*, and *Hox11*, the representative markers of the posterior IM. Subsequent treatment with low levels of Wnt agonists and Fgf9 activated the expression of NP markers. When these NPs were co-cultured with the embryonic spinal cord, acting as exogenous Wnt signaling sources in place of the UB, they differentiated into self-organized epithelial nephron structures that included glomerular podocytes, parietal epithelia (Bowman capsule), and renal tubules. The signaling factors used in mouse ESC differentiation were also effective in inducing human nephron structures from human-induced PSCs (iPSCs), with adjustments made for the interspecies developmental rate. For example, Wnt agonist treatment was extended from 60 hours in mice to 144 hours in humans, corresponding to the approximately 2.5 times longer period of early gastrulation processes. While there are recognizable variations in individual induction protocols, most subsequent reports have adopted sequential induction steps involving prolonged high-Wnt exposure followed by withdrawal of Wnt and addition of FGF signaling to induce the NP lineage^{87,88} (see Fig. 1.9).

LINEAGE SPECIFICATION TOWARD THE NEPHRIC DUCT/URETERIC BUD

Several components of the genetic network supporting the development of the nephric duct and the UB have been identified (see Fig. 1.11). *Pax2* and *Pax8* are required to maintain the expression of *Lhx1*.¹⁶⁸ Nephric duct specification fails in *Pax2/Pax8* mutants, while conditional deletion of *Lhx1* impairs maintenance of the nephric duct epithelia. *Pax2*, *Pax8*, and *Lhx1* altogether likely coordinate the expression of *Gata3*, which is necessary for elongation of the nephric duct.¹⁶⁹ *Gata3* and *Emx2*, which are required for the expression of *Ret* in the nephric duct,

are both regulated by β -catenin (*Ctnnb1*), an effector of the canonical Wnt signaling pathway.^{41,170,171} Unlike these genes that are expressed in the nephric duct epithelia, *Aldh1a2* (*Raldh2*), a gene in the retinoic acid synthesis pathway, is expressed in the surrounding mesenchyme and likely acts in parallel with *Gata3* to maintain *Ret* expression in the nephric duct epithelia.¹⁷² Therefore the absence of *Gata3* or *Aldh1a2* causes misguided elongation of the nephric duct, terminating into either blind-ended ureters or abnormal connections between the bladder and urethra.¹⁶⁹ The absence of *Aldh1a2* also leads to formation of ectopic ureters and hydronephrotic kidneys.¹⁷² *Ret* does not affect the nephric duct fate but has importance in later UB development and insertion of the nephric duct to the cloaca.¹⁷²⁻¹⁷⁴ Furthermore, scRNA-seq analysis of the developing nephric duct shows that it is divided into several subdomains from rostral to caudal and *Tfap2b* is expressed in the midregion.¹⁷⁵ Deletion of *Tfap2a* and *Tfap2b* leads to defects in nephric duct morphogenesis and reduction of *Gata3+* caudal domains, while *Gata3* deletion results in the absence of the *Aldh1a3+* *Ret+* caudalmost domain including the nephric duct tips (Fig. 1.12).

IN VITRO INDUCTION OF THE URETERIC BUD

The Wolffian duct is derived from the E8.5 anterior IM, which differentiates from the *T+* mesoderm earlier than the NP lineage (see Fig. 1.10). Consistent with the in vivo lineage separation process, a shorter exposure (36 hours vs. 60 hours in NP induction) of mouse ESCs to high concentrations of Wnt signaling followed by administration of retinoic acid, Fgf, and a transforming growth factor-beta (Tgfb) inhibitor induced the anterior IM to express *Osr1/Pax2/Lhx1/Gata3*⁸⁹ (see Fig. 1.9). The effect of Tgfb inhibitor in inducing anterior IM was in sharp contrast to the requirement of Tgfb superfamily agonist: activin signaling in posterior IM differentiation. Subsequent treatment with a moderate concentration of Wnt agonist, retinoic acid, and Fgf induced committed Wolffian duct precursors expressing *Lhx1/Pax2/Ret* and exhibiting cell surface molecules *Cxcr4+*/*Kit+*. Further continuous treatment with retinoic acid, Wnt signaling, and increased administration of Gdnf signaling induced Wolffian duct maturation and eventual formation of bud structures.⁸⁹ Induction of the UB lineage in human iPSCs revealed that a 36-hour Wnt signaling treatment (vs. 144 hours for NP induction) for *T+* mesoderm induction and the addition of a Bmp signaling inhibitor to the mouse anterior IM induction factors were critical. These observations suggest a commonly critical feature of the Wnt signal exposure period for the selective induction of either anterior or posterior IM-derived kidney lineages, as well as slight differences in signal requirements between species. Accordingly, subsequent reports on human UB lineage induction commonly used an initial short exposure (24 to 48 hours) of the Wnt signaling for the primitive streak/mesoderm differentiation, and then the anterior IM was induced with the combination of Fgf, retinoic acid, and dual inhibition of Tgfb and Bmp signaling. The induced ureteric buds in both mice and humans show branching capacity in vitro when embedded in the extracellular matrix in the presence of Wnt agonist and Gdnf (see Fig. 1.9).¹⁷⁶⁻¹⁷⁹

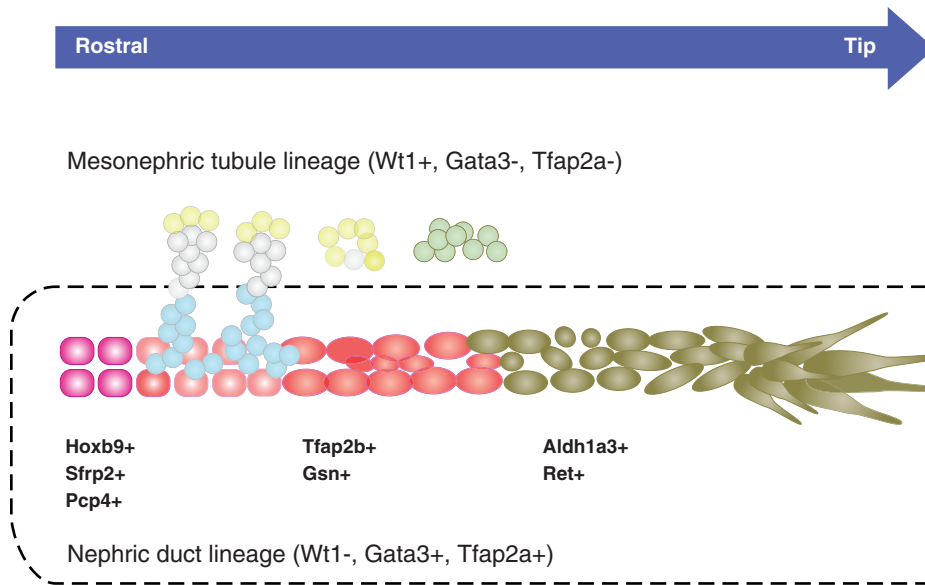


Fig. 1.12 Spatial organization of nephric duct progenitors. Spatial development of the nephric duct. Representative genes expressed in each region are shown. (Modified with permission from Sanchez-Ferras O, Pacis A, Sotiropoulou M, et al. A coordinated progression of progenitor cell states initiate urinary tract development. *Nat Commun.* 2021;12:2627.)

METANEPHRIC MESENCHYMAL GENES THAT INDUCE URETERIC BUDDING

In many cases of renal agenesis, a failure of the GDNF-*Ret* signaling axis has been identified.¹⁸⁰ GDNF, a member of the TGF- β superfamily and secreted by the MM, activates the Ret-GFR α 1 receptor complex that is expressed by cells of the nephric duct and the UB. Activation of the Ret tyrosine kinase is of central importance to ureteric budding and branching. Most mutant embryos lacking *Gdnf*, *Ret*, or *Gfra1* exhibit partial or complete renal agenesis due to severe impairment of UB sprouting while exogenous GDNF is sufficient to induce sprouting of ectopic buds from the nephric duct.^{148-150,154,173,181-183} Consistently, other genes linked to renal agenesis are known to regulate the normal expression of GDNF. These include transcription factors (e.g., *Eya1*, *Pax2*, *Six1*, *Hox11* paralogues, and *Sall1*) and proteins required to stimulate or maintain GDNF expression (e.g., GDF11, Kif26b, nephronectin, α 8 β 1-integrin, and *Fras1*) (see Fig. 1.11).^{142,145,146,151,184-192}

As described earlier, *Eya1* mutants fail to form the MM. *Pax2* is a transcriptional regulator of the paired box family and is expressed widely during the development of both UB and mesenchymal components of the urogenital system.^{189,162} Through a combination of molecular and in vivo studies, it has been demonstrated that *Pax2* appears to act as a transcriptional activator of *Gdnf* and regulates the expression of *Ret*.^{190,193} *Pax2* also appears to regulate kidney formation through epigenetic control because it is involved in the assembly of a histone H3-lysine 4 methyltransferase complex through the ubiquitously expressed nuclear factor PTIP, which regulates histone methylation.¹⁹⁴ The *Hox* genes are conserved in all metazoans and specify positional information along the body axis. *Hox11* paralogues include *Hoxa11*, *Hoxc11*, and *Hoxd11*. Mice carrying mutations in any one of these genes do not have kidney abnormalities; however, triple-mutant mice for these genes demonstrate a complete absence of metanephric kidney induction.¹⁹¹ *Eya1*, *Pax2*, and *Hox11* appear to form a complex to

coordinately regulate the expression of *Gdnf*.¹⁹⁵ However, the GDNF reduction may not always be mediated by direct transcription. For example, total amount of secreted GDNF can be reduced secondarily when the maintenance of NPs is affected due to the impaired gene regulatory network consisting of multiple transcription factors. Indeed, there are many “super-enhancers,” which are co-occupied by *Six2*, *Hoxd11*, *Osr1*, *Wt1*, and *Sall1*, for the genes involved in the progenitor program,¹⁹⁶⁻¹⁹⁸ and these transcription factors appear to activate one another to maintain the progenitors.

Other possibilities include impaired adhesive interactions within the NPs and between the NP and ureteric bud. For example, *Sall1* is necessary for the expression of the kinesin *Kif26b* by the MM cells,¹⁸⁶ and in the absence of either *Sall1* or *Kif26b*, the α 8 β 1-integrin expressed by the MM is downregulated. The loss of *Itga8* (α 8 integrin) or *Itgb1* (β 1-integrin), as well as deletion of their ligand *Npnt* (nephronectin) expressed in the UB tips, compromises the adhesion of the MM cells to the UB tips, ultimately causing loss of *Gdnf* expression and failure of UB outgrowth.^{187,188,199} A MM-derived secreted protein Isthmin-1 (*Ism1*) also acts on integrin α 8 β 1 and promotes cell-cell adhesion of the MM, thereby regulating GDNF expression UB attraction.²⁰⁰ Genetic inactivation of basement membrane proteins associated with Fraser syndrome (*Fras1*, *Frem1*, and *Frem2*) also leads to renal agenesis characterized by severe downregulation of *Gdnf* expression.^{184,185,201-204} It has been proposed that the *Fras1*/*Frem1*/*Frem2* ternary complex anchors nephronectin to the UB basement membrane, thus stabilizing engagement with α 8 β 1 integrin expressed by the MM (Fig. 1.13).²⁰¹ *Grip1*, a PDZ-domain protein known to interact with *Fras1*, is required to localize the *Fras1*/*Frem1*/*Frem2* complex on the basal aspect of the UB epithelium.²⁰⁵ *Grip1* mutations phenocopy Fraser syndrome, including renal agenesis, thus further highlighting the importance of the strategic localization of nephronectin on the UB surface toward the opposing MM.²⁰⁵⁻²⁰⁷

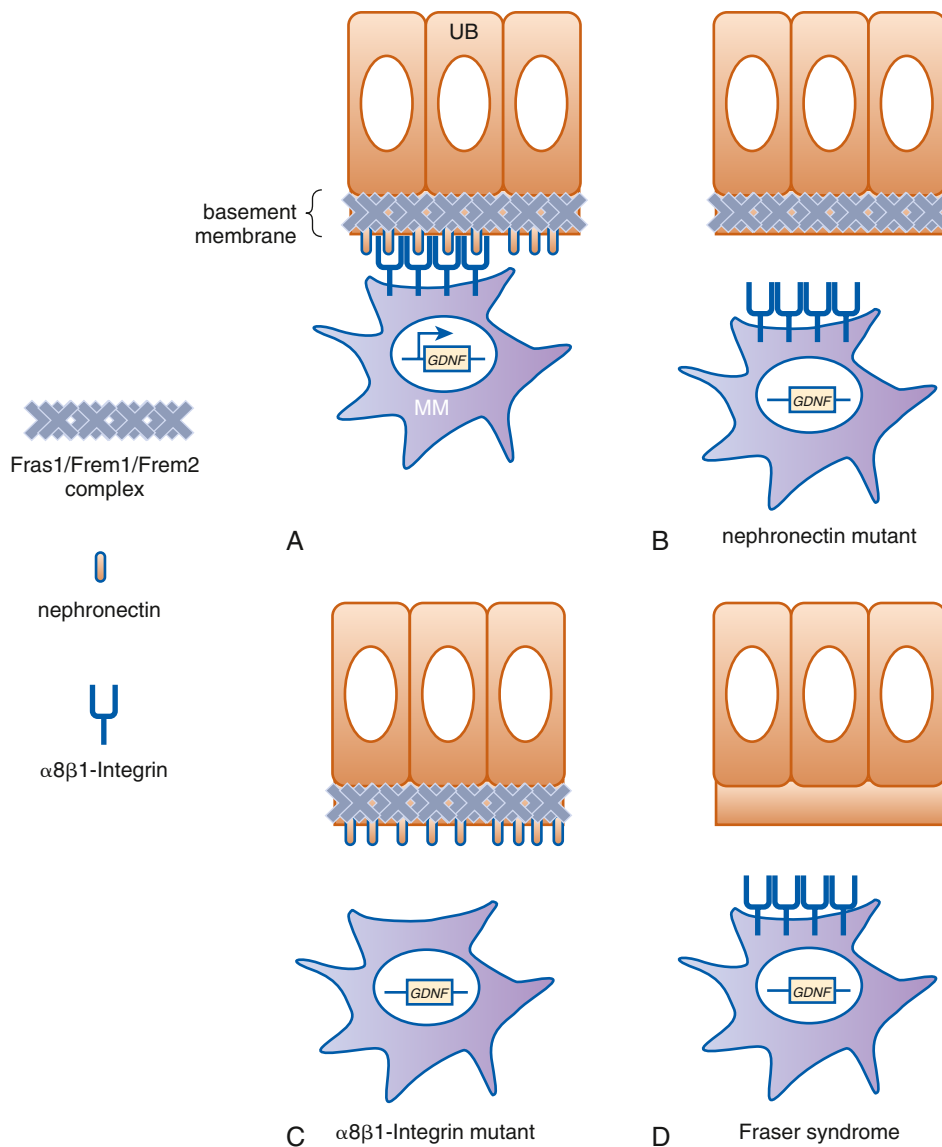


Fig. 1.13 Molecular model of renal defect in Fraser syndrome. (A) Adhesion to the ureteric bud (UB) epithelium positively regulates the expression of glial cell-derived neurotrophic factor (GDNF) by the metanephric mesenchyme (MM). Adhesion and GDNF expression are impaired in the absence of (B) nephronectin (expressed by the UB), (C) $\alpha 8 \beta 1$ integrin (expressed by the MM), and (D) or the Fras1/Frem1/Frem2 complex. Fras1, Frem1, and Frem2 are implicated in Fraser syndrome and are believed to coordinately anchor nephronectin to the UB basement membrane and stabilize the conjugation with $\alpha 8 \beta 1$ integrin. (Modified from Kiyosumi, Takeichi M, Nakano I, et al. Basement membrane assembly of the integrin $\alpha 8 \beta 1$ ligand nephronectin requires Fraser syndrome-associated proteins. *J Cell Biol.* 2012;197:677–689.)

URETERIC BUD GENES THAT REGULATE BUDDING AND BRANCHING

UB sprouting and subsequent branching requires a unique spatial organization of Ret signaling. The bulbous UB tip is a region enriched with proliferative ureteric epithelial cells, in contrast to the emerging stalk regions of the developing ureteric tree.^{42,208} It is well appreciated that receptor tyrosine kinase signaling primarily through Ret is key to the proliferation of UB tip epithelia. Exogenous GDNF supplemented in explanted embryonic kidneys can cause expansion of the UB tip region toward the source of the ligand.^{208–210} ERK kinase activation is prominent within the ampullary UB terminals where Ret expression is elevated.⁴² Consistently, chimera analysis in mice reveals that Ret-deficient cells do not contribute to the formation of the UB tips.¹⁷³ Altogether, these studies underscore the importance of strategic levels of Ret expression and activation of proliferative signaling pathways in the stereotypical sculpting of the nascent collecting duct network.

A ligand-receptor complex formed by GDNF, GFR α 1, and Ret is necessary for autophosphorylation of Ret on its intracellular tyrosines (Fig. 1.14). A number of downstream adaptor molecules and effectors have been identified to interact with active phosphorylated Ret, including Grb2, Grb7, Grb10, ShcA, Frs2, PLC γ 1, Shp2, Src, and Dok adaptor family members (Dok4/5/6).^{211–222} These downstream Ret effectors altogether are likely contributors to the activation of the Ras/SOS/ERK and PI3K/Akt pathways supporting the proliferation, survival, and migratory behavior of the UB epithelium.^{42,44,223} Knock-in mutations of the interaction site for Shc/Frs2/Dok adaptors on the short isoform of Ret lead to the formation of rudimentary kidneys.^{136,224–226} Specific mutation of the PLC γ 1 docking site on Ret leads to renal dysplasia and ureter duplications.²²⁴ The loss of *Shp2* and upstream ERK regulators *Map2k1* (*Mek2*) and *Map2k2* (*Mek1*) in the UB lineage also cause severe renal hypoplasia phenocopying that is observed in occasional Ret-deficient kidneys.^{227,228} UB-specific inactivation of *Pten*, a target of the PI3K/Akt pathway, disrupts UB branching.²²⁹ These

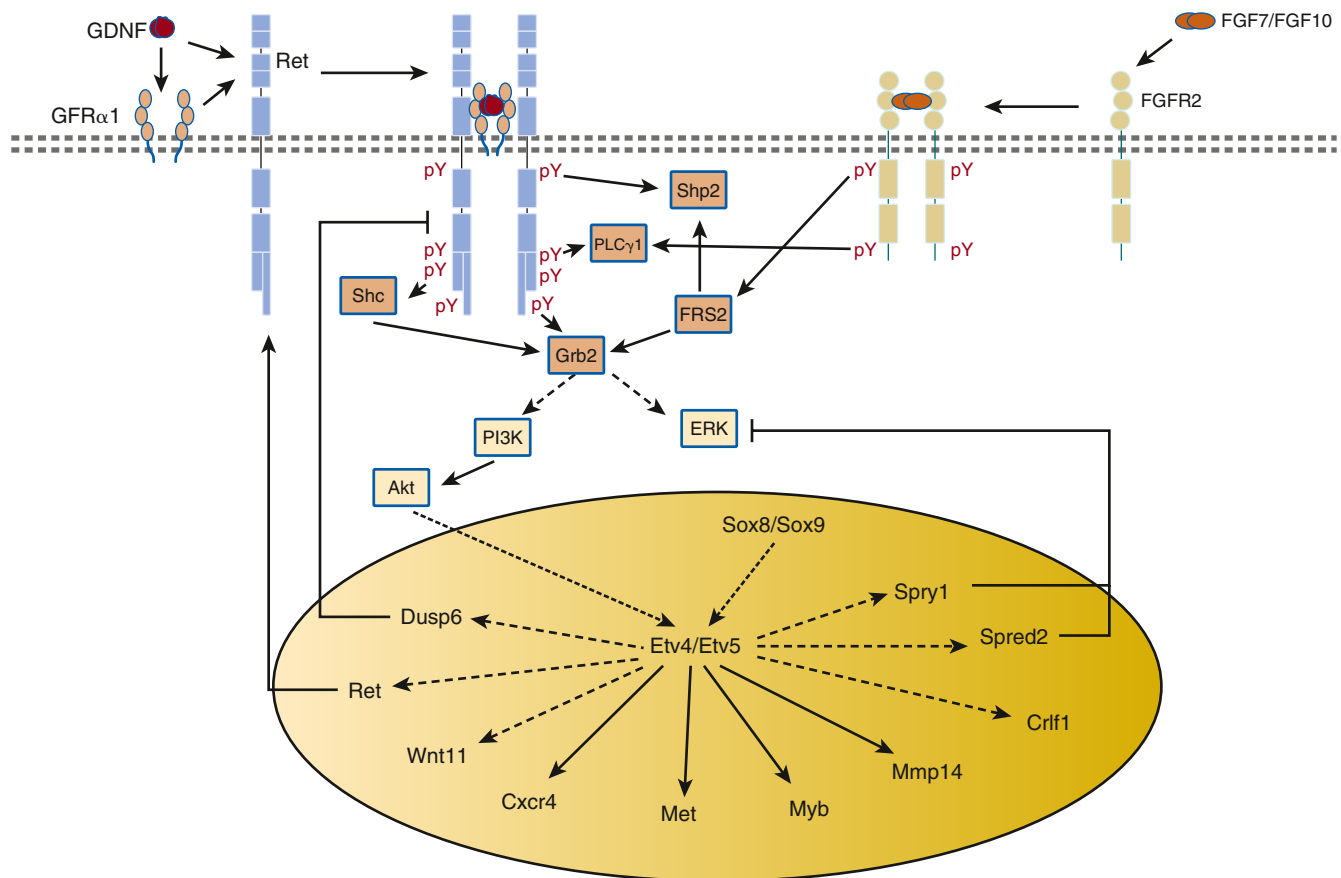


Fig. 1.14 Ret signaling pathway. Ret is activated and becomes autophosphorylated on intracellular tyrosine residues upon association with glial cell-derived neurotrophic factor (GDNF) and GFR α 1. Signaling molecules such as Grb2, Shc, FRS2, PLC γ 1, and Shp2 bind directly to the phosphorylated tyrosine residues within the intracellular domain of Ret. Recruitment of Shc, FRS2, and Grb2 leads to activation of extracellular signal-regulated kinase (ERK) and PI3K/Akt pathways. GDNF-Ret signaling leads to specific activation of a host of genes, some of which are strongly dependent on the upregulation of the transcription factors Etv4 and Etv5 (solid arrows). Etv4/Etv5 activation requires activation of the PI3K/Akt but not the ERK pathway. Sox8 and Sox9 are believed to act in parallel to reinforce transcriptional responses to GDNF-Ret engagement. Some of these pathways are shared with the FGF7/10-FGFR2 receptor signaling system. Spry1 and Spred2 negatively regulate ERK signaling, whereas Dusp6 likely mitigates the dephosphorylation of the Ret receptor, thus acting as part of a negative feedback regulatory loop.

findings underscore the significance of Ret signaling in normal UB branching.

A number of transcriptional targets of Ret activation in UBs stimulated with GDNF have been elucidated (see Fig. 1.14).²³⁰ Among these are *Ret* itself and *Wnt11*, which stimulates *Gdnf* expression in the MM,²³¹ suggesting that a positive feedback loop exists for the GDNF-Ret signaling pathway. Ret activation also positively regulates the ETS transcription factors *Etv4* and *Etv5*, which are also necessary for normal UB branching morphogenesis. *Etv4*-null homozygous mutants and compound heterozygous mutants for *Etv4* and *Etv5* manifest severe renal hypoplasia or renal agenesis, suggesting that these transcription factors are indispensable targets of Ret for proper UB development.²³⁰ In chimeric animals, *Etv4/Etv5*-deficient cells, like *Ret*-deficient cells, fail to integrate within the UB tip domain.^{173,232}

The gene *Sprouty* was identified as a general antagonist of receptor tyrosine kinases and was discovered for inhibiting the FGF and EGF signaling pathways.²³³⁻²³⁵ Of the four mammalian *Sprouty* homologs, *Spry1* is expressed at the UB tips.²³⁶ *Sprouty* molecules are thought to uncouple receptor tyrosine kinases with the activation of ERK pathways either

through competitive binding with the Grb2/SOS complex or the kinase Raf, effectively repressing ERK activation. Interestingly, its expression is upregulated upon GDNF-mediated Ret activation.²³⁰ This suggests that Ret activates a negative feedback mechanism via *Spry1* to control activated ERK levels and modulate cell proliferation in the UB.^{79,237-241} Thus *Spry1* deficiency leads to ectopic UB induction and it can partially rescue renal development in the absence of either GDNF or Ret.^{241,242} The transcriptional targets of Ret, such as *Etv4*, *Etv5*, and *Wnt11*, are retained in *Gdnf/Spry1* or *Ret/Spry1* compound null mutants.^{241,242} Other signaling including FGF7/FGF10 and FGFR2 receptors is likely to partially compensate the GDNF-Ret pathway.

NEGATIVE REGULATORS OF THE MM-UB INTERACTION

A crucial aspect of kidney development that is of great relevance to renal and urologic congenital defects in humans relates to the positioning of the UB (see Fig. 1.11). Incorrect positioning of the bud, or duplication of the bud, results in abnormally shaped kidneys and incorrect insertion of the ureter into the bladder, with a resultant ureteral reflux that can predispose to infection and scarring of the kidneys and urologic tract.

Foxc1 is a transcription factor of the Forkhead family, expressed in the IM and the MM adjacent to the Wolffian duct. In the absence of *Foxc1*, the expression of GDNF adjacent to the Wolffian duct is less restricted than in wild-type embryos. *Foxc1* deficiency results in ectopic UBs, hypoplastic kidneys, and duplicated ureters.²⁴³ Additional molecules that regulate the location of UB outgrowth are *Slit2* and *Robo2*, signaling molecules best known for their role in axon repulsion in the developing nervous system. *Slit2* is a secreted factor, and *Robo2* is its cognate receptor. *Slit2* is mainly expressed in the Wolffian duct, whereas *Robo2* is expressed in the mesenchyme.²⁴⁴ UBs form ectopically in embryos deficient in either *Slit2* or *Robo2*, similar to the *Foxc1* mutant. However, in contrast to the *Foxc1* phenotype, none of the ureters in *Slit2/Robo2* mutants failed to undergo the normal remodeling that results in insertion in the bladder.²⁴⁴ Instead, the ureters remained connected to the nephric duct in *Slit2* or *Robo2* mutants. The domain of *Gdnf* expression is expanded anteriorly in the absence of either *Slit2* or *Robo2*. Indeed, mutations in *Robo2* have been identified in patients with vesicoureteral junction defects and vesicoureteral reflux.²⁴⁵ The absence of *Robo2* most likely compromises normal separation of the Wolffian duct from the nephrogenic mesenchyme, ultimately broadening the nephrogenic zone and provoking ectopic UB induction.

Another negative regulator of branching is BMP4, which is expressed in the mesenchyme surrounding the Wolffian duct. *Bmp4* heterozygous mutants have duplicated ureters,

and, in organ culture, BMP4 blocks the induction of ectopic UBs by GDNF-soaked beads.²⁴⁶ In contrast, knockout of *Grem1*, which encodes for the secreted BMP inhibitor gremlin, causes renal agenesis, supporting a role for BMP in the suppression of UB formation.²⁴⁷

MAINTENANCE OF THE NEPHRON PROGENITOR VERSUS NASCENT NEPHRON FORMATION

The generation of a sufficient number of nephrons requires a highly regulated balance between the expansion of NP compartments and the differentiation toward epithelialized renal vesicles (Fig. 1.15). This has important clinical implications as the impaired renewal of nephron precursors or their perturbed differentiation can ultimately cause a wide range of renal pathologies due to significant paucity of functional nephrons. Here, the extracellular signaling molecules such as Wnt, Bmp, and Fgf, as well as the downstream transcription factor *Six2*, regulate the delicate balance between progenitor self-renewal and differentiation.

NPs differentiate into epithelial nephron structures through mesenchymal-to-epithelial transition. The UB-secreted factor Wnt9b activates the canonical β -catenin-dependent pathway and induces the expression of Wnt4 and Fgf8 in part of the cap mesenchyme, called pretubular aggregates. The Wnt4 expressed in pretubular aggregates, in turn, cell-autonomously acts to form epithelialized renal vesicles³⁷ (see Fig. 1.15). Canonical Wnt signaling pathway

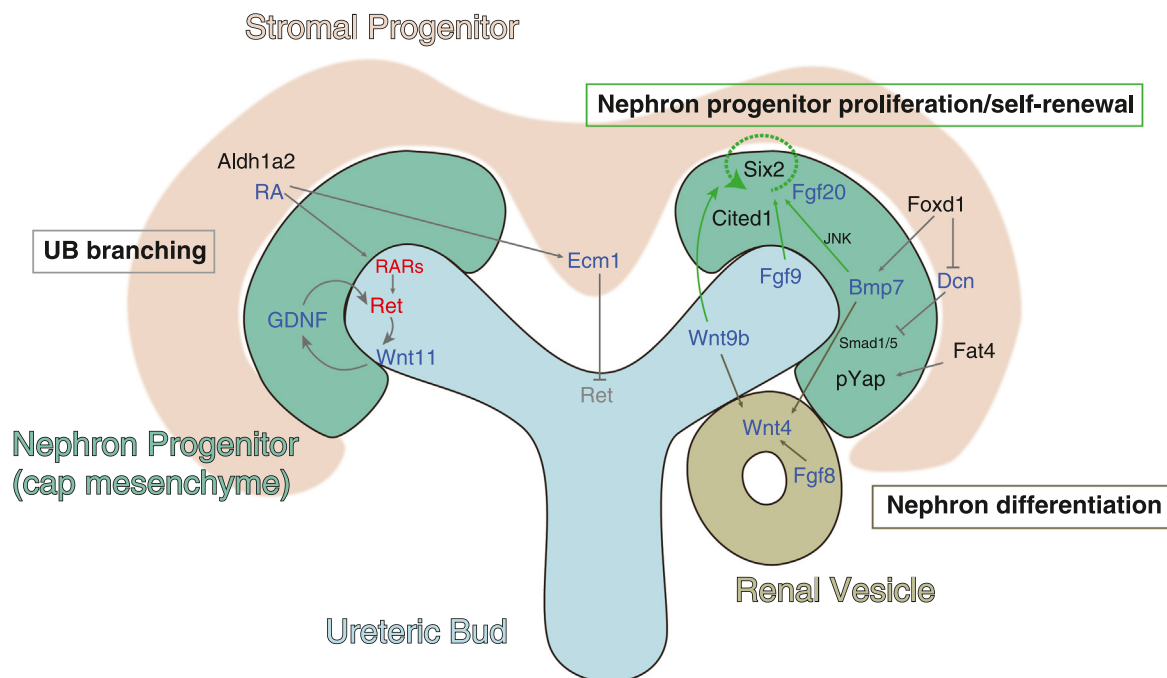


Fig. 1.15 Tripartite inductive interactions regulating ureteric branching, nephron progenitor self-renewal, and nephrogenesis. *Six2* and *Cited1* are expressed in the self-renewing nephron progenitors within the cap mesenchyme (CM) surrounding the ureteric bud. The UB tip domains express high levels of *Ret*, which is activated by glial cell-derived neurotrophic factor (*GDNF*) from the surrounding CM. *Wnt11* is up-regulated in response to *Ret* activation and stimulates *GDNF* synthesis in the CM. *Wnt9b* expressed by the UB and *Fat4* by the *Foxd1*-positive stroma are required to initiate nephrogenesis from a subset of the CM. This results in the formation of a transient renal vesicle expressing FGF8 and *Wnt4*, factors that sustain epithelialization. The stroma expresses *Aldh1a2*, a gene required for retinoic acid synthesis, and genes for the retinoic acid receptors (*Rara* and *Rarb*). Retinoic acid signaling stimulates elevated expression of *Ret* in the UB tip domain while at the same time suppressing *Ret* expression in the UB cleft domain via *Rara/Rarb2* and *Ecm1* in the stroma to initiate bifurcation of the UB tip to generate new branches. *Foxd1* in the cortical stroma also represses *Dcn*, thus relieving the *Dcn*-mediated suppression of BMP7-dependent signaling, which results in phosphorylation of SMAD1/5/8 (pSMAD1/5/8) and epithelialization of the cap mesenchyme.

is necessary and sufficient for the early inductive actions of Wnt9b and Wnt4, although it is also known that Wnt4 can activate a noncanonical alternative pathway during the final phase of epithelialization.^{48,248-252}

Although the Wnt signal was originally recognized as the differentiation-inductive signal in the NPs more recently, its role in the maintenance and proliferation of NPs especially through the interaction with transcription factor *Six2*, has become apparent in both vivo and in vitro studies.

The transcription factor *Six2* is exclusively expressed in the undifferentiated NPs, called cap mesenchyme, which condenses adjacent to the UB and is downregulated once this cap mesenchyme differentiates into pretubular aggregates. Complete loss of *Six2* causes premature ectopic formation of renal vesicles and the untimely depletion of nephron precursors.^{60,253} In contrast, overexpression of *Six2* prevented epithelialization of the cap mesenchyme.²⁵³ Therefore *Six2* is required to keep these NP cells in a naïve, proliferative precursor state.

At the molecular level, *Six2* has been demonstrated to function as both a transcriptional activator that maintains NPs and as part of a repressor complex silencing differentiation-related genes.^{254,255} For the maintenance of NPs, *Six2* synergizes with multiple transcription factors including *Osr1*, *Hoxd11*, *Wt1*, and *Sall1* to promote the transcription of relevant downstream genes (e.g., *Six2*, *Eya1*, and *Sall1*), forming a positive feedback loop.¹⁹⁶⁻¹⁹⁸ To prevent nephron differentiation, *Six2* interacts with *Tcf* (*Lef*) and *Aes* (*Groucho/TLE*) to form a repressor complex that suppresses genes related to epithelialization (e.g., *Fgf8* and *Wnt4*).^{256,257}

The interaction between Wnt signaling pathways and *Six2* appears to be intricately dosage, duration, and context dependent.^{49,53,255,257} When *Six2* expression remains high, Wnt signaling promotes progenitor renewal.²⁵⁸ In this situation, the stabilization of β -catenin promotes the association of *Six2* downstream target *Myc* with β -catenin, favoring precursor proliferation.^{49,259} Indeed, deletion of the agonistic Wnt signaling molecules R-spondins (*Rspo1* and *Rspo3*) leads to a rapid decline of NPs in mice.²⁶⁰ In vitro studies using primary cultures of mouse NPs further indicate the dose-dependent role of Wnt signaling, where weak activation favors proliferation, while strong action drives differentiation.⁵⁰⁻⁵³

On the other hand, with a sustained canonical Wnt-signaling pathway, β -catenin accumulates and displaces Groucho family repressors converting the Tcf complex into a differentiation driver.²⁵⁵ The stabilization of β -catenin eventually interferes with *Six2* expression, thus attenuating *Six2* expression. Although the continued activation of β -catenin can initiate early tubulogenesis, as evidenced by the expression of Wnt4 and *Fgf8*, it fails to complete the mesenchymal-to-epithelial transition.²⁵² Therefore the proper nephron differentiation requires the shutdown of canonical Wnt signaling after a transient activation. The more precise resolution of this time window has been addressed by in vitro nephron differentiation studies using chemical Wnt signaling activators, such as Gsk3 β inhibitors.^{45,46,54,255}

Notch signaling is required to prime NPs for differentiation and contributes to the silencing of *Six2* expression.²⁶¹ All

nephron segments fail to form when Notch signaling is lost within the *Six2*⁺ precursor lineage.²⁶²

The growth factor BMP7 is also required for the formation of the nephrogenic compartment.²⁶³⁻²⁶⁵ Activation of the Jnk pathway mediates the proliferative effect of BMP7 in uncommitted NPs.^{266,267} BMP7, through activation of the p38-MAPK, causes upregulation of the transcriptional repressor *Trps1*.²⁶⁸ Loss of *Trps1* severely impairs the formation of renal vesicles. It has been speculated that *Trps1* may indirectly relieve repression of *Cdh1* expression. Additionally, BMP7-dependent phosphorylation and nuclear translocation of SMAD1/5/8 are required for Wnt9b-induced epithelialization.²⁶⁹ Thus BMP7 has dual essential roles in promoting both progenitor replenishment and priming for epithelialization. How these pathways integrate with *Six2*-dependent signaling complexes remains poorly understood.

FGF8, FGF9, and FGF20, and their cognate receptors FGFR1 and FGFR2 are essential to form nephrons. FGF9 and FGF20 are required to maintain the multipotency and proliferative state of NPs.²⁷⁰ FGF8 is not needed for the NP maintenance or formation of renal vesicles but is required for the survival of the newly formed nephrogenic epithelia. Renal vesicles lacking *Fgf8* fail to express *Wnt4* and *Lhx1* and do not progress into S-shaped intermediate nephrons.^{249,250} Potential downstream targets of FGFR1/FGFR2 relevant to nephrogenesis are the closely related MAGUK family proteins encoded by the genes *Cask* and *Dlg1*.²⁷¹ The absence of *Cask* and *Dlg1* causes impaired proliferation and cell death in nephron precursors, with a distinctive dampening of the Ras/ERK signaling pathway.²⁷²

On the basis of these observations, the protocols to expand the mouse NPs in vitro have been established and applied to human organoid-derived progenitors. Many of the resumes include BMP, FGF, a Notch inhibitor, and a Wnt activator at low concentrations because too much Wnt stimulation leads to progenitor differentiation.⁵⁰⁻⁵² LIF is also shown to be effective for in vitro expansion, but the physiological role of LIF in progenitor maintenance in vivo remains unclear.

NEPHRON SEGMENTATION AND TUBULOGENESIS

Nephron segments are organized along a proximal–distal axis, from the most proximal renal corpuscle or glomerulus, followed by the proximal tubule, the loop of Henle, the distal tubule, and the most distal connecting tubule that links directly to the UB-derived collecting duct (see Fig. 1.4). The segmental patterning of nephrons involves a complex series of events instructed by inductive cues between neighboring cells and controlled by epigenetic signaling mechanisms. A variety of human diseases result from the mispatterning of nephrons.²⁷³

Elegant imaging studies combined with high-throughput single-cell gene expression analysis demonstrate that nephron patterning is determined as early as the recruitment of mesenchymal NPs from the cap mesenchyme.²⁷⁴ The renal vesicle arises not from a single event in time. Instead, NPs progressively incorporate into the nascent nephrons, with the timing of their recruitment predicting their acquisition of proximal–distal fates (i.e., initial recruits

commit to distal fates while the last recruits contribute to the formation of more proximal fates, including glomerular podocytes) (Fig. 1.16).

By the renal vesicle stage, gene expression asymmetry highlights an early establishment of proximal and distal domains. Genes such as *Fgf8*, *Lhx1*, *Dll1*, *Dkk1*, *Hnf1b*, *Sox9*, and *Pou3f3* are markedly elevated in the distal portion of the renal vesicles, whereas *Wt1*, *Foxc2*, and *Mafb* are largely restricted in the proximal end.^{56,275} Some genes are expressed in both regions, such as *Wnt4*, *Jag1*, *Cdh6*, and *Ccnd1*, albeit nonuniformly and more elevated in the distal domain.²⁷⁵ By the S-shaped body stage, nephron segmentation has become more evident, with several more marker genes having distinctively regionalized expression patterns.

As the earliest nephron precursors that interact with the UB become destined to acquire distal fates, it can be speculated that localized Wnt9b signaling orients the proximal–distal axis. During nephrogenesis, a gradient of β -catenin activity is established along the longitudinal axis of the nephron as early as the stage of renal vesicle formation, with the highest β -catenin activity found in the distal end and progressively decreasing toward the proximal end.⁴⁶ In organ cultures, pharmacologic manipulation of β -catenin activity can alter proximal and distal fate acquisition. Attenuation of β -catenin activity accelerates glomerular development. Conversely, augmentation of β -catenin activity favors the expression of *Lgr5*, a marker of distal fate, while repressing proximal identity. Indeed, a brief exposure of mouse embryonic or human PSC-derived NPs to a Wnt agonist, combined with subsequent

Tgfb β inhibition, results in a biased induction of glomerular podocytes.⁵⁴

After Wnt-mediated mesenchymal-to-epithelial transition, the Notch pathway comes into play, as evidenced by the expression of Notch ligands such as *Dll1* and *Jag1* in S-shaped bodies. Notch was initially reported to be involved in the proximalization of nephron segments,^{43,276} but scRNA-seq analysis of mature markers fails to detect a fate bias. This is because Notch promotes proliferation, survival, and maturation of nascent nephron epithelia independent of proximodistal segments, although Notch may support early proximal tubule fate selection.²⁷⁷

Although many genes are now known as marking nephron segments, only a few other genes and pathways that strongly influence proximal versus distal fate determination have been characterized. Loss of *Hnf1b* in the nephrogenic precursors causes marked loss of proximal and median domain markers at the S-shaped body stage, causing the formation of immature and cystic glomeruli connected to the collecting duct by a severely truncated renal tubule.^{278–280} Genes required to generate distal tubules and the loop of Henle are *Pou3f3*, *Sall1*, and *Adamts1*.^{281,282–284}

PODOCYTE DEVELOPMENT

Presumptive podocytes are located at the proximal end of the S-shaped body, lining the emerging vascular cleft (Fig. 1.17). Immature podocytes are simple columnar epithelia but do not express E-cadherin. Mature podocytes still do not express E-cadherin but atypically express vimentin, an intermediate filament protein more common among mesenchymal cells

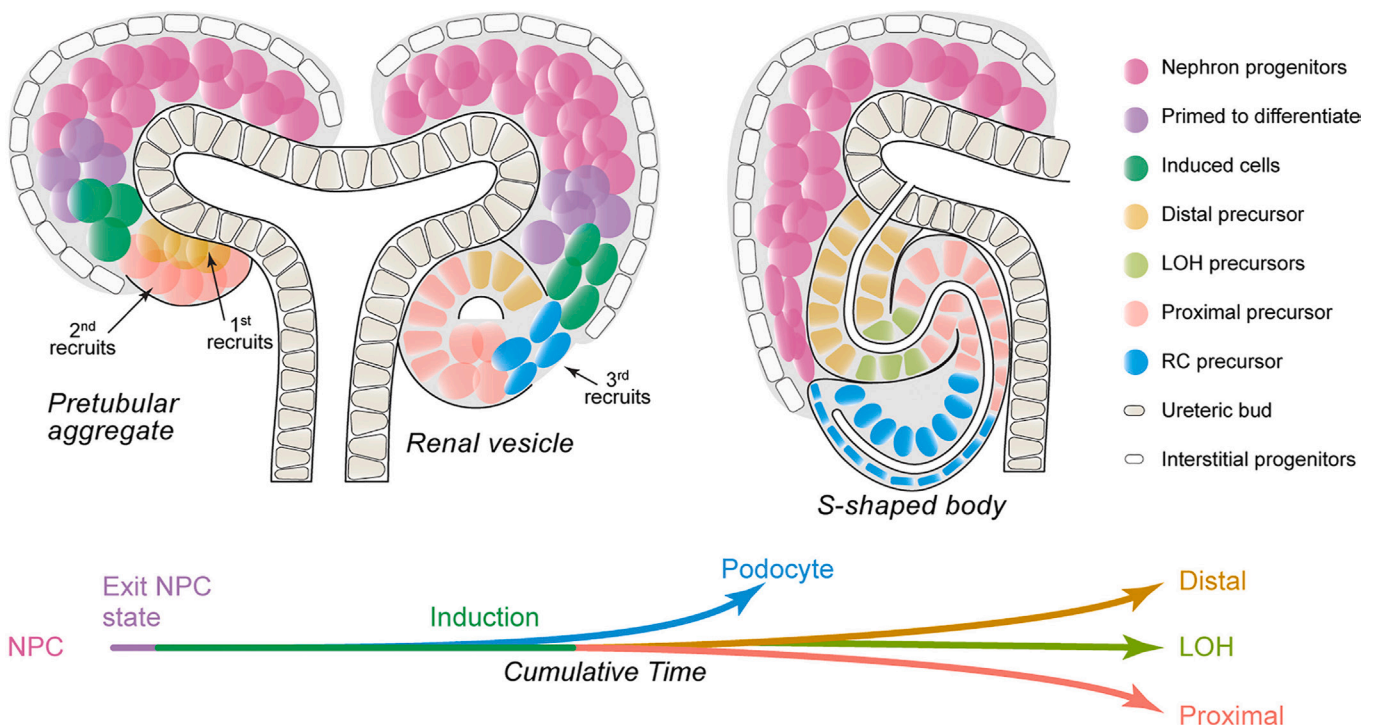


Fig. 1.16 Gradual recruitment of progenitor cells during nephrogenesis. Early recruited nephron progenitors clustering around the ureteric bud give rise to pretubular aggregates. New progenitors continue to be recruited and incorporate into the proximal end of the renal vesicle. Distal, medial, and proximal domains are established by the S-shaped body stage. LOH, Loop of Henle; NPC, nephron progenitor cell; RC, renal corpuscle. (Adapted from Lindström NO, De Sena Brandine G, Tran T, et al. Progressive recruitment of mesenchymal progenitors reveals a time-dependent process of cell fate acquisition in mouse and human nephrogenesis. *Dev Cell*. 2018;45:651–660.)

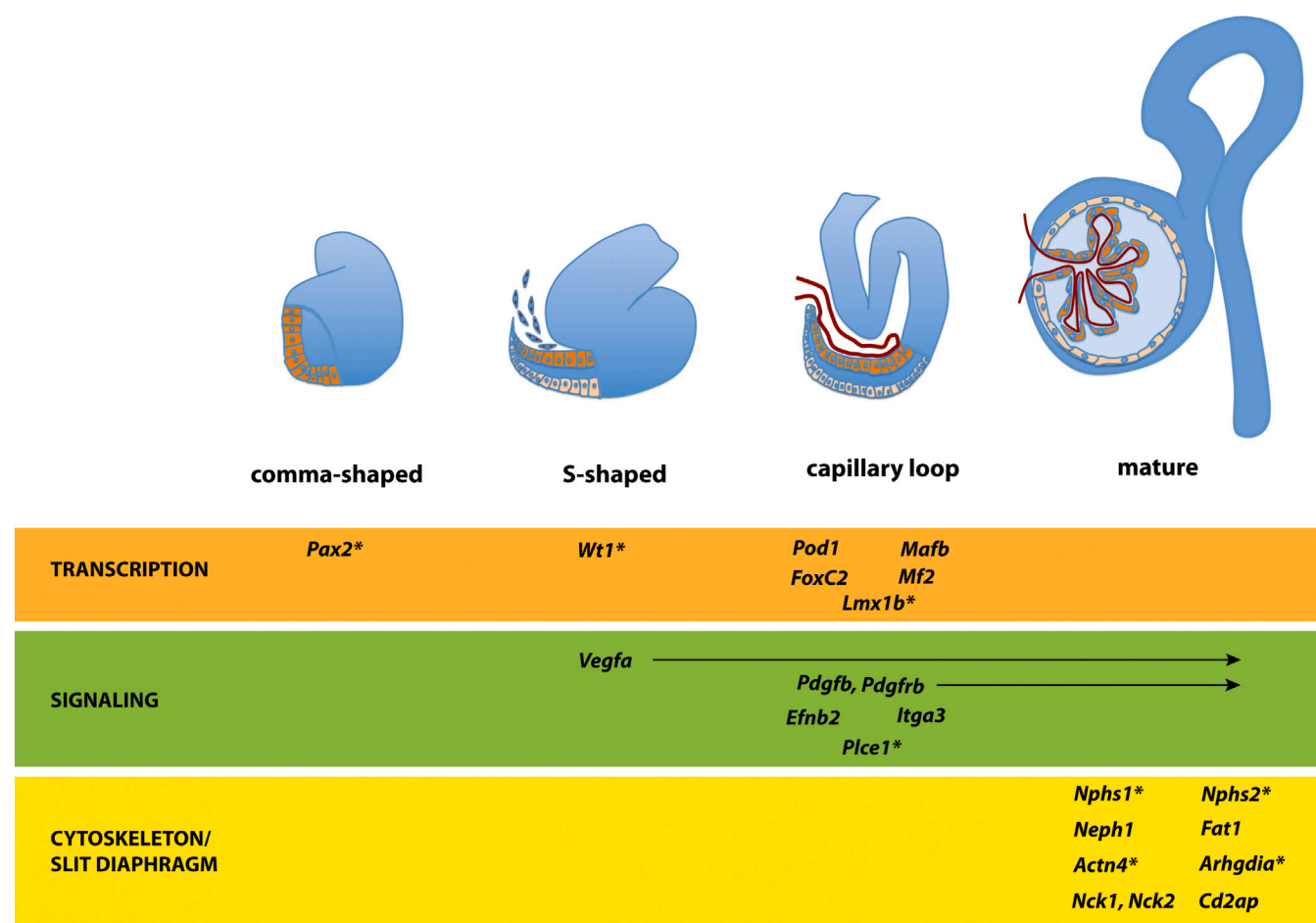


Fig. 1.17 Molecular basis of glomerular development. Key factors are shown along with the time point where major effects were observed in knockout or transgenic mouse studies. Many factors play roles at more than one time point. Genes identified as mutated in patients with glomerular disease are marked by asterisks.

but weak in most epithelial cells. Therefore podocytes may be in an intermediate state between the mesenchyme and epithelia. The most distinctive morphologic feature of a fully differentiated podocyte is its arborized and stellate appearance. Podocytes ensheath the glomerular capillaries with their foot processes, effectively forming the final layer of the glomerular filtration barrier. Foot processes emanating from adjacent podocytes interdigitate in *trans* and form a unique and porous intercellular junction called the slit diaphragm through which primary urinary filtrate passes. Three-dimensional reconstruction of podocyte ultrastructure obtained by block-face scanning electron microscopy reveals the morphologic transformation of podocytes during development and the formation of interdigitating foot processes.^{25,285} Columnar-shaped immature podocytes are linked by tight and adherens junctions that progressively migrate from the apical to the basal side. Once the junctional complex has descended close to the basement membrane, podocytes begin to flatten, spread, and interdigitate with short primitive foot processes underneath the junctions. As the primitive processes grow, the tight and adherens junctions relocate from the cell body to between the processes forming the immature foot processes. Finally, the junctional complexes are gradually replaced with slit diaphragms, resulting in mature foot processes (Fig. 1.18).

The transcription factors *Wt1*, *Tcf21*, *Mafb*, *Foxc2*, and *Lmx1b* are highly expressed by developing podocytes and are important for the elaboration of podocyte foot processes and the establishment of slit diaphragms. Complete loss of *Wt1* leads to renal agenesis.⁴⁰ However, specific loss of a *Wt1* splice isoform results in poor development of podocyte foot processes.²⁸⁶ The *Wt1*-null phenotype in mice can also be rescued using a yeast artificial chromosome containing the human *WT1* gene and, depending on the level of expression of *WT1*, the mice developed a range of glomerular pathologies ranging from crescentic glomerulonephritis to mesangial sclerosis, clinical features observed in Denys-Drash syndrome arising from a mutant *WT1* allele in humans.²⁸⁷ Podocyte-specific *Wt1* deletion in mice results in reduced numbers of mature glomeruli and proximal tubules, as well as increased foot process effacement in the remaining podocytes, leading to death within 1 day of birth.²⁸⁸ Genome-wide analysis of the *Wt1* targets in podocytes reveals that *Wt1* autoregulates its own transcription and acts as master regulator of a complex transcriptional network that regulates podocyte development, structure, and function, including transcription factors (*Lmx1b*, *Tcf21*, *Mafb*, *Tead1*, *Foxc1*, and *Foxc2*) and genes strongly linked to podocyte dysfunction and nephrotic diseases (*Actn4*, *Arhgap24*, *Cd2ap*, *Col4a3*, *Col4a4*, *Lamb2*, *Nphs1*, *Nphs2*, and *Plce1*).²⁸⁹ Among them, *Nphs2*, *Mafb*, and *Magi-2* were significantly downregulated

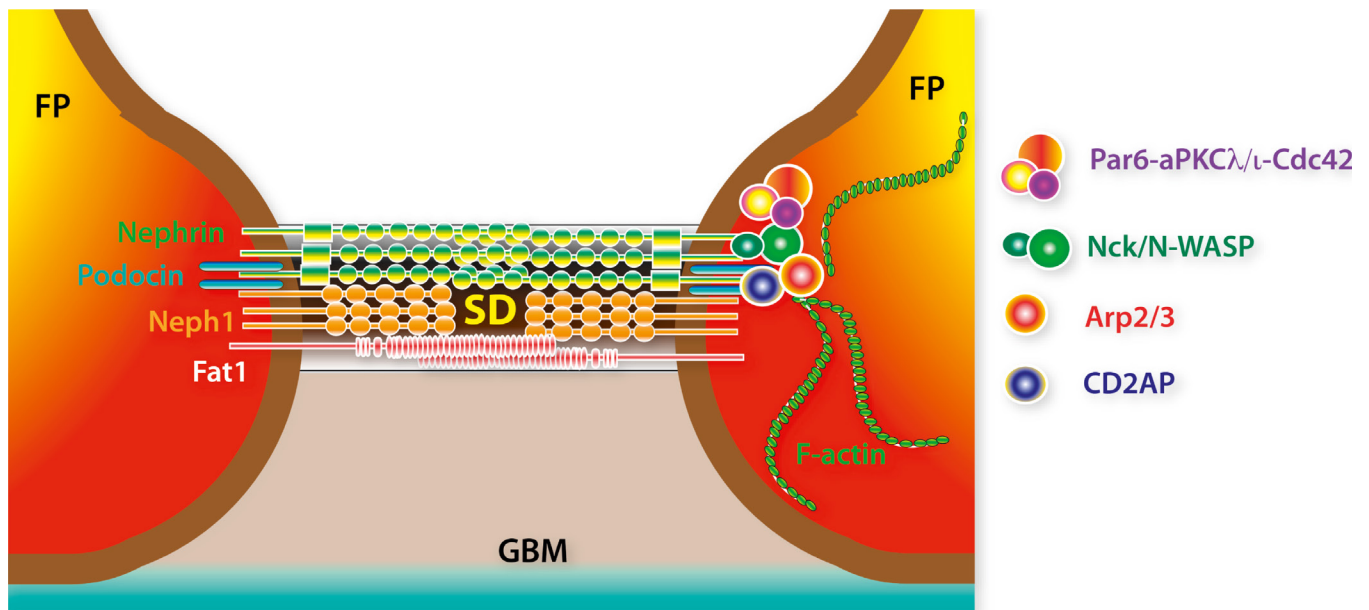


Fig. 1.18 Structural overview of the slit diaphragm (SD). An oversimplified diagram depicting the major adhesion receptors comprising the SD and how they are possibly integrated with the actin cytoskeleton of podocyte foot processes. *FP*, Podocyte foot process; *GBM*, glomerular basement membrane. (Adapted from Scott RP, Quaggin SE. Formation and maintenance of a functional glomerulus. In Little MH (ed.). *Kidney Development, Disease, Repair and Regeneration*. San Diego: Academic Press; 2016.)

in podocyte-specific *Wt1* deletion in mice,²⁸⁸ indicating that the other transcription factors cannot compensate for the *Wt1* loss for expression of these genes. Inactivation of *Lmx1b*, *Tcf21*, *Mafk*, and *Foxc2* causes podocytes to remain as cuboidal epithelia and failure to spread on the glomerular capillary bed.^{75,290-292} *Tcf21* likely acts upstream of *Mafk* as the latter is downregulated in *Tcf21*-null mice.²⁹¹ Loss of *Mafk* and *Lmx1b* reduces the expression of *Nphs1* (nephrin) and *Nphs2* (podocin), the major components of the slit diaphragm of the podocytes, whereas the absence of *Foxc2* causes the specific downregulation of *Nphs2* and $\alpha3\alpha4\alpha5$ (IV) collagen.^{75,291,293} *Lmx1b* mutations are linked to nail-patella syndrome in humans, with a subset of affected individuals manifesting nephrotic disease.^{294,295} *Wt1*, *Tcf21*, *Mafk*, *Foxc2*, and *Lmx1b* are expressed from the S-shape stage onward and remain constitutively expressed in adult glomeruli. Proteinuria develops from loss of these genes, thus underscoring the importance of normal podocyte maturation in the establishment of the glomerular filtration barrier.

Podocytes also produce VEGF-A and other angiogenic growth factors to attract vasculature into glomeruli, as described later in this chapter. It is likely that endothelial cells also produce hitherto unknown trophic factors that promote terminal differentiation and survival of podocytes.

GLOMERULAR BASEMENT MEMBRANE DEVELOPMENT

The mature glomerular basement membrane (GBM) is a fusion of the extracellular matrices (ECM) of podocytes and glomerular endothelial cells. The GBM is a highly organized and compositionally complex matrix whose abundant components include collagens (types I, IV, VI, and XVIII); laminins ($\alpha5$, $\beta2$, and $\gamma1$); nidogen-1; heparan sulfate

proteoglycans (agrin and perlecan); and tubulointerstitial nephritis antigen-like protein.²⁹⁶⁻²⁹⁸ The GBM is an essential part of the glomerular filtration barrier, functioning as an intermediary sieving matrix and a sink for secreted trophic and signaling factors, as well as mediating cellular communication between the glomerular endothelium and podocytes. Furthermore, adhesive cell-ECM interactions among the GBM, podocytes, and glomerular endothelial cells maintain the structural integrity of the glomerular filtration barrier.

Among the major components of the GBM, type IV collagens and laminins have been shown to be the most indispensable, highlighted by proteinuric kidney diseases such as Alport syndrome, Goodpasture disease, and Pierson syndrome. Alport syndrome is linked to a growing list of mutations in the genes *COL4A3*, *COL4A4*, and *COL4A5* encoding for type IV collagen subunits $\alpha3$, $\alpha4$, and $\alpha5$, respectively.^{299,300} Maturation of the GBM requires the replacement of juvenile $\alpha1\alpha1\alpha2$ with $\alpha3\alpha4\alpha5$ type IV collagen trimers, a developmental switch improving the structural resilience of the GBM.³⁰¹ In Alport syndrome, the assembly of heterotrimeric $\alpha3\alpha4\alpha5$ type IV collagen complex is compromised while the $\alpha1\alpha1\alpha2$ type IV collagen complexes persist. Because $\alpha3\alpha4\alpha5$ type IV collagen trimers constitute about half the total proteins in the mature GBM, it is not surprising that Alport syndrome GBM is severely distorted.³⁰² The importance of type IV collagens in the GBM is further underscored in Goodpasture disease, an autoimmune disorder targeting $\alpha3$ type IV collagen subunit.³⁰³ In Pierson syndrome, mutations in *LAMB2* encoding the laminin $\beta2$ subunit impair the assembly of the laminin complex LM-521 (a trimer formed among laminin- $\alpha5$, $\beta2$, and $\gamma1$ subunits).^{304,305} Deformation of the GBM and proteinuria results from loss of *Lamb2* and *Lama5* (laminin- $\alpha5$) in mice.³⁰⁶⁻³⁰⁹

DEVELOPMENT OF THE BOWMAN CAPSULE

The outermost cells of the proximal end of the S-shaped nascent nephron are the presumptive parietal epithelial cells that eventually form the Bowman capsule, which encloses the glomerular tuft and where primary urinary filtrate collects. Similar to podocytes, parietal precursors are originally cuboidal epithelial cells that progressively flatten and become squamous. Parietal epithelia express the proteins claudin-1 (*Cldn1*) and claudin-2 (*Cldn2*) that form part of tight junctions, which help contain urinary filtrate.^{310,311} Parietal epithelial cells like podocytes express *Wt1*, although at a reduced level. Whether this differential level of *Wt1* expression affects cell fate commitment to either the podocyte or parietal cell lineage warrants further investigation. Specific loss of *Cttnb1* (β -catenin) in nephrogenic epithelia from the S-shaped stage causes formation of glomerular cysts, where the Bowman capsule lacks parietal epithelia and is instead formed by podocytes notably containing foot processes and slit diaphragms.³¹² Parietal cell specification is therefore strongly dependent on canonical Wnt- β -catenin signaling, which sets it apart from podocytes where *Cttnb1* expression is dispensable.³¹³ In the absence of parietal epithelial cells, glomerular capillaries develop poorly while the mislocalized *Vegfa*-expressing podocytes cause formation of ectopic capillaries next to the abnormal Bowman capsule.³¹² It can therefore be inferred that parietal epithelial cells may function in compartmentalizing the podocyte–glomerular endothelial cell crosstalk, ensuring that glomerular endothelial cells form a well-elaborated capillary tuft covered by an adequate number of podocytes.

CESSATION OF NEPHROGENESIS

Nephrogenesis is a time-limited event that is not reactivated post-injury to the kidneys of humans and mice at adulthood. The last wave of nephrogenesis is observed around the 36th week of gestation in humans and shortly after birth in mice.^{314,315} Cessation of nephron generation is characterized by the exhaustion of NP cells and completion of epithelial differentiation of the remaining nephrogenic precursor. Although various morphologic and molecular changes occurring at this time have been characterized in mouse kidneys, the exact trigger is not fully understood.^{314,316,317} For example, increased oxygen tension after birth could possibly act as an active trigger that ends nephrogenesis,³¹⁷ but it may not be a shared mechanism in humans where nephrogenesis ceases before parturition.

In mice, the nephrogenic zone progressively shrivels after birth and is replaced by mature tubules. The NP markers *Six2* and *Cited1* are significantly downregulated after birth and become undetectable by postnatal day 3 (P3), concomitant with the disappearance of the cap mesenchyme. Multiple newly induced nephrons are found associated with each UB tip at P3 that is no longer found at P7. Ureteric branching also ends between birth and P3 accompanied by a significant reduction of *Ret* and *Wnt11* and loss of the UB ampullary shape.^{314,316} In contrast, *Wnt9b* expression in the UB remains high even at P4,³¹⁴ which means NP depletion precedes *Wnt9b* reduction in the UB.

Of course, deletion of mouse genes required for the maintenance of NPs leads to their premature exhaustion before birth. Metabolism also regulates progenitor lifespan. Glycolysis regulates the progenitor state: High glycolytic flux supports self-renewal of young (E13.5) progenitors, whereas inhibition of glycolysis induces their differentiation.³¹⁸ Methionine deficiency also results in progenitor depletion.³¹⁹ In contrast, reports of mutant mice with delayed arrest are limited. The RNA-binding protein Lin28 inhibits *let-7* microRNA expression, thereby regulating developmental timing across species. Transient overexpression of *Lin28b* or direct reduction of *let-7* in mice results in prolonged nephrogenesis,³²⁰ although the precise mechanism remains unknown. Deletion of one copy of *Tsc* in NPs results in a 25% increase in the number of nephrons.³²¹ This phenotype is independent of Mtor activity, although *Tsc* encodes hamartin, an inhibitor of Mtor. Further analysis of these mice suggests that stage-dependent differential translation, but not transcription, of Wnt agonists may tip the balance between FGF9-mediated maintenance and Wnt-mediated differentiation of nephron progenitors.^{322,323} Specifically, Wnt agonists including R-spondin are poorly translated in young (E14.5) nephron progenitors, while self-renewal promoting FGF20 expression is high, leading to progenitor maintenance. In contrast, in old progenitors (P0), translation of Wnt agonists is increased while FGF20 expression becomes low, leading to differentiation. This tipping point model proposes that when the number of old cells rises above a threshold, all remaining progenitors differentiate en masse.³²³ A thorough understanding of the mechanism of cessation of nephrogenesis is necessary to better evaluate the potential of regenerative therapies for the kidney.

DEVELOPMENT OF THE COLLECTING DUCT

The overall shape, structure, and size of the kidneys are largely guided by stereotypical branching of the UB and subsequent patterning of the collecting duct system. During late gestation, past embryonic stage E15.5 in the mouse, the trunks of the UB tree undergo extensive elongation to establish the array of collecting ducts found in the renal medulla and papilla. The radial arrangement of elongated collecting ducts together with the loops of Henle (derived from the NP) establishes the corticomedullary axis by which nephron distributions are patterned. After birth, further elongation of the newly formed collecting duct network is partly responsible for the postnatal growth of the kidney.

Elongation of the collecting duct involves oriented cell division characterized by the parallel alignment of the mitotic spindle of proliferating ductal epithelia with the longitudinal axis of the duct.³²⁴ Oriented cytokinesis therefore guarantees that daughter cells contribute to lengthening of the duct with minimal effect on tubular lumen diameter. The renal medulla and pelvis are nonexistent in mice lacking *Wnt7b*.³²⁵ Notably, the collecting ducts and loops of Henle are stubbier due to reorientation of cell division toward a radial instead of a longitudinal axis. *Wnt7b* expression is restricted within the nonbranching stalk of the ureteric tree and is absent in the ampullary UB tips. Failed

development of the renal medullary and papillary regions is also recapitulated in mice where *Ctnnb1* is ablated in the renal stroma, suggesting that Wnt7b activates the canonical β -catenin-dependent Wnt signaling pathway involving the ureteric epithelia and surrounding stroma.³²⁵⁻³²⁷ However, the relevant reverse signal from the interstitial stroma to the collecting duct that drives oriented cell division in the duct epithelia remains unknown.

Normal development of collecting ducts also depends on cell survival cues provided by diverse ligands, such as Wnt7b, EGF, HGF, and interactions with the extracellular matrix.^{325,328,329} Papillary collecting ducts display higher incidences of apoptosis in mice lacking Wnt7b or EGFR.^{325,328} Conversely, loss of Dkk1 (Dickkopf1), a secreted antagonist of Wnt7b, results in overgrowth of the renal papilla.³³⁰ The HGF-receptor Met, $\alpha 3\beta 1$ integrin (*Itga3/Itgb1*), and laminin $\alpha 5$ (*Lama5*) are all required to maintain expression of Wnt7b and thus likely support the viability of collecting duct cells.^{199,329,331}

Poor development of the renal medulla and papilla are also observed in mutant mice lacking *Fgf7*, *Fgf10*, *Fgfr2*, *Bmpr1a* (*Alk3*), and *Shh* (Sonic hedgehog). FGF7 and FGF10 are the cognate ligands of FGFR2. Renal hypoplasia is observed when *Fgfr2* is conditionally removed from the ureteric lineage and is more severe than in mutants lacking *Fgf7* or *Fgf10*, suggesting that these related ligands may have some functional redundancy in the development of the UB and collecting ducts.^{59,332,333} *Shh* is expressed in the more distal derivatives of the UB, the medullary collecting ducts, and the ureter.⁵⁸ The germline deletion of *Shh* results in either bilateral renal agenesis or a single ectopic dysplastic kidney.^{334,335} It has been shown that *Shh* controls the expression of early inductive and patterning genes (*Pax2* and *Sall1*), cell cycle regulators (*Mycn* and *Ccnd1*), and signaling effectors of the Hedgehog pathway (*Gli1* and *Gli2*). Interestingly, genetic removal of *Gli3* in an *Shh*-null background restores the expression of *Pax2*, *Sall1*, *Ccnd1*, *Mycn*, *Gli1*, and *Gli2*, providing physiologic proof for the role of *Gli3* as a repressor of the Shh pathway in renal development.³³⁵ Frameshift mutations resulting in truncation of the expressed *Gli3* protein are linked to Pallister-Hall syndrome and the presence of hydronephrosis and hydroureter in both humans and mice.^{336,337}

The mature collecting duct epithelia consist of two major cell subtypes: the abundant principal cells, which strongly express the aquaporins, ion channels, and pumps mediating Na^+ and K^+ transport, and the fewer intercalated cells, which are responsible for secretion of protons and bicarbonate ions. UB lineage-specific deletion of mouse *Mib1*, which encodes an E2 ubiquitin ligase required for initiation of Notch signaling, results in a reduced principal cell/intercalated cell ratio and nephrogenic diabetes insipidus.³³⁸ ADAM10 (a disintegrin and metalloproteinase 10) activates Notch signaling by cleaving Notch ligands, and ureteric bud-specific deletion of this gene in mice also resulted in the reduced principal cell/intercalated cell ratio.³³⁹ Thus Notch signaling regulates the balance of principal and intercalated cells in the developing collecting duct. A model has been proposed in which Notch signaling is important for fate determination at the level of individual cells³⁴⁰ (Fig. 1.19). The expression of the transcription factor *Tfcp2l1* is upregulated in some cell populations during early differentiation, leading to increased expression of intercalated cell-specific genes including the

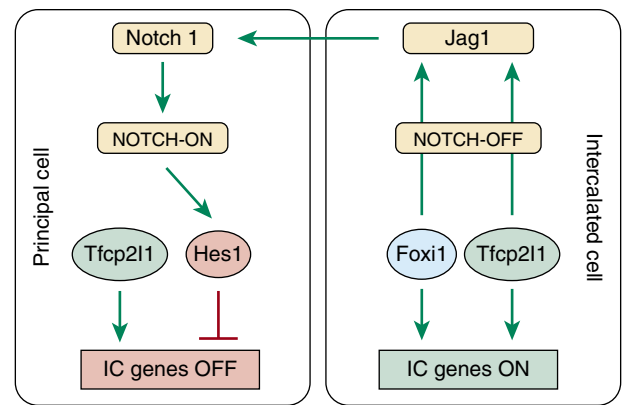


Fig. 1.19 Development of cellular diversity in the collecting duct. Lateral inhibition in collecting duct differentiation. The combination of Foxi1 and Tfcp2l1 induces the expression of intercalated (IC) genes including Jag1, which triggers Notch signaling in adjacent principal cells. This results in the activation of Hes1 (Notch-ON) and the silencing of IC genes in the principal cells. Notch activity in Jag1-expressing ICs remains low (Notch-OFF). (Adapted with permission from Werth M, Schmidt-Ott K, Leete T, et al. Transcription factor TFPC2L1 patterns cells in the mouse kidney collecting ducts. *eLife*. 2017;6:e24265.)

Notch ligand *Jag1*. In neighboring cells receiving the Notch signal, *Jag1* expression is repressed by negative feedback. On the other hand, negative feedback does not work in the signaling cell, so *Jag1* expression is maintained. The propagation of such feedback between neighboring cells determines the activity of Notch signaling at the single-cell level, which ultimately determines the fate of principal and intercalated cells. This model can explain the mosaic distribution of the two cell types. Downstream targets of Notch signaling in the principal cells include the transcription factor *Elf5*, which positively regulates the expression of the principal cell-specific genes *Aqp2* and *Avpr2*.³⁴¹ Conversely, mice lacking another transcription factor Foxi1 have defects in intercalated cells and are unable to secrete acid in response to acid load leading to metabolic acidosis.³⁴² Histologic analysis revealed the absence of differentiated principal cells and intercalated cells in these mice, and instead, cells with intermediate characteristics were present in collecting ducts. Such intermediate cells are present in approximately 2% of the collecting ducts of wild-type mice, and these cells are proposed to be immature “collecting duct progenitors.” Indeed, some intercalated cells have been shown to differentiate from AQP2-positive cells.³⁴³ Single-cell RNA sequencing (scRNA-seq) of adult mouse kidneys has also revealed the presence of cells coexpressing both markers,^{344,316} suggesting plasticity between principal cells and intercalated cells during not only development but also in adult animals. Indeed, genetic activation of Notch is sufficient to drive the transition from interstitial cells to principal cells in adult mice.³¹⁶ It remains to be determined whether both genetic changes and environmental cues can influence the plasticity of the two cell types in the collecting duct.

ELONGATION OF RENAL TUBULES AND COLLECTING DUCTS

In addition to cell differentiation, the spatial orientation of cells is essential for elongation of renal tubules and collecting ducts. In epithelia, cells are uniformly organized

along an apical-basal plane of polarity. However, in addition, cells in most tissues require positional information in the plane perpendicular to the apical-basal axis. This type of polarization is referred to as planar cell polarity and is critical for the morphogenesis of metazoans.^{345,346} Using cell lineage analysis and close examination of the mitotic axis of dividing cells, it has been shown that lengthening of renal tubules and collecting ducts is associated with mitotic orientation of cells along the tubule axis, demonstrating intrinsic planar cell polarity.³²⁴ Dysregulation of oriented cell division can give rise to cysts due to abnormal widening of tubule diameters.³⁴⁷ To date, molecules implicated in planar cell polarity and tubule elongation include *Wnt9b*, *Hnf1b*, *Pkhd1*, *Fat4*, *Celsr1*, and *Vangl2*.^{324,326,348-355}

DEVELOPMENT OF THE STROMAL CELL LINEAGE

In embryonic kidneys, the stroma is roughly organized into two distinct zones: an outer cortical stroma (interstitium) and a medullary stroma (interstitium). However, scRNA-seq analyses have revealed that there are at least more than five cell layers in the developing renal stroma along the corticomedullary axis.³⁵⁶ The outermost layer expresses *Foxd1*, while the innermost layer expresses *Wnt4*.

Foxd1 is a winged helix transcription factor and expressed in stromal cells adjacent to NPs and ureteric buds in the developing renal cortex (see Fig. 1.15). Fate mapping studies reveal that the *Foxd1*⁺ cells located at the periphery of the developing kidney represent the progenitors for intrarenal stroma (i.e., cortical and medullary interstitial cells, pericytes, smooth muscle cells, as well as renin cells, and glomerular mesangial cells).³² Genetic loss of *Foxd1* results in the thickening of the renal capsule and the formation of large metanephric mesenchymal condensates.^{30,357} The morphologically altered renal capsule in *Foxd1*-mutant kidneys has notably lost expression of *Aldh1a2* (*Raldh2*) and *Sfrp1* (a regulator of Wnt signaling), and is abnormally interspersed with endothelial cells. Transcriptome analysis reveals that the gene *Dcn*, which encodes for the collagen-binding proteoglycan decorin, is a specific target that is repressed by *Foxd1* in the cortical interstitium.³⁵⁸ *Dcn* expression is normally localized within the medullary stroma but is normally absent in the cortical stroma of wild-type kidneys. In the absence of *Foxd1*, *Dcn* becomes abundantly expressed in the presumptive cortical stromal region, and the *Foxd1*-null phenotype is partially rescued through genetic inactivation of *Dcn*.

The *Foxd1*⁺ stromal progenitors differentiate into multiple types of interstitial cells and also regulate proper differentiation of the NPs³⁵⁹⁻³⁶³ (see Fig. 1.5). When the stromal lineage is selectively annihilated by *Foxd1-Cre*-driven expression of diphtheria toxin, the zone of condensing mesenchymal cells capping the UBs is abnormally broadened while the development of pretubular aggregates is strongly hindered. This reiterates findings from the *Foxd1*-null mice, suggesting that regulation of nephrogenesis involves a crosstalk between stroma and UB-derived inductive signals. In particular, it was shown that *Fat4*-dependent Hippo signaling initiated by the stroma integrates with canonical Wnt signaling derived from the ureteric lineage in order to balance the nephron precursor propagation

and differentiation. The absence of *Fat4* in the stromal compartment or its ligands *Dchs1* and *Dchs2* from the cap mesenchyme phenocopies the expansion of the nephrogenic precursor domain and failed epithelial differentiation of NPs seen in stroma-deficient kidneys.^{360,363} It was postulated that *Fat4*, acting through the Hippo pathway, promotes the differentiation of the epithelial transition of NPs.³⁶⁰⁻³⁶³ This was further reiterated by the rescue of depletion of NPs by *Fat4* deficiency in *Wnt9b* knockouts.³⁵⁹ Interestingly, the loss of *Vangl2*, a signaling partner of *Fat4* known to regulate renal tubular diameter, fails to rescue the loss of NPs in *Wnt9b* knockouts, suggesting that *Fat4*-mediated signaling is likely independent of the planar cell polarity pathway.^{353,359} Transcriptional regulation of stroma-dependent restriction of NP expansion involves the transcription factors *Sall1*, *Foxd1*, and *Pbx1*.^{30,357,47} *Sall1* is likely an upstream regulator of many of the stroma functions as *Sall1* binds directly to several stroma-related gene loci, including *Fat4*, *Dcn*, *Pbx1*, *Tcf21*, *Meis1*, and *Hoxd10*.³⁶⁴ Loss of *Sall1* in the stroma downregulates *Fat4* expression and results in an excess of *Six2*⁺ NPs.

The *Foxd1*⁺ stromal progenitors also regulate ureteric branching. Retinoids secreted by the renal stroma are recognized as important for the maintenance of a high level of *Ret* receptor expression in the UB tip, promoting the proliferation of UB epithelial cells and growth of the ureteric tree.^{19,365-367} The defective UB branching seen in *Foxd1*-null mutants is most likely a direct consequence of the loss of cortical expression of *Aldh1a2*, a gene involved in retinol synthesis.³⁵⁷ It has been shown that renal stroma immediately around the UB tips is also important in regulating the bifurcation of the tips and creation of new UB branches.³⁶⁸ Autocrine retinoid signaling in the stromal cells juxtaposed to the UB tips stimulates the expression of extracellular matrix 1 (*Ecm1*). *Ecm1* is specifically expressed at the UB cleft, where it suppresses and restricts *Ret* expression domains within the UB tips. In the absence of *Ecm1*, *Ret* expression in the UB tips broadens, effectively attenuating UB branching due to impaired formation of UB bifurcation clefts. Thus stromal retinoids promote and confine *Ret* expression domains and more likely cell proliferation patterns within the UB tips. In addition, *Fat4* deletion results in excessive *Ret* signaling and duplex kidney phenotypes. Mechanistically, *FAT4*, expressed in the NPs and/or stromal progenitors, interacts with the extracellular domain of *RET* and reduces *RET* signaling, thereby fine-tuning the ureteric bud development.³⁶⁹

A basic helix-loop-helix (bHLH) transcription factor *Tcf21* is expressed in the medullary stroma and the condensing MM.^{290,370} *Tcf21* is also expressed in a number of differentiated renal cell types including developing podocytes of the renal glomerulus, cortical and medullary peritubular interstitial cells, pericytes surrounding small renal vessels, and adventitial cells surrounding larger blood vessels.³²⁶ *Tcf21*-null mice show a disruption of branching morphogenesis with an associated arrest and delay in nephrogenesis.^{290,357,371} The analysis of chimeric mice³⁷¹ shows that the glomerulogenesis defect is rescued by the presence of wild-type stromal cells (i.e., mutant cells will epithelialize and form nephrons normally as long as they are surrounded by wild-type stromal cells). In addition, there is

a cell-autonomous requirement for *Tcf21* in stromal lineage: *Tcf21*-null cells were unable to differentiate into interstitial and pericyte cell lineages of the cortex and medulla.

scRNA-seq analyses have identified many more genes that are expressed in the stroma, and functional studies of these genes are needed to understand developmental mechanisms of the renal stroma. It is also important to determine whether such mechanisms are conserved across species. For example, *Foxd1* is selectively expressed in the stromal progenitors in mice, but human *FOXD1* is also expressed in NPs¹⁰¹; thus it does not serve as a marker for stromal progenitors in humans. scRNA-seq analysis of human embryonic kidneys, combined with kidney organoid studies, will be useful to resolve these discrepancies.

DEVELOPMENT OF THE MESANGIUM

Mesangial cells are derived from *Foxd1*+ stromal progenitors³² and grow into the developing glomerulus during the S-shaped body stage, eventually locating between the capillary loops. Transcription factor GATA3 mutations in humans cause hyperparathyroidism, renal dysplasia, and deafness (HDR) syndrome.³⁷² Although most of the renal symptoms in this disease are likely caused by ureteric bud defects, mesangial abnormalities may also be involved. In mice, *Gata3* is expressed in not only the ureteric bud but also a subset of stromal progenitors, cells migrating into the vascular cleft of the S-shaped bodies, and mesangial cells integrated into glomeruli.³⁷³ *Gata3* continues to be expressed in adult mesangial cells. *Gata3* heterozygous mice have fewer mesangial cells, suggesting the importance of this gene in mesangial development. A stromal cell lineage-specific deletion of *Gata3* is awaited.

Mice carrying a hypomorphic allele of *Notch2* are born with a reduced number of glomeruli that lack both endothelial and mesangial cells.^{374,375} The downstream Notch signaling target, *Rbpj*, has also been described as crucial for the proper development of the renal vasculature and the glomerular mesangium. Conditional inactivation of *Rbpj* in the *Foxd1*-expressing stromal lineage leads to profound renal maldevelopment and early postnatal death.³⁷⁶ These mice show glomerular dilatation and lack of mesangial cells and renin cells. These studies highlight the significance of Notch signaling within the stromal lineage in the establishment of the mesangial cells.

Interactions with the vascular endothelial cells are critical for mesangial cell development. Gene deletion studies have demonstrated an essential role for PDGF-B/PDGFR- β signaling in this process (Fig. 1.20). Deletion of the *Pdgfb* gene, which is expressed by glomerular endothelia, or the PDGFR- β receptor gene (*Pdgfrb*), which is expressed by mesangial cells, results in glomeruli with a single balloon-like capillary loop, instead of the intricately convoluted glomerular capillaries of wild-type kidneys. Furthermore, the glomeruli contain no mesangial cells.³⁷⁷ Endothelial cell-specific deletion of *Pdgfb* results in the same glomerular phenotype and shows that production of PDGF-B by the endothelium is required for mesangial migration.³⁷⁸ PDGF-B-dependent recruitment of mesangial cells into nascent glomeruli also requires the coreceptor neuropilin-1 (*Nrp1*).³⁷⁹ *Nrp1*-deficiency within the *Pdgfrb*+ cell lineage in mice phenocopies the glomerular aneurysm seen upon loss

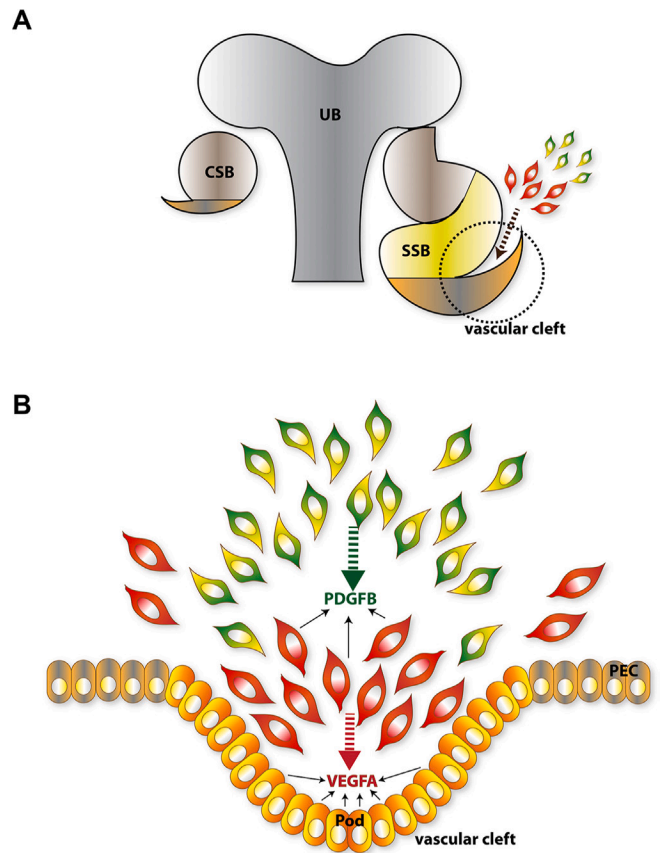


Fig. 1.20 Sequential recruitment of endothelial and mesangial cells into the developing glomerulus. (A) The glomerulus forms around the vascular cleft at the proximal end of S-shaped intermediate nephrons. (B) Within the vascular cleft, presumptive podocytes (*Pod*) secrete VEGFA that attracts VEGFR2-expressing angioblasts (red), which are the precursors of GECs. The angioblasts are followed by mesangial cell precursors expressing PDGFR- β (green), which are attracted by PDGFB secreted by the endothelial cells. CSB, Comma-shaped body; PEC, parietal epithelial cell; SSB, S-shaped body; UB, ureteric bud. (Adapted from Scott RP, Quaggin SE. Formation and maintenance of a functional glomerulus. In Little MH (ed.). *Kidney Development, Disease, Repair and Regeneration*. San Diego: Academic Press; 2016.)

of either *Pdgfb* or *Pdgfrb*. In vitro, mesangial cell chemotaxis, but not cell proliferation and survival in response to PDGF-B, strongly requires the mesangial expression of *Nrp1*. Furthermore, loss of the mesangium resulting from *Nrp1* ablation results in delamination of the glomerular endothelium and inward buckling of the GBM. These findings altogether underscore the key supportive role of the mesangium for development and maintenance of a highly convoluted glomerular capillary tuft. Indeed, targeted injury to the mesangium, with either snake venom or antibodies against *Thy1*, results in secondary damage to the glomerular endothelium and glomerular aneurysms.³⁸⁰⁻³⁸⁴ Mesangial cells and the matrix they produce are required to pattern the glomerular capillary system. Loss of podocyte-derived factors such as VEGF-A also leads to failure of mesangial cell ingrowth, likely through the primary loss of endothelial cells and subsequent failure of PDGF-B signaling.³⁸⁵

Developing mesangial cells also express integrin alpha 8, a receptor for nephronectin, which is produced by podocytes and deposited into the GBM. Deletion of

nephronectin in the nephron lineage or podocytes leads to a marked increase in mesangial cell number and mesangial sclerosis.³⁸⁶ Therefore mesangial cell adhesion to the GBM mediated by the nephronectin-integrin $\alpha 8\beta 1$ interaction exerts an antiproliferative effect that likely occurs after the establishment of mesangial cell–endothelial cell interactions.

scRNA-seq analyses of developing glomeruli show that mesangial cells have unique gene expression profiles and are distinct from other stromal cell types.^{387–390} Functional studies of these genes will further elucidate the detailed developmental mechanisms of mesangial cells.

RENIN CELLS AND THE JUXTAGLOMERULAR APPARATUS

The juxtaglomerular apparatus (JGA) consists of cells that line the afferent arteriole, the macula densa cells of the distal tubule, and the extraglomerular mesangial cells that are in contact with intraglomerular mesangium.³⁹¹ Renin-expressing cells are located in afferent arterioles and are derived from *Foxd1*-expressing stromal progenitors,³⁹² as are mesangial cells. scRNA-seq analyses show that renin cells have similar but distinct gene expression profiles to mesangial cells.^{387–390} As described earlier, stroma-specific deletion of Notch signaling leads to loss of renin cells and mesangial cells,³⁹³ suggesting a common developmental pathway for these two cell types. Gene promoter analysis indicates that renin expression is dependent on the Notch signaling pathway. The conditional ablation of *Rbpj* in renin cells results in severe paucity of juxtaglomerular cells with consequential decrease in overall renin expression and the development of lower blood pressure.³⁹⁴

In contrast, macula densa cells are part of the distal tubule epithelia and are derived from Six2+ NPs. PAPP2 has been shown to be expressed in the presumptive macula densa precursors even from the S-shaped body stage, at least in humans.¹⁰² These cells are near-developing podocytes, suggesting that the overall JGA configuration may be established early in nephron development.

Renin converts angiotensinogen to angiotensin I, which is further converted to angiotensin II by angiotensin-converting enzyme (ACE).³⁹⁵ The renin-angiotensin-aldosterone axis is required for normal renal development. In humans, the use of ACE inhibitors during pregnancy has been associated with congenital defects including renal anomalies.^{396,397} Two subtypes of angiotensin receptors exist: AT1 receptors are responsible for most of the classically recognized functions of the renin-angiotensin system (RAS), including pressor effects and aldosterone release mediated through angiotensin; the type 2 receptors oppose the actions of the AT1 receptors.³⁹⁸ Genetic deletion of angiotensinogen or ACE results in hypotension and defects in formation of the renal papilla and pelvis.^{399–402} Humans have one *AT1* gene, whereas mice have two: *Agtr1a* and *Agtr1b*. Mice carrying a knockout for either AT1 receptor alone exhibit no major defects,^{403,404} whereas combined deficiency phenocopies the angiotensinogen and ACE phenotypes.^{405,406} Lineage-specific deletion is required to determine which cell types are responsible for receiving the angiotensin signal. Genetic deletion of the AT2 receptor leads to increased blood pressure but does not cause major impairment of renal development.^{407,408} However, an association between

Agtr2-deficiency and malformations of the collecting duct system, including vesicoureteral reflux and ureteropelvic junction obstruction, has been reported.⁴⁰⁹

IN VITRO INDUCTION OF THE STROMAL LINEAGE AND GENERATION OF THE HIGHER-ORDER KIDNEY STRUCTURE

The kidney has many branching collecting ducts in the center that are connected to numerous nephrons in the periphery. This type of “higher order” structure is essential for kidney function to produce and excrete urine. In addition, at the periphery of the developing kidney, NPs cap dichotomously branched ureteric bud tips, forming the nephrogenic niche. As described earlier, *Foxd1*-positive stromal progenitors, located in the dorsal outer layer adjacent to NPs within the nephrogenic niche, play a critical role in ureteric bud branching and proper nephron differentiation in the developing kidney. Indeed, the addition of mouse embryo-derived stromal progenitors is required to generate such a higher order structure in vitro from mouse ESC-derived NPs and ureteric buds⁸⁹ (Fig. 1.21). Since stromal progenitors are derived from the posterior IM¹² (see Fig. 1.10), the culture condition that can generate stromal progenitors from the posterior IM was investigated, which revealed that the combination of retinoic acid, Fgf, Shh, and a Bmp antagonist enhanced the induction of *Foxd1*-positive dorsal stromal progenitors (see Fig 1.9). The requirement for Fgf signaling at this stage is consistent with the NP lineage.⁴¹⁰ Conversely, retinoic acid acts as a critical lineage-specifying signal to differentiate stromal progenitors, the presence of which in turn suppresses NP differentiation from the posterior IM. In contrast, the Bmp antagonist and Shh presumably contribute to priming the dorsally expressed gene regulatory network within the metanephric stromal populations. The resulting mESC-derived stromal progenitors reconstituted the higher-order kidney structure when combined with ESC-derived NPs and ureteric buds (Fig. 1.22). When transplanted into mice, the ESC-derived stromal progenitors differentiated not only cortical and medullary interstitial cells but also mesangial cells and renin-producing cells.⁴¹⁰ Application of this technology to human PSCs is awaited.

URETER DEVELOPMENT

The tip of the ureteric bud that emerges from the Wolffian duct repeatedly branches and elongates to form a collecting duct in the kidney. On the other hand, the extrarenal part of the ureteric bud becomes the ureteric epithelium and the development of the ureter proceeds through interactions between the ureteral epithelia and surrounding stroma. This periureteral stroma is derived from the Tbx18+ stromal progenitors at E11.5 in mice, but not from *Foxd1*+ intrarenal stromal progenitors.⁴¹¹

Two major signaling pathways are known to be important for ureteral development in mice. FGF7/10 secreted from the surrounding stroma enhances Shh expression in the ureteral epithelium.^{412,413} SHH signaling from the ureteral epithelium induces BMP4 expression in the surrounding stroma through the transcription factor *Foxf1*. BMP4 acts on the stroma itself to promote smooth muscle cell differentiation. BMP4 also acts back on the

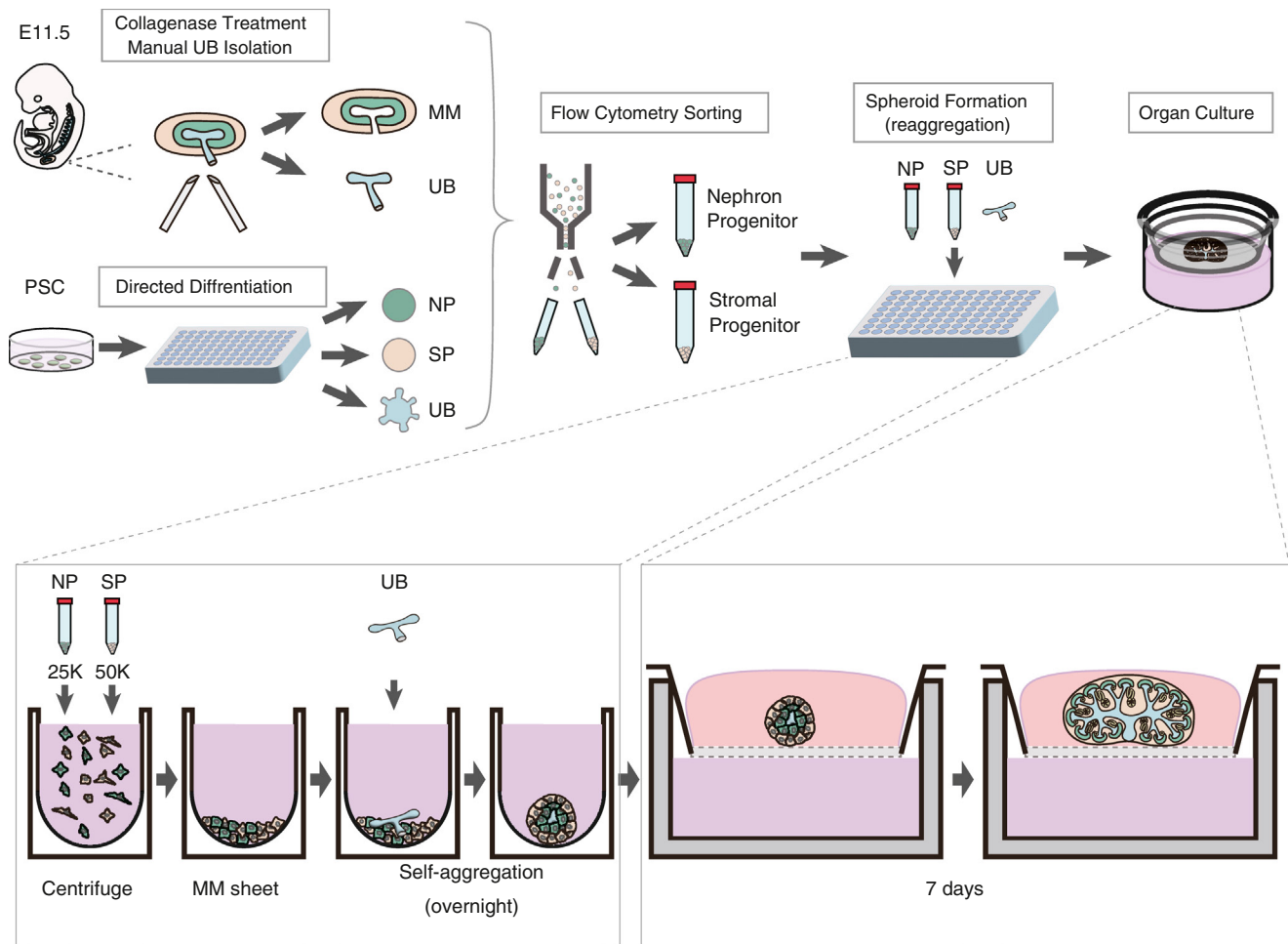


Fig. 1.21 In vitro reconstitution of higher-order kidney structures. Nephron progenitors (NP) and stromal progenitors (SP) are either isolated from E11.5 embryonic kidney or selectively induced from pluripotent stem cells (PSC). The obtained cells are further purified by flow cytometry sorting and counted. 25,000 NP cells and 50,000 SP cells are seeded into 96-well low cell binding plates and centrifuged to form sheetlike structures. Isolated embryonic ureteric bud (UB) or PSC-derived UB is placed in the center of this cell sheet, allowing the spontaneous formation of spheroids overnight. The reassembled spheroids can then differentiate into self-organized embryonic kidney structures including branched collecting ducts, peripheral nephron progenitors, and differentiated epithelial nephrons within a week in the organ culture setup.

ureteral epithelium to promote its development.^{58,414} Another important pathway is the stromal expression of the transcription factor *Tbx18* in response to Wnt (Wnt7b/9b) signaling from the ureteral epithelium.⁴¹⁵ Mutations in *Tbx18* are known to cause CAKUT in both humans and mice.^{411,416} The most overt phenotype of *Tbx18* inactivation in mice is the onset of hydronephrosis and hydroureter as a result of impaired development of the ureteral smooth muscle cells.^{411,417}

In summary, ureteral development is regulated by two signaling pathways, SHH-FOXF1-BMP4 and WNT7B/9B-TBX18, in the ureteral epithelium and surrounding stroma. The overall ureteral development is complete at approximately E18.5 in mice (Fig. 1.23). These in vivo findings will be useful for in vitro ureter generation, which has yet to be achieved.

MOLECULAR GENETICS OF VASCULAR FORMATION

We still know little how the complex vascularization of the kidney is coordinated with epithelial and stromal

development. During kidney development, the formation and elaborate patterning of an arterial, venous, and capillary blood vascular network involves a combination of both vasculogenesis and angiogenesis. In addition, lymphangiogenesis underlies the development of the lymphatic vasculature from veins. The major renal vessels align close to the branching UB and likely elaborate through angiogenesis into large-caliber afferent and efferent distributaries.⁴¹⁸ Vasculogenesis in the kidney likely arises from sporadic endothelial cells within the embryonic kidney that organize to form a primitive vascular network and then give rise to most of the peritubular capillaries.⁴¹⁸⁻⁴²¹ Additionally, vasculogenesis within the S-shaped nephron intermediates establishes the glomerular capillaries.⁴²²⁻⁴²⁵

Grafting studies demonstrate that transplanted embryonic kidneys can become vascularized from the invasion of host-derived extrinsic blood vessels.^{426,427} However, genetic labeling studies provide compelling evidence for the existence of endogenous endothelial precursors within the kidney.^{418,419,422,423,428} Most of the vascular endothelial cells in the kidney are considered to be derived from the lateral plate mesoderm, while the adjacent stromal cells, including

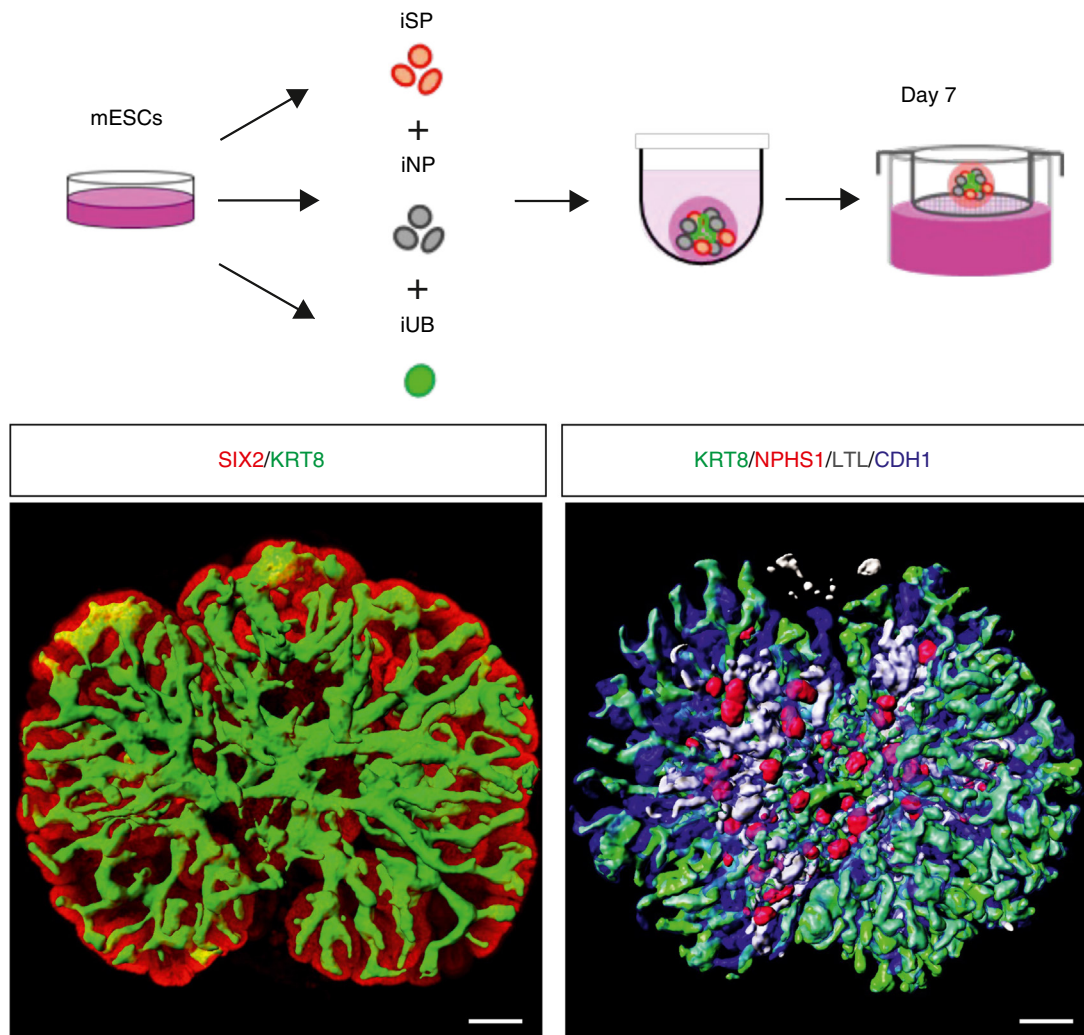


Fig. 1.22 In vitro reconstitution of higher-order kidney structures from pluripotent stem cells. “Higher-order” kidney structures are generated in vitro when stromal progenitors (iSP), nephron progenitors (iNP), and a uterine bud (iUB) derived from mouse embryonic stem cells (ESCs) are assembled and cultured at the air/liquid interface for 7 days. Note the multiply branched collecting ducts (KRT8) surrounded by nephron progenitors (SIX2) at the periphery and the formation of numerous glomeruli (NPHS1), proximal renal tubules (LTL) and distal tubules (CDH1). (Adapted with permission from Tanigawa S, Tanaka E, Miike K, et al. Generation of the organotypic kidney structure by integrating pluripotent stem cell-derived renal stroma. *Nat Commun.* 2022;13(1):611.)

pericytes and smooth muscle cells, are derived from the posterior IM via stromal progenitors.

At E11.5 in mice, the kidneys are located close to the common iliac arteries (Murakami⁴²⁹). At E12.5, short stalks bud out from the aorta and start to branch into the kidney. One of the branches elongates caudally and connects to the common iliac artery, of which connection disappears at E13.5. The overall framework of the extrarenal vasculature is established by E15.5.

Intrarenal vascularization is initiated at E11 in the mouse when systemic vessels from embryonic circulation circumscribe around the UB.^{430,431} From E13.5, the endothelial network surrounds the cap mesenchyme and UB in a cyclical manner by which endothelia form across and in close contact with the bifurcating UB. Endothelial cells may not be required for the initial morphogenesis of nephrons and ureteric buds because combination of the NPs, ureteric buds, and stromal progenitors is sufficient to generate the primitive kidney structure in vitro.^{89,410} However, given

the importance of oxygen levels in nephrogenesis,⁴³²⁻⁴³⁴ the intimate integration of the renal vascular plexus, which carries oxygen-delivering erythrocytes, is likely to be essential for kidney maturation.

Conditional gene targeting experiments and podocyte-specific deletion of *Vegfa* demonstrate that VEGF-A signaling is required for formation and maintenance of the glomerular filtration barrier.^{385,435} Glomerular endothelial cells express VEGFR2 as they migrate into the vascular cleft. Although a few endothelia migrate into the developing glomeruli of *Vegfa* podocyte conditional knockout mutants (likely due to a small amount of VEGF-A produced by presumptive podocytes before Cre-mediated genetic deletion), the endothelia fail to develop fenestrations and rapidly disappear, leaving capillary “ghosts” (Fig. 1.24). Deletion of a single *Vegfa* allele from podocytes leads to glomerular endothelial defects known as endotheliosis, a phenomenon involving hypertrophy of glomerular endothelial cells and loss of fenestrations, progressing to capillary occlusion and

thrombosis (thrombotic microangiopathy).⁴³⁵ Inactivation of *Vegfr2* phenocopies endotheliosis and thrombotic microangiopathy, indicating that VEGFA primarily

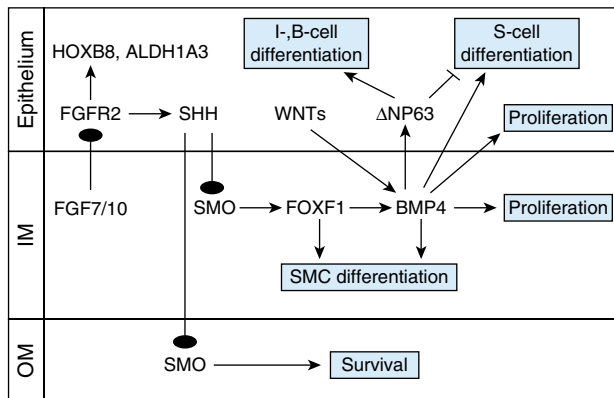


Fig. 1.23 FGF-SHH-BMP4 signaling axis in early ureter development. FGF signaling in the ureteric epithelium augments Shh expression, which in turn activates the SHH-FOXF1-BMP4 signaling pathway in the ureteric mesenchyme, leading to mesenchymal and epithelial proliferation and differentiation. *IM*, inner mesenchyme; *OM*, outer mesenchyme; *SMC*, smooth muscle cell. (Adapted with permission from Meuser M, Deuper L, Rudat C, et al. FGFR2 signaling enhances the SHH-BMP4 signaling axis in early ureter development. *Development*. 2022;149:dev200021.)

signals through VEGFR2 in supporting the development and maintenance of the glomerular endothelium.⁴³⁶ Furthermore, tight regulation of the dose of VEGF-A is essential for proper formation of the glomerular capillary system. The molecular basis for the dosage sensitivity is unclear at present and is particularly intriguing given the inducible regulation of VEGF-A by hypoxia-inducible factors (HIFs) at a transcriptional level. Similarly, VEGF-A signaling is essential for the development of the peritubular capillaries. Ablation of *Vegfa* in renal tubules did not cause gross histologic disturbance in the kidney but led to a dramatic reduction of peritubular capillaries and an abnormal elevation in renal erythropoietin production (see Fig. 1.24).⁴³⁷ This uncovered an important tubulovascular crosstalk involved in the promotion of erythrocyte development by the kidney.

A second major receptor tyrosine kinase (RTK) signaling pathway required for maturation of developing blood vessels is the angiopoietin-Tie signaling system. Angiopoietin 1 (*Angpt1*) stabilizes newly formed blood vessels and is associated with loss of vessel plasticity and concurrent recruitment of pericytes or vascular support cells to the vascular wall.⁴³⁸ The molecular pathway leading to vessel maturation through the activation of Tie2 (encoded by the gene *Tek*, the major receptor for Angpt1) is not known and appears to be independent of

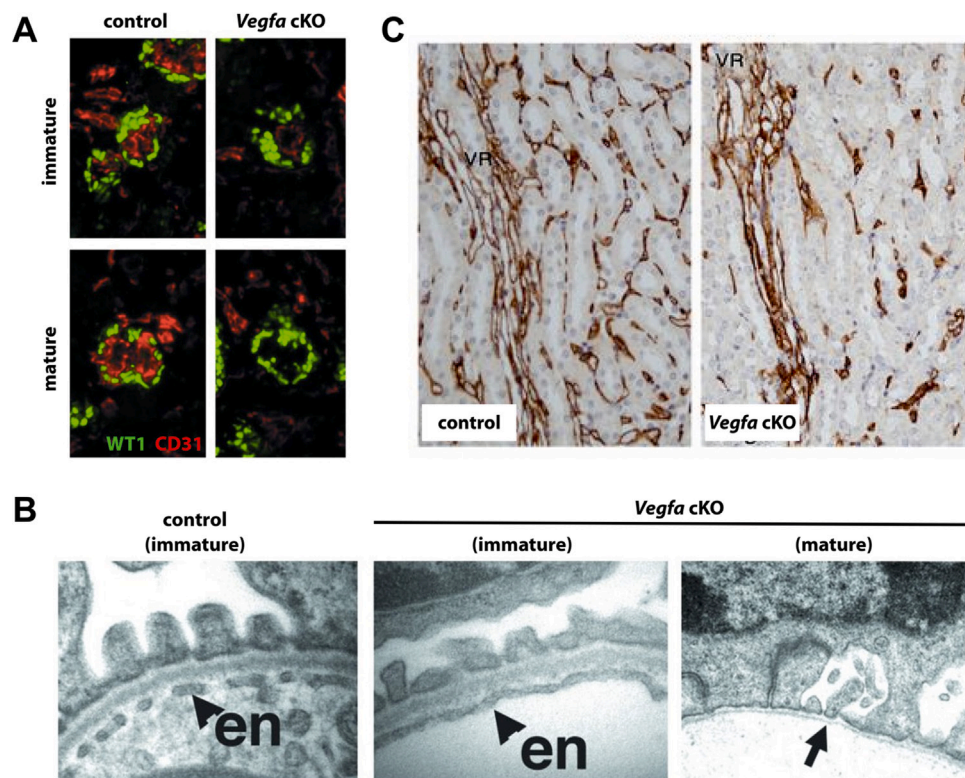


Fig. 1.24 VEGF-A is essential for development of glomerular and peritubular capillaries. (A) *Vegfa* inactivation in podocytes leads to recruitment of fewer endothelial cells in the glomerulus, which are subsequently lost. Podocytes and endothelial cells are stained with WT1 (green) and CD31 (red), respectively. (B) Transmission electron micrograph showing that in podocyte-specific *Vegfa* knockouts, the glomerular endothelium (*en*) lacks fenestrae (arrowheads) and eventually detach leaving the glomerular basement membrane bare (arrow). (C) *Vegfa* ablation in renal tubules leads to a significant reduction in peritubular capillaries (stained brown for CD34). *VR*, Vasa recta. (Panels (A) and (B) were adapted from Eremina V, Sood M, Haigh A, et al. Glomerular-specific alterations of VEGF-A expression lead to distinct congenital and acquired renal diseases. *J Clin Invest*. 2003;111:707–716. Panel (C) adapted from Dimke H. Tubulovascular cross talk by vascular endothelial growth factor A maintains peritubular microvasculature in kidney. *J Am Soc Nephrol*. 2014;26:1027–1038.)

the platelet-derived growth factor (PDGF) signaling system that is required for pericyte recruitment. The importance of *Angpt1* in promoting the development of the renal microvasculature was first suggested on the basis of observations that exogenous *Angpt1* enhances the growth of interstitial capillaries in mouse metanephric organ cultures.⁴³⁹ Because *Angpt1*-null mice perish embryonically at around E12.5, the *in vivo* role of *Angpt1* during renal development was gleaned using an inducible knockout strategy.^{438,440} Ablation of *Angpt1* deletion at around E10.5 results in general dilation of renal blood vessels, including the glomerular capillaries sometimes observed as simplified single enlarged loops.⁴⁴⁰ A marked reduction of mesangial cells was also observed in *Angpt1*-deficient mutants. Without *Angpt1*, a few endothelial cells are seen detached from the glomerular basement membrane. In contrast, it is proposed that angiopoietin 2 (*Angpt2*) functions as a context-dependent antagonist of the Tie2 receptor (*Tek*).^{393,441} Consistent with this hypothesis, overexpression of *Angpt2* in transgenic mice results in a phenotype similar to the *Angpt1* or *Tek* knockout mice.⁴⁴² *Angpt2*-null mice are viable but exhibit defects in peritubular cortical capillary development.⁴⁴³ Inducible deletion of an angiopoietin Tie2 receptor or both angiopoietin-1 and angiopoietin-2 prevents ascending vasa recta formation in mice, leading to urinary concentration defects and interstitial fluid retention that culminates in the formation of interstitium-derived medullary cysts.⁴⁴⁴

Renal vascular development also relies on a third tyrosine kinase-dependent signaling mediated by ephrins and Eph family receptors, which are better known for their involvement in axon guidance and specification of arterial and venous cell fates.^{445,446} Ephrins and their cognate receptors are expressed widely during renal development. Overexpression of *Ephb4* leads to defects in glomerular arteriolar formation, whereas conditional deletion of *Efnb2* (*EphrinB2*) from perivascular smooth muscle cells and mesangial cells leads to glomerular vascular abnormalities.^{447,448} How this occurs is not entirely clear.

An additional pathway that is likely to play a role in glomerular endothelial development and perhaps of the entire renal vasculature is the SDF1-CXCR4 axis. CXCR4, a G-protein-coupled chemokine receptor, is strongly expressed in endothelial cells. SDF1 (encoded by the gene *Cxcl12*), the only known ligand for CXCR4, is expressed in a dynamic segmental pattern in podocytes and later in the mesangial cells of the glomerulus.⁴⁴⁹ Embryonic deletion of either *Cxcl12* or *Cxcr4* does not preclude nephrogenesis but results in the defective formation of blood vessels, notably an abnormal patterning of the renal vasculature and development of a simplified and dilated glomerular capillary tuft.⁴⁵⁰ Genetic loss of CXCR7, which is thought to act as a decoy receptor for SDF1, interestingly phenocopies defective development seen in SDF1- and CXCR4-mutant mice. Unlike CXCR4, CXCR7 is specifically expressed by podocytes and not endothelial cells.⁴⁵¹ It has been proposed that CXCR7, acting as a scavenger receptor, establishes an SDF1 morphogen gradient preventing feedback inhibition of CXCR4 receptor expression in target cells. Consistent with this, inactivation of *Cxcr7* distinctively causes downregulation of *Cxcr4* expression in the renal cap mesenchyme and glomerular tuft. Thus the spatial regulation of SDF1-CXCR4

signaling appears to be important for normal development of the glomerular vasculature.

Two transcription factors belonging to the large Sry-related HMG box (Sox) gene family, named *Sox17* and *Sox18*, have distinctive and overlapping expression in vascular endothelial cells.⁴⁵² A point mutation in *SOX18* in humans has been implicated in hypotrichosis-lymphedema-telangiectasia (HLT) syndrome, which affects hair, lymph, and blood vessel vasculature.⁴⁵³ Kidneys from *Sox17/Sox18* double-null mutant mice have hypoplastic outer medulla, and the radiating outer medullary vascular bundles of the vasa recta are missing.

Despite these findings, how the complex renal vasculature network is formed still remains unknown. scRNA-seq analyses have identified heterogeneity in renal endothelial cells,⁴⁵⁴ and representative genes for each cell type, including glomerular endothelial cells and vasa recta, are being identified. Functional *in vivo* validation of these genes will be necessary to fully understand the development of renal vasculature.

LYMPHATIC DUCT DEVELOPMENT

Renal lymphatics play important roles in interstitial fluid homeostasis and regulation of the immune response. Lymphatic vessels have been identified surrounding the renal artery in the hilum, in the arcuate and interlobular arteries in the cortex, and are also found in the renal capsule.⁴⁵⁵⁻⁴⁵⁷ These lymphatic vessels are known to express the hyaluronan receptor *Lyve1*.^{444,458} The medulla was previously thought to lack lymphatic vessels, but a study of angiopoietin-Tie2 signaling in the kidney reveals that the ascending vasa recta represent hybrid vessels that express markers of both blood (*Cd34*, *Emcn*, *Pecam1*, and *Plvap*) and lymphatic endothelial cells (*Prox1* and *Vegfr3*) (see Fig. 1.24).⁴⁴⁴ This underscores a novel lymphatic circuit for the drainage of medullary interstitial fluid as part of the osmoregulation of urine. VEGF-C signaling via the receptor VEGFR3 is essential for lymphangiogenesis.⁴⁵⁹⁻⁴⁶² *Vegfr3* deficiency in adult mice causes extravasation of fibrinogen into the renal interstitium.⁴⁶³ How the renal lymphatic vasculature is particularly remodeled upon inactivation of VEGF-C/VEGFR3 signaling remains unknown.

NEURAL DEVELOPMENT

Renal vascular tone and urinary functions are regulated by a dense neural network in the kidney that relays bidirectional signals to the brain.⁴⁶⁴ Glomerular filtration rate, renal blood flow, tubular resorption of fluid, electrolytes, and urinary solutes, as well as the secretion of renin, are regulated by sympathetic innervations of the glomeruli, renal tubules and blood vessels.⁴⁶⁵ Over the past decade, the renal sympathetic innervation has attracted considerable attention after it has been recognized that persistently elevated renal sympathetic nerve activity contributes to the pathogenesis of renal hypertension.^{466,467} In particular, it has been shown that surgical excision of sympathetic input to the kidneys can alleviate refractory hypertension.^{468,469} The distribution of afferent and efferent nerves to the kidney has been partially mapped, but the developmental program involved in establishing them is largely unknown.⁴⁷⁰⁻⁴⁷²

Fate mapping and molecular studies specifically addressing the origin and development of renal nerves have yet to be reported, although many lessons have been learned about guidance pathways in play in both neural and vascular development in other systems. Neurons, like early blood vessels, appear to closely track the branching UB in cultured embryonic kidneys.^{418,430,473} It can be speculated that the sympathetic innervation and vascularization of the kidney are coordinately synchronized with inductive events between the UB and the MM.

POSTNATAL MATURATION OF THE KIDNEY

Compared with the perinatal kidney, the adult kidney has more sophisticated structures and functions, including glomerular podocytes with intricately interdigitating foot processes, proximal tubules with greater reabsorption capacity and vitamin D metabolism, elongated collecting ducts capable of tight water and pH control, and specialized stromal cells that secrete hormones (e.g., erythropoietin and renin). How the perinatal kidney acquires these characteristics after birth is an important area for future research. The transcription factor HNF4A is known to regulate maturation-associated genes in the perinatal proximal tubules.^{474,475} Nongenetic parameters such as oxygen, metabolites, hormones, and mechanical forces may also contribute to the maturation process. Spatial metabolomics has identified a shift from glycolysis to fatty acid beta-oxidation in proximal renal tubules.⁴⁷⁶ Flow in devices has improved some aspects of function in kidney organoids.⁴⁷⁷ However, the processes of kidney maturation remain largely unknown. Elucidating the mechanisms of kidney maturation will contribute to further maturation of kidney organoids and establishment of treatment in immature infants.

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